

Ownology

SOP Library

Standard Operating Procedures library — 38 SOPs across 12 categories. Generated 2026-06-28.

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Additions & Chemistry

Acid Adjustment — Tartaric Addition

CELLAR-READY QUICK STEPS

["Confirm MLF is complete or not required.", "Run bench trials at 0.5–2.0 g/L.", "Calculate dose and weigh tartaric accurately.", "Add slowly during strong tank mixing.", "Re-test pH, TA, and taste next day."]

FULL PROCEDURE

Step 1 — Confirm the need for tartaric adjustment

Review the finished wine's pH, titratable acidity (TA), sensory profile, and style target before dosing. This SOP is for **tartaric acid addition** to raise perceived brightness and microbiological stability, not for fixing faults caused by oxidation, VA, or poor ripeness. As a practical cellar target, most table wines sit best with a pH in the **3.20–3.60** range, but adjust to variety and style. Do **not** add acid blindly: verify the wine is dry, settled, and not undergoing malolactic fermentation (MLF) unless the winemaking plan specifically allows the addition before MLF.

Step 2 — Check MLF status and timing

If MLF is planned or likely, delay tartaric addition until MLF is complete unless the winemaking plan and winemaker approve otherwise. Tartaric additions before or during MLF can upset bacterial activity, increase the risk of instability, and make sensory outcomes harder to predict. Confirm completion with paper chromatography, enzymatic analysis, or validated cellar records. Do not rely on taste alone. If the wine is post-MLF, rack off gross lees, settle, and ensure no active CO₂ release or temperature rise is occurring.

Step 3 — Calculate the addition

Base the dose on bench trials wherever possible. As a guide, **1.0 g/L** tartaric acid increases TA by approximately **1.0 g/L** and typically lowers pH slightly, though the pH response depends on wine composition and buffer capacity. Common additions are **0.5–2.5 g/L** in one treatment; avoid large single jumps unless the winemaking plan requires it. Calculate total mass using:

kg required = dose (g/L) × volume (hL) ÷ 10

Example: 50 hL at 1.5 g/L = $1.5 \times 50 \div 10 = 7.5$ kg tartaric acid.

Step 4 — Run a bench trial before tank dosing

Prepare at least three bench samples, typically **100–200 mL** each, with candidate additions such as **0.5, 1.0, 1.5 and 2.0 g/L**. Mix thoroughly, allow the samples to sit for **12–24 h** at cellar temperature, then re-check pH and taste. Select the dose that gives the best balance of freshness, mouthfeel, and varietal expression without making the wine sharp or thin. Record the chosen dose, sample volumes, and sensory comments in the cellar log.

Step 5 — Prepare the tartaric solution

Weigh tartaric acid accurately on calibrated scales. Dissolve in clean wine or demineralised water to make a manageable slurry or solution; a practical stock is **100–200 g/L**, stirred until fully dissolved. Use food-grade, clean, dry vessels. Wear gloves, eye protection, and dust protection when handling powder. Check that all transfer hoses, pumps, and fittings are clean and free of caustic or detergent residue.

Step 6 — Add to tank with strong mixing

Introduce the solution into a well-mixed wine stream, preferably during pump-over, recirculation, or tank transfer. Add slowly to avoid local over-acidification and uneven distribution. Mix for **15–30 minutes** after addition, then take a representative sample from the tank's top, middle, and bottom if possible. If the wine is in barrel, rack to tank for dosing where practical; if not, roll barrels or use inert-gas-assisted circulation to improve distribution.

Step 7 — Verify the result

After **12–24 h**, re-test pH, TA, free SO₂, and sensory balance. Confirm the wine remains stable and that the adjustment has not pushed pH into an undesirable range. Typical cellar follow-up is to aim for a modest pH drop and a TA increase consistent with the bench trial. If the wine still tastes flat, repeat with a small correction rather than a large second dose. If the wine tastes hard or brittle, hold further additions and reassess after cold stabilisation and blending options are reviewed.

Step 8 — Record, protect, and monitor

Record lot ID, date, product, supplier, calculated dose, actual dose, operator, mixing method, and analytical results before and after addition. Protect the wine from oxygen during and after dosing by maintaining inert gas cover where available. If the wine is to be filtered, bottled, or heat treated, confirm acid adjustment is complete and stable before moving to the next process. Retain samples for QA reference and trend future additions against style outcomes.

DECISION LOGIC — WHY THIS APPROACH

Tartaric acid is the preferred acidulant in wine because it is already a natural grape acid, gives a clean sensory lift, and is predictable in finished wine. It is usually better than using citric or malic acid in cellar correction because citric can feed spoilage organisms and malic can be metabolised by bacteria during MLF, changing the wine again after dosing. Tartaric also supports lower pH, which helps colour stability, microbial control, and overall freshness. The main trade-off is over-acidification: too much tartaric can make the wine angular, reduce mid-palate weight, and trigger cold instability if the wine is not managed properly. Bench trials are the safest way to avoid that. Timing matters as well; if MLF is pending, tartaric additions can complicate bacterial performance and outcome, so the default is to wait until MLF is confirmed complete unless the winemaker has a specific reason not to. In practice, small measured corrections beat one big swing every time.

TRIBAL KNOWLEDGE

Use a clean stainless mix tank or a vigorous recirc loop if you want the addition to land evenly; static additions often stratify, especially in larger volumes. Tartaric dissolves more cleanly if you make a warm slurry first, but don't use hot water on a fragile wine lot — keep the solution modest and add it slowly. Some lots show a bigger pH response after 24 hours than they do straight after mixing, so don't chase the number immediately. If you're near bottling, remember cold stability: a wine that's already borderline can drop crystals later if you push tartaric too hard. Always taste the bench trial at cellar temperature; cold samples can hide the sharp edge and tempt you into over-dosing.

Sugar Adjustment & Chaptalisation

CELLAR-READY QUICK STEPS

["Check must °Brix and target alcohol.", "Calculate sugar at 17 g/L per 1% v/v.", "Dissolve sucrose in 40–60 °C water.", "Add syrup during active mixing.", "Recheck Brix, temperature, and aroma daily."]

FULL PROCEDURE

Step 1 — Confirm the need for sugar adjustment

1. Review the must analysis before any addition: °Brix, potential alcohol, pH, titratable acidity, cultivar style target, and intended final alcohol.
2. For a dry red table wine, keep the target rise in potential alcohol conservative; do not chase high alcohol just because the fruit came in low. As a rule, adjust only enough to reach the house style and yeast tolerance.
3. Confirm the fermentation stage. Chaptalisation is best done at the start of fermentation or very early in active fermentation, when yeast can assimilate the added sugar without a major osmotic shock.
4. Do not add sugar to a stuck or sluggish ferment without first checking temperature, nutrient status, and yeast viability.

Step 2 — Calculate the addition

1. Determine the required sugar increase from the target final alcohol or target starting °Brix.
2. Use sucrose as the standard chaptalisation material unless the winemaking specification says otherwise.
3. Practical dosing guide: approximately 17 g/L sugar raises potential alcohol by about 1.0% v/v, or

approximately 1.0 °Brix in must. Verify against your winery's conversion table if a tighter calculation is required.

4. Record the batch volume in hL and convert the addition to kg: grams per litre multiplied by batch litres, then divide by 1,000.

5. Cap additions conservatively. If the required uplift is large, split the addition across two dose points rather than putting all the sugar in at once.

Step 3 — Prepare the sugar solution

1. Use clean potable water only, heated to 40–60 °C to aid dissolution. Do not boil unless your site procedure explicitly allows it.

2. Dissolve the measured sucrose completely before adding to wine or must. Undissolved crystals are a common cause of inaccurate dosing and local fermentation spikes.

3. If the tank or fermenter is open, protect the solution and the wine from contamination and excessive oxygen pickup during the transfer.

4. Verify the solution is clear and homogenous before use.

Step 4 — Add to the fermenting must

1. Add the syrup with good mixing, preferably during an active pump-over or punch-down so the sugar distributes evenly through the cap and liquid phase.

2. Avoid dumping concentrated syrup onto the cap surface or tank wall. Localised high sugar can stress yeast and create uneven fermentation.

3. After addition, mix thoroughly, then confirm the must temperature remains within the yeast's working range, typically 20–30 °C for red fermentations, with the house target commonly 24–28 °C.

4. Continue standard cap management: punch down several times daily to keep the cap wet, support extraction, and disperse heat.

Step 5 — Recheck fermentation performance

1. Measure and record °Brix, temperature, and sensory signs of fermentation at least daily after addition; more often in warm, fast ferments.

2. Expect active fermentation to recommence or continue within 1–3 days of yeast inoculation, depending on temperature and yeast health.

3. Watch for signs of stress: rotten egg aroma, sluggish CO₂ production, excessive heat, or a cap that dries out or compacts.

4. If the ferment slows unexpectedly, check nutrient status before considering further sugar addition. Yeast often fail on nutrition or temperature, not on sugar shortage.

Step 6 — Manage completion and press timing

1. As sugars drop, test to confirm the fermentation is complete before pressing. Do not rely on cap collapse alone.

2. Typical dry-red endpoint is near 0–2 g/L residual sugar, or stable dry readings over 24–48 hours, depending on site method.

3. Press only when fermentation is complete or nearly complete and the wine style allows it. Transfer fermented grapes into the press, then move wine to temporary storage to settle gross lees.

4. Record all additions, calculations, timings, and final results in the batch log for compliance and traceability.

DECISION LOGIC — WHY THIS APPROACH

Chaptalisation is used to correct deficient musts and hit style-aligned alcohol, not to mask poor fruit. Sucrose is preferred because it dissolves cleanly, is predictable to calculate, and ferments reliably once dispersed.

The key control point is timing: early fermentation or inoculation stage is safest because yeast are active and can adapt to the extra osmotic load. Late additions are higher risk, especially if the ferment is already slowing from heat stress, low nitrogen, or poor yeast health. Splitting large additions reduces stress and improves distribution. Water-based syrup is more practical than dry sugar thrown into the tank, which tends to clump, settle, or create hotspots. The main trade-off is precision versus simplicity: exact lab correction is ideal, but in the cellar a disciplined g/L-to-alcohol estimate is usually sufficient, provided temperature,

nutrition, and progress are checked closely. If the ferment is sluggish, fix the underlying cause first rather than adding more sugar.

TRIBAL KNOWLEDGE

Warm syrup dissolves fast, but don't overheat it or you'll create a sticky mess and slow the cellar down. Make the syrup in a bucket or small tote that can be rinsed straight back into the batch so you don't lose kilograms to the floor. Always pre-mark the transfer vessel; guessing a hL volume in the middle of harvest is how errors happen. If you're dosing into an open red ferment, add during a pump-over so the cap breaks up the syrup naturally. Some tanks stratify badly after sugar additions, especially if the cap is still tight, so take the top, middle, and bottom readings before signing off. Keep an eye on yeast nutrient timing as well — a well-fed ferment handles sugar corrections much better than one that's already starved.

Tannin Additions

CELLAR-READY QUICK STEPS

- Bench trial the dose before tank addition.
- Slurry tannin in 20× its weight.
- Add during active circulation or punch-down.
- Taste again after 24–72 hours.
- Record product, rate, and sensory result.

FULL PROCEDURE

Step 1 — Confirm the winemaking objective

Assess the parcel, wine style and current ferment stage before any tannin is added. Tannin additions are used to build mid-palate, support colour stability, improve structure, and help manage oxidation risk in red musts and young wines. They are most useful where fruit is soft, colour is light, skins are thin, or the finished wine needs firmer phenolic backbone. Record grape variety, ferment temperature, Baume/Brix, cap condition, and tasting notes on the daily cellar sheet.

Step 2 — Choose the tannin type and timing

Select an enological tannin suited to the job: gallotannin for oxidation protection and structure, ellagitannin where greater mouthfeel and polymerisation support is needed, or a blended product if the manufacturer provides a clear use case. For ferment additions, apply early-to-mid fermentation, ideally after yeast activity is established and before the cap becomes overheated or excessively dry. For press or post-ferment additions, use only when the must or wine has been racked clean enough to assess the effect. Do not add blindly during a stressed ferment.

Step 3 — Determine the trial dose

Start with bench trials before tank dosing. Typical working ranges are 0.2–1.0 g/L, depending on product style and target impact. As a practical cellar starting point:

- delicate reds: 0.2–0.4 g/L
- mid-weight reds: 0.4–0.6 g/L
- robust reds or press fractions: 0.6–1.0 g/L

Never exceed the supplier's maximum recommended dose. Record the exact product name, batch number, and dose rate. If the wine already has high seed tannin, hard press character, or drying palate, reduce or avoid addition.

Step 4 — Prepare the addition correctly

Weigh the tannin accurately on a calibrated scale. Pre-slurry the powder in 10–20 times its weight in clean wine, juice, or warm potable water at 20–35 °C to prevent clumping. Stir until fully wetted and free of dry pockets. Add the slurry to a clean mixing vessel, pump-over line, or tank top where good circulation is available. Avoid throwing powder directly onto skins or into a dead zone in the tank.

Step 5 — Mix thoroughly and safely

Add during a pump-over, délestage, punch-down, or tank recirculation so the tannin is dispersed evenly

through the must or wine. For fermenting reds, combine the addition with cap management so the tannin is incorporated into the active mass. Verify even distribution by taking samples from top, mid and bottom of the vessel after mixing. Wear PPE: dust mask or P2 respirator, safety glasses, gloves, and long sleeves. Fine tannin powders are a nuisance dust and must not be inhaled.

Step 6 — Taste, measure, and monitor

Reassess the wine 24–72 hours after addition. Compare against a control or retained bench sample. Check for:

- reduced oxidation sensitivity
- improved palate line and grip
- no hardening of finish
- no loss of fruit expression
- no increase in bitterness or dryness

If the wine becomes austere, reduce future doses or blend with softer fractions. If the effect is too light, repeat with a smaller incremental dose rather than making one large correction.

Step 7 — Manage additions through pressing and settling

For red ferments nearing completion, be cautious with tannin additions immediately before pressing. Press fractions already carry more seed and skin phenolics, so additions should be minimal unless the juice is clearly thin or vulnerable. After pressing, transfer the wine to temporary storage to settle gross lees before making further chemistry decisions. Once gross solids drop out, retaste; what seemed necessary on cloudy wine may be excessive once clean.

Step 8 — Document and release hold

Record date, tank, lot, product, supplier, dose g/L, method, operator, and sensory outcome. Note whether the addition was made during fermentation, at press, or post-ferment. Hold the lot for final tasting before any blending or bottling decision. If the wine shows instability, haze, or excessive astringency, escalate to the winemaker before further processing.

DECISION LOGIC — WHY THIS APPROACH

Tannin additions work best when they are targeted, trialled, and matched to the wine's natural phenolic load. The reason for starting with bench trials is simple: tannin can improve structure and oxidation resistance, but it can just as easily harden the finish or strip fruit if overdone. Early ferment additions integrate more cleanly with extraction and cap management, while post-ferment additions are better for fine-tuning once gross lees and suspended solids are reduced. Pre-slurrying and mixing thoroughly prevents “hot spots” that create bitter pockets. We avoid routine blanket dosing because grape variety, ripeness, press fraction, and existing seed tannin change the outcome dramatically. If the ferment is stressed, fix yeast nutrition and temperature first; tannin is not a cure for poor fermentation health. Deviation is justified when fruit is already phenolically strong, when press fractions are aggressive, or when the wine is intended for a softer, earlier-drinking style.

TRIBAL KNOWLEDGE

Powder tannins bridge air and cling to tank walls if you dump them in dry, so always make a slurry first. Hydrate in a clean bucket and whisk hard; most clumping problems are self-inflicted. If you add during ferment, do it just after a pump-over so you get immediate circulation. Small pump lines can blind with tannin fines if the slurry is too thick, so keep it pourable. Some products taste fine on day one and get much firmer after 48 hours, especially in warmer ferments. Keep a retained control sample every time; cellar memory is poor, and tannin effects are easy to over-credit when the wine is already changing from press to settle.

Oak Programme — Barrel & Adjunct Selection

CELLAR-READY QUICK STEPS

["Confirm wine style and MLF status first.", "Bench trial three oak treatments.", "Select barrel or adjunct by target impact.", "Dose accurately and record lot numbers.", "Monitor, top, and taste every 2–4 weeks."]]

FULL PROCEDURE

Step 1 — Confirm the wine is ready for oak planning

Review the wine style, structure, and cellar timing before ordering any oak. Confirm the wine is dry, stable, and has finished alcoholic fermentation. If malolactic fermentation (MLF) is intended, do not add or carry excessive SO₂ beforehand; MLF bacteria are inhibited by elevated free SO₂ and can stall. Target wine conditions for MLF-compatible oak planning are: temperature above 16 °C, pH at or above 3.1, and free SO₂ preferably below 20 ppm for sensitive strains, with some tolerance up to about 50 ppm depending on strain choice. Check current free SO₂, pH, VA, and sensory status. Record the batch, volume, varietal, vintage, and intended oak style in the cellar log.

Step 2 — Define the oak objective

Choose the oak outcome before selecting product. Decide whether the wine needs spice, toast, vanilla, coconut, structure, micro-oxygenation, or only subtle mouthfeel. Match the oak programme to wine weight and ageability:

- Light, delicate wines: low-impact French oak, older barrels, or restrained adjunct use.
- Medium-bodied wines: a mix of seasoned and new oak, or medium toast adjuncts.
- Full-bodied reds: higher proportion of new barrels or more assertive adjuncts.

Set a target oak impact window, usually 3–18 months depending on varietal, toast level, and fill history. Avoid trying to “fix” poor fruit with heavy oak; oak should support, not mask.

Step 3 — Select barrel type and fill percentage

Select barrel origin, grain, toast, and age based on the style brief.

1. **Barrel species/origin:** French oak for finer spice and tighter grain; American oak for stronger vanilla/coconut and sweeter palate impression.

2. **Toast level:**

- Light toast: preserves fruit and adds subtle tannin.
- Medium toast: general-purpose balance for most reds.
- Medium-plus/heavy toast: use only when the wine can carry more char, smoke, or roasted notes.

3. **Barrel age:**

- New barrel: maximum extraction and oxygen ingress.
- 1–2 year barrel: moderate impact.
- 3+ year barrel: structural polishing and gentle oxygen exposure.

Plan fill volume with headspace control. Fill barrels to the bottom of the bung hole, typically leaving minimal ullage. Mark each barrel with varietal, vintage, cooper, toast, date filled, and expected rack date.

Step 4 — Select adjuncts when barrel is not the best fit

Use chips, staves, cubes, or spirals only when the target is clear and the contact regime can be controlled. Adjuncts suit tank trials, short turnaround wines, or when barrel inventory is constrained. Match format to extraction speed:

- Chips: fastest extraction, strongest immediate impact.
- Staves/cubes/spirals: slower, more integrated oak delivery.

Dose conservatively and trial first where possible. Typical starting points are 1–4 g/L depending on adjunct format, toast, and wine sensitivity. For fine tuning, run bench trials in 1–2 L samples before full-scale addition. Always check that adjuncts are food-grade, sulphur-free if required by programme, and traceable by lot number.

Step 5 — Trial, dose, and document

Run a bench trial before committing to the cellar addition. Compare at least three treatments: control, low rate, and target rate. Evaluate after 24–72 hours for aroma, palate weight, toast integration, and drying tannin. For barrels, compare cooper, toast, and fill history against the wine's structure. For adjuncts, calculate addition rate precisely and mix thoroughly to avoid localised over-oaking. Record all dosing in g/L or barrel count, along with contact time and sampling dates.

Step 6 — Manage contact, topping, and monitoring

Monitor barrel lots every 2–4 weeks at the start, then monthly once stable. Check fill level, SO₂, aroma, and

any signs of oxidation, volatile acidity, or sulphidic reduction. Top barrels promptly to minimise oxygen pickup and evaporation losses. Maintain free SO₂ in line with pH and style, and keep winery hygiene tight; oak surfaces and bung areas are common contamination points. If using adjuncts in tank, rack or stir only as required by the product regime, and verify homogeneity at each tasting point.

Step 7 — Assess removal timing and integration

Taste, analyse, and decide removal based on balance rather than calendar alone. Remove adjuncts when the wine shows the desired aromatic lift and tannin integration; this may be as short as a few weeks for chips or several months for staves/cubes. For barrels, rack when the oak is integrated and no longer improves the wine. Do not extend contact simply because oak is available. If the wine becomes overly woody, drying, or smoky, remove oak exposure immediately and consider blending or fining only after sensory and analytical review.

Step 8 — Final quality check and release readiness

Before moving the wine to the next stage, confirm the oak treatment is documented, the lot remains microbiologically and chemically stable, and the sensory profile matches the style brief. Recheck free SO₂, pH, and, where relevant, completion of MLF. Ensure barrels are clean and stored correctly for reuse or retirement. File all supplier details, lot numbers, dosing records, and tasting notes in the quality system.

DECISION LOGIC — WHY THIS APPROACH

Barrels and adjuncts are selected to match wine structure, style, and timing rather than applying a one-size-fits-all oak treatment. Barrels provide slower extraction plus controlled oxygen ingress, which generally gives better integration and complexity for wines that can age. Adjuncts are useful where cost, inventory, or turnaround demands a faster and more controllable solution. The key trade-off is precision versus texture: barrels cost more and vary by cooper, while adjuncts are cheaper and more repeatable but can look blunt if overused. The SOP emphasises bench trials because oak impact is highly varietal- and batch-dependent. It also prioritises early MLF compatibility checks: high SO₂, low temperature, or poor pH conditions can stall MLF and complicate oak integration. Deviate when the wine is very delicate, highly reduced, or already strongly oaked in prior fills; then reduce new oak, shorten contact, or shift to older barrels or lower adjunct doses.

TRIBAL KNOWLEDGE

Seasoned barrels can still bite if they've been stored dry or bung-up too long; hydrate and inspect before use. New barrels often look matched on paper but differ in real extraction, so keep cooper lots separate until you've tasted them. Chips extract fast and can go from helpful to rustic in a narrow window, especially in warmer cellar conditions. If you're using adjuncts in tank, suspend them properly or they'll settle and create patchy results. Always smell the bung and the headspace before topping — that's often the first place oxidation, vinegar, or taint shows up. Keep a dedicated tasting sheet for oak lots; memory gets unreliable once you're juggling multiple fills.

Bottling & Packaging

Bottling & Packaging

CSU Reference: WSC202, WSC318, WSC401

CELLAR-READY QUICK STEPS

["Confirm final lab results and lot release.", "Sanitise all product-contact surfaces and lines.", "Run a fill-and-closure trial on the first bottles.", "Monitor closure integrity and fill heights regularly.", "Pack, label, and store in cool conditions."]

FULL PROCEDURE

Step 1 — Confirm wine is ready for pack-out

1. Verify the wine has completed all planned fermentations and is chemically stable. If malolactic fermentation (MLF) was intended, confirm completion before bottling; if MLF was not intended, confirm the wine has been protected with SO₂ and has remained microbiologically quiet.
2. Take final measurements on the lot: pH, titratable acidity (TA), free SO₂, total SO₂ if required by your lab program, residual sugar, temperature, and clarity. Target the bottling temperature at 15–20 °C for consistent fill and to reduce dissolved gas issues.
3. Taste the wine and assess for fault, oxidation, reduction, haze, or sediment risk. Do not bottle until the wine is clean, bright enough for the house style, and stable to the degree required for the intended shelf life.
4. Record the batch ID, volume, lab results, intended closure, bottle type, and any deviations. Bottling is a final-release step: once sealed, the wine is much harder to correct.

Step 2 — Stabilise and finish the wine

1. If the wine requires polish for appearance, use the least intervention possible. Run bench trials before any fining, and dose at the lowest rate that achieves the target result. Over-fining strips mouthfeel, aroma, and flavour.
2. If filtration is used, choose the lightest filter load that gives the required clarity and microbial security. Confirm compatibility with the style; overly aggressive filtration can flatten texture and strip aroma.
3. If the wine contains residual sugar, confirm that it is intentionally stable. Either bottle dry and sweeten later under a controlled program, or ensure any sweet wine has been properly stabilised before bottling. Do not rely on sugar presence without stability checks.
4. Adjust SO₂ only after final blending, fining, and filtration decisions are complete. Mix thoroughly and allow the wine to equilibrate before re-checking free SO₂.

Step 3 — Prepare bottles, closures, and line

1. Use only clean, sound, food-grade bottles and closures matched to the wine style and intended cellaring life. Check bottles for chips, cracks, mould, odours, and labels or glue residues.
2. Store closures in their original packaging in a clean, dry area. Keep corks, screwcaps, and crown caps sealed until needed to avoid contamination and moisture gain.
3. Clean and sanitise the bottling line, hoses, filler heads, valves, corker, and any product-contact surfaces. Verify the sanitiser is at working concentration and safe for contact surfaces according to the winery chemical register.
4. Run a water or sanitised water trial to check fill heights, drip control, and closure application. Confirm there are no leaks, air draw, or inconsistent fills.

Step 4 — Bottle the wine

1. Transfer the finished wine gently to the bottling tank or bottling vessel, minimising splashing and oxygen pick-up. Keep hoses full where possible.
2. Maintain a consistent fill level across the run. Check the first 12 bottles closely, then inspect regularly during production. Adjust filler height or flow if fill variation exceeds house tolerance.
3. If using an inert gas regime, maintain a protective blanket in the tank headspace and keep turbulence low. Where available, use nitrogen or argon sparging only as required by the equipment and wine style.
4. Cull any bottles that show contamination, broken glass, poor fill, or closure faults. Never top up a faulty

bottle from another bottle on the line.

Step 5 — Seal and protect the bottles

1. Apply the selected closure immediately after filling to limit oxygen exposure. Match closure compression and insertion depth to the supplier specification.
2. For cork closure, verify straight insertion, no cork fracture, and no cork proudness beyond tolerance. For screwcap, confirm proper thread engagement and torque. For crown cap or other closures, confirm complete seal and correct capsule fit where used.
3. Invert or tip test a sample from each run start and then at regular intervals to check for leaks.
4. Keep bottled wine upright long enough for the closure to settle, then move carefully to avoid agitation.

Step 6 — Package, label, and release

1. Allow bottles to rest before final packaging, especially after closure application and any transport vibration.
2. Apply labels, back labels, capsules, and export marks cleanly and in the correct legal format. Check vintage, varietal, lot code, alcohol statement, sulphite declaration where required, and any market-specific compliance text.
3. Pack bottles into cartons with correct orientation and quantity. Verify carton strength, divider use, and pallet stability.
4. Sample the finished lot for post-bottling QA: closure integrity, fill height, label accuracy, and sensory check. Hold the lot if any defect trend is identified.

Step 7 — Store and document the finished lot

1. Move finished goods to a cool, dark, vibration-free storage area, ideally 12–18 °C.
2. Record bottle count, losses, lot numbers, lab results, packaging materials used, and any corrective actions taken.
3. File all packaging approvals, chemical checks, and release sign-off before dispatch. No wine leaves cellar control without documented final approval.

DECISION LOGIC — WHY THIS APPROACH

This procedure prioritises stability, hygiene, and oxygen control because bottling is the point where a good wine can be damaged fastest and least reversibly. The main alternatives are to bottle earlier and fix later, or to rely on heavy filtration and high SO₂ as catch-all insurance. Both are poor cellar practice: early bottling increases the risk of continued microbial activity or haze formation, while heavy-handed filtration and excessive sulphur can strip character and still not guarantee stability. The better approach is to finish fermentations, confirm chemical status, and make only the minimum intervention needed for the style.

MLF completion matters because residual malic acid and low protection can allow unwanted bacteria to establish during storage. Likewise, residual sugar wines need deliberate stability planning rather than assumptions. Bench trials before fining keep dosage tight and reduce the risk of over-correction. We also bottle at moderate temperature and with low turbulence because dissolved oxygen and fill inconsistency are common causes of premature ageing and closure faults. Deviate only for a defined style requirement, market regulation, or a documented technical issue supported by lab data and sign-off.

TRIBAL KNOWLEDGE

A lot of bottling problems look like wine faults but start as machine faults: a lazy filler valve, a tired corker head, or a hose that's holding air. Always run the first few bottles slowly and keep one person watching fill height and one watching closures. If the line has been sitting, flush it properly; stale water in a filler is a contamination trap.

Corks and screwcaps behave differently in warm weather. Corks can swell and jam, while some caps show torque variation if the line gets hot. Keep closures in the room for long enough to match cellar temperature, not the truck temperature. For smaller lots, a clean bottling bucket and simple filler often gives better control than a complicated setup that is out of tune. Most rejected bottles are caught fastest by a sharp eye and a good light.

Pre-Bottling Filtration

CELLAR-READY QUICK STEPS

["Confirm wine is stable and dry before filtering.", "Trial 5–10 L through the chosen filter set-up.", "Start slow and watch pressure every 10–15 min.", "Blanket receiving tank with inert gas throughout.", "Clean and record the run immediately after use."]

FULL PROCEDURE

Step 1 — Confirm wine is ready for final filtration

1. Verify the wine has completed alcoholic fermentation and, if required by style, malolactic fermentation is complete or intentionally not pursued.
2. Check residual sugar and stability indicators before any filter is connected. Target is dry wine: final sugar below 2 g/L, with no active fermentation signs.
3. If the wine has completed MLF, confirm malic acid is absent or at the winery's defined stable level, and free SO₂ is within the winery target for the wine's pH.
4. Take a bench sample and assess for haze, tartrate crystals, protein instability, and visible lees. Filtering will not fix major instability; it only removes suspended material.
5. Do not filter if the wine is still throwing CO₂ heavily, warm from recent processing, or not settled enough to avoid rapid pad blockage.

Step 2 — Set the filtration target and method

1. Choose the lightest filtration that will achieve pack-down, clarity, and bottling stability. Avoid over-filtration where possible, as this strips texture and aroma.
2. Use coarse pre-filtration first if the wine is turbid or has fine lees. Move to final membrane filtration only when the pre-filter is stable and the wine remains bright enough.
3. Typical final bottling targets are:
 - Brightness: no visible haze in a full sample bottle under strong light.
 - Turbidity: as low as practical for the style, with a winery-defined maximum for bottling.
 - Microbial stability: appropriate pore size for the wine's risk profile and package format.
4. Confirm the filter cartridge, pads, or membrane are rated for wine use, in date, and compatible with the wine temperature and alcohol content.

Step 3 — Prepare the wine and equipment

1. Chill or warm the wine to a stable filtration temperature, ideally 10–18 °C. Keep temperature consistent; viscosity changes will affect flow and clarity.
2. Rack the wine off gross lees before filtration. Minimise oxygen pickup at every transfer.
3. Purge hoses, filter housing, and receiving tank with inert gas where available.
4. Assemble the system clean and sanitised. Check gaskets, seals, pressure gauges, and sight glasses for wear or leaks.
5. Pre-wet or rinse filter media according to supplier instructions using clean water or wine, then discard rinse liquid unless the system is specifically approved to retain it.

Step 4 — Run a small-scale trial first

1. Before filtering the full volume, run 5–10 L through the chosen filter set-up as a trial.
2. Check for:
 - Excessive pressure rise.
 - Premature pad blinding.
 - Aromatic stripping.
 - Haze breakthrough.
3. If pressure climbs quickly, step back to a coarser grade or add a pre-filter stage.
4. If the wine tastes thin or over-processed after the trial, reconsider the pore size and the necessity of the last filtration pass.

Step 5 — Filter the wine

1. Start the pump slowly and bring the system up to steady flow without surging.

2. Maintain low, even pressure. Do not exceed the equipment or media manufacturer's maximum working pressure.
3. Monitor differential pressure, flow rate, and clarity every 10–15 min at the start, then at least each tank transfer.
4. Keep the receiving vessel closed and blanketed with inert gas. Limit headspace to reduce oxygen pickup.
5. If the filter begins to blind, stop and change media or backflush if the system is designed for it. Do not force wine through blocked media.
6. Collect the first and last litres separately and inspect for haze or sediment. Re-filter any suspect portion.

Step 6 — Verify bottling readiness

1. Take a final polished sample after filtration.
2. Check clarity against a black-and-white background and confirm no visible particulate or haze.
3. Re-test free SO₂ if the wine has had significant contact time, movement, or oxygen exposure during filtration.
4. Confirm the wine is within bottling specification for dissolved oxygen, temperature, fill stability, and microbial security.
5. If the wine is not within spec, hold it in topped-up tank under inert gas and rework before bottling.

Step 7 — Clean down and record

1. Flush wine from the system immediately after use.
2. Clean with approved detergent, rinse to neutral, sanitise, and dry or drain as required by the equipment.
3. Record wine ID, filtration media used, start and finish times, pressure profile, temperature, operator, and any faults.
4. Retain trial results and final release notes as part of the bottling batch record.

Step 8 — Safety and compliance checks

1. Wear PPE: safety glasses, gloves, and suitable footwear.
2. Treat pressurised housings as a pinch and burst hazard.
3. Handle cleaning chemicals and sanitiser under site chemical procedures.
4. Ensure all filtration equipment meets winery food-contact and pressure safety requirements before use.
5. Dispose of spent pads and rinses according to local environmental and wastewater rules.

DECISION LOGIC — WHY THIS APPROACH

Final filtration is about balancing clarity, microbial security, and wine preservation. A staged approach — settling, coarse pre-filtration, then final filtration only if required — gives the best chance of bottling a stable wine without stripping aroma or texture. Going straight to a tight membrane on a turbid wine usually causes rapid blockages, pressure spikes, and unnecessary oxygen exposure. That is poor cellar practice and expensive.

The SOP favours the lightest effective filtration because every extra pass adds risk: aroma loss, foam, oxidation, and time out of tank. However, if a wine is unstable, high in solids, or destined for a high-risk package format, a tighter final filtration is justified. The key trade-off is between sensory preservation and microbiological security. Deviate only when the wine style, residual sugar, MLF status, or release spec demands it, and always base that decision on bench trials and measured pressure behaviour, not guesswork.

TRIBAL KNOWLEDGE

A wine that looks bright in tank can still blind a filter fast if it has fine tartrate or protein haze, so always trial a small volume first. Old pads and tired gaskets are the usual culprits when operators chase pressure instead of fixing the setup. Keep the receiving tank well blanketed with inert gas; the last litre into bottle is where oxidation shows up later as dull fruit and browning.

If a filter starts pulsing or surging, check the suction side for air leaks before blaming the media. On tangy whites and delicate reds, a slower flow nearly always gives better results than pushing for throughput. Good operators watch the pressure trend, not just the final number.

Closure Selection (Screwcap / Cork)

CELLAR-READY QUICK STEPS

- Confirm wine stability and closure target.
- Check bottle finish, line setup, and torque settings.
- Validate first 20–30 bottles for seal quality.
- Reject damaged closures and contaminated lots immediately.
- Record closure lot, machine settings, and defects.

FULL PROCEDURE

Step 1 — Confirm the wine style and bottle risk profile

Assess the finished wine before choosing the closure. Record variety, vintage, pH, free SO₂, dissolved oxygen (DO), residual sugar, alcohol, and intended market. As a practical rule, choose screwcap for wines where freshness, low oxygen ingress, and consistency are the priority: aromatic whites, rosé, early-drinking reds, and wines with modest bottle-ageing plans. Consider cork where the style benefits from slower oxygen exchange and traditional presentation, typically structured reds intended for medium ageing. Confirm the wine is microbiologically stable and finished dry before either closure is approved.

Check for sensory risks that closures can mask or worsen. Any wine showing volatile acidity, mercaptans, reductive funk, oxidation, or instability needs correction before bottling, not a “better” closure choice. Record the target cellar temperature at bottling: 15–20 °C is preferred for stable fill volumes and reliable closure performance.

Step 2 — Match closure to package format and quality target

Select screwcap when batch-to-batch consistency matters, cork taint risk must be minimised, or the wine has low tolerance for oxygen pickup. Specify closure liner type to suit the wine: standard liners for most fresh styles, and higher oxygen-transmission liners only when a controlled development curve is desired.

Select natural cork or technical cork only when the wine style, bottle format, and market expectation justify the added variability. Use reputable suppliers with documented rejection rates, sensory screening, and diameter/length suitability for the bottle neck and planned shelf life. Do not use cork as a cure for an unstable wine or poor bottling hygiene.

Set the target closure decision in the bottling order and sample approval sheet, including closure type, size, supplier, lot number, and target extraction force or application torque.

Step 3 — Verify bottle, line, and closure compatibility

Confirm bottles are from one approved specification: identical finish, neck bore, and internal surface quality. Inspect random bottles for chips, finish distortion, and contamination. For screwcap, confirm the cap head, spindle, and torque control are matched to the closure size and bottle finish. For cork, confirm the corker settings suit the cork diameter, compression, insertion depth, and bottle neck diameter.

At least 24 hours before bottling, run a dry check on the line using sample bottles and closures. Verify:

1. Screwcap seats squarely with no cross-threading.
2. Cork is inserted evenly with no shoulder damage.
3. No visible leaks, blow-backs, or excessive closure deformation.
4. Application force or torque falls within the equipment and closure supplier specification.

Step 4 — Control oxygen and hygiene at bottling

Closures do not compensate for poor bottling hygiene. Bottle only when the wine is clear, stable, and prepared according to the bottling plan. Keep dissolved oxygen as low as practical through inert gas management, minimal splashing, and tight transfer lines. Target bottling DO should be as low as the winery can consistently achieve, and never ignored as “closure-dependent”.

Sanitise all closure-contact parts, cap bowls, cork magazines, jaws, chutes, and feed tracks before use. Keep

closures sealed in original packaging until required. Handle cork and screwcaps with clean gloves or sanitised hands. Reject any closure lot showing dust, damage, mould, oil contamination, or package breach.

Step 5 — Apply the closure and verify seal integrity

Run the first 20–30 bottles as a validation set. Check every bottle for:

1. Fill height within target tolerance.
2. Closure seating or torque within specification.
3. No leaking, weeping, or bottle lip contamination.
4. Correct headspace and finish alignment.

For screwcap, inspect for consistent thread engagement and no over-torque. Over-tightening can damage the liner or finish and may cause later leakage. For cork, verify insertion depth and that the cork top sits flush or to the approved projection. Reject any bottle with damaged finish, split cork, wobbly cap, or abnormal resistance during application.

Step 6 — Monitor post-bottling performance and release product

Hold finished goods under controlled storage for at least 7–14 days before release testing. Recheck a sample for leakage, ullage change, taste, and closure appearance. For cork, keep lots upright only if the supplier or logistics plan requires it; otherwise store as specified by the winery's packaged goods standard. For screwcap, confirm no panel bulging, liner failure, or seal creep.

Retain closure lot records with bottling date, batch code, operator, line settings, and any faults. If one closure lot or one line setting produces more than a minor isolated defect rate, stop and investigate before continuing production. Release only when the wine remains sound, the closure has performed cleanly, and all bottle records are complete.

DECISION LOGIC — WHY THIS APPROACH

Screwcap is the preferred default for most boutique wineries because it delivers the most repeatable oxygen barrier, reduces cork taint exposure, and suits wines meant to stay fresh and varietally correct. That consistency matters more than tradition when you are bottling small lots and cannot afford wide bottle-to-bottle variation. Cork remains valid where the wine style genuinely benefits from a slower, less uniform oxygen ingress and where the market expects that format, especially for structured reds with ageing potential.

The key trade-off is development versus control. Cork can allow more complex evolution, but also more variability, higher taint risk, and greater dependence on supplier quality and line setup. Screwcap gives cleaner retention of primary fruit and simpler QC, but can be too restrictive for some long-ageing red styles if liner choice is wrong. In practice, the closure should follow the wine's sensory target, not the label trend. Deviate only when trial data, supplier testing, or past bottle-ageing performance supports a different closure for that exact wine style.

TRIBAL KNOWLEDGE

Most closure failures are line failures, not closure failures. If bottles start leaking, check capper alignment, bottle finish variation, and operator speed before blaming the supplier. Screwcaps need clean, square bottle mouths; one chipped finish can throw a whole run off if the machine is drifting.

For cork, don't let the closure sit open to the room. Even a good cork lot picks up dust and cellar odours quickly. Keep the cartons closed and bring out only what you need. On hot bottling days, cork compression changes noticeably, so watch insertion depth and bottle shoulder damage. If the first few bottles look "tight", don't force the setting up blindly; verify neck dimensions and corker compression first.

Fault Diagnosis

Stuck Fermentation — Diagnosis & Restart

CELLAR-READY QUICK STEPS

- Confirm sugar, temperature, and density trend.
- Correct temperature and wet the cap.
- Rehydrate and acclimatise restart yeast.
- Add complex nutrient at measured dose.
- Track density twice daily until dry.

FULL PROCEDURE

Step 1 — Confirm the fault and record the baseline

1. Verify the ferment is genuinely sluggish or stuck, not simply in a late, slowing phase. Take duplicate samples from the tank top and drain valve.
2. Measure and record:
 - Temperature: target 22–28 °C for reds in active ferment; below 18 °C increases risk of arrest.
 - Baume or °Brix and, if available, residual sugar in g/L.
 - Density trend over 24 hours. If there is no measurable drop for 24–48 hours and sugar remains above 4–6 g/L, treat as a stuck ferment.
 - Wine sensory checks: reduction, volatile acidity, H₂S, solvent notes, and seed/tannin harshness.
3. Check tank history: yeast strain, inoculation rate, rehydration method, addition of nutrients, cap management, must temperature, and any pump-over delays or temperature spikes.
4. Confirm there is no equipment fault such as failed cooling/heating, blocked jacket, incorrect tank sensor, or air ingress through fittings.

Step 2 — Stabilise fermentation conditions

1. Set the ferment at 24–28 °C if the wine is sound and the yeast is still viable. Avoid sudden changes greater than 3 °C in 2 hours.
2. Restore cap management immediately. Punch down or pump over 2–4 times per day, depending on cap size and extraction target. Keep the cap wet to improve yeast contact and oxygen transfer.
3. If the ferment is hot (>32 °C), cool it promptly to below 30 °C before any restart action. If it is cold (<18 °C), warm it gradually into the 22–28 °C band.
4. Check for nutrient depletion. If YAN was low at the start, or if the wine was heavily clarified, high sugar, or very ripe, prepare for a nutrient rescue as part of restart.

Step 3 — Assess whether the wine needs a rescue or a re-inoculation

1. If sugar is still falling slightly and yeast appears active, first correct temperature and nutrition, then reassess after 12–24 hours.
2. If the ferment is clearly stuck with residual sugar above 6 g/L, no CO₂ release, and low viability, plan a restart with fresh active yeast.
3. If reduction is present, aerate only carefully and only early in the restart window. Do not splash aggressively if volatile acidity, oxidation risk, or microbial spoilage is already evident.
4. If the wine is near dryness and only a small sugar tail remains, consider whether completion is safer than intervention. Only intervene if the security of the final product is at risk.

Step 4 — Prepare the restart yeast

1. Select a robust restart strain suited to high alcohol and low nutrient conditions. Use a commercial yeast with documented restart performance.
2. Rehydrate according to supplier instructions in clean water at 35–40 °C. Keep hygiene strict; use sanitised vessels and tools.
3. Use a protective rehydration aid if available, and acclimatise the yeast gradually with small additions of the stuck wine to avoid shock.
4. Target a restart slurry temperature within 5 °C of the wine temperature before inoculation.

5. If the wine is low in assimilable nitrogen, add a measured nutrient dose at restart. Typical practice is 20–40 g/hL of a complex yeast nutrient; do not overfeed.

Step 5 — Inoculate and support the restart

1. Introduce the acclimatised yeast into the tank with gentle mixing to distribute evenly.
2. Restore a modest oxygen input only if the wine is still in the early restart phase and sensory assessment allows it. Oxygen is a tool for yeast recovery, not a cure for spoilage.
3. Keep temperature steady at 24–28 °C for the first 48 hours.
4. Check density every 12 hours for the first 2 days, then daily once movement is evident.
5. Maintain cap management if solids remain. If the ferment has been pressed, transfer the wine to a clean vessel, keep gross lees out, and monitor closely.

Step 6 — Verify completion and manage follow-on fermentation

1. Consider primary fermentation complete only when residual sugar is at dry threshold for the winery standard, typically <2 g/L for red table wine, with stable density over 48 hours.
2. Rack off gross lees promptly to a clean temporary vessel to reduce reductive risk and microbial load.
3. If malolactic fermentation is intended, do not delay once alcohol fermentation is complete and the wine is sound. Prepare and inoculate the bacterial culture under controlled conditions.
4. Monitor MLF progress with chromatographic check or equivalent approved method. Maintain suitable cellar temperature and sulphur management to avoid arrest.
5. Record the intervention, doses, temperatures, timings, and final outcome in the lot file.

Step 7 — Escalate if restart fails

1. If no fermentation movement occurs within 48 hours of restart, re-check viability, sugar, temperature, pH, alcohol level, and spoilage indicators.
2. If alcohol is very high, pH is low, or the wine shows serious spoilage, escalate to the winemaker before further additions.
3. If the wine cannot be safely restarted, assess whether sterile filtration, blending, or declassification is the most compliant option.
4. Dispose of any contaminated restart culture and clean/sanitise all vessels, hoses, and fittings immediately after use.

DECISION LOGIC — WHY THIS APPROACH

A stuck fermentation is best treated as a yeast recovery problem first, not a chemical correction problem. Temperature, nutrition, and oxygen balance usually explain most failures, so the SOP focuses on confirming the fault, stabilising conditions, and then using a properly acclimatised restart yeast. This is safer than immediately dosing more nutrient or forcing heat, which can worsen solvent notes, volatile acidity, or microbial instability. A robust restart strain is preferred over trying to revive a weak native population because it gives better control in high alcohol, low nutrient, or cool conditions. The procedure also distinguishes between a ferment that is merely slowing and one that is genuinely arrested, preventing unnecessary intervention. After restart, completion must be confirmed by stable density and dry sugar rather than by appearance alone, because red ferments can look active even when sugar remains. MLF is handled only after alcoholic fermentation is complete to avoid competing stresses.

TRIBAL KNOWLEDGE

A stuck red ferment often needs gentle persistence, not drama. Too much pump-over can strip heat and drive oxygen too deep; too little leaves the cap dry and the yeast starved. Keep a spare calibrated thermometer and density meter at the tank — bad readings cause bad decisions. When rehydrating restart yeast, the biggest killer is temperature shock, so acclimatise in small steps and never dump it straight into a cold tank. In the cellar, the wines that restart best are usually the ones kept clean from the start: low gross solids, good cap wetting, and no lingering H₂S. If the wine smells reductive, fix that smell early; it rarely improves on its own. Some yeast suppliers are better for restart jobs than others — use the strain the winery has already validated rather than experimenting mid-crisis.

Brettanomyces — Detection & Control

CELLAR-READY QUICK STEPS

["Quarantine the lot and stop all transfers.", "Check pH, free SO₂, and residual sugar.", "Sample cleanly; confirm with sensory and lab testing.", "Raise protection, cool the wine, and minimise oxygen.", "Sanitise all shared equipment before reuse."]

FULL PROCEDURE

Step 1 — Define the trigger and isolate the suspect wine

1. Act immediately if you detect any of the following in tank, barrel, or bottle: horse sweat, band-aid, leather, medicinal, mousey, smoky, spicy phenolics, or a drying, harsh finish that is increasing over time.
2. Confirm the lot identity, vessel number, last racking date, SO₂ history, pH, alcohol, residual sugar, temperature, and any recent exposure to oxygen.
3. Quarantine the wine: stop transfers, blending, topping, filtration, or bottling until the risk is assessed.
4. Mark the vessel clearly as suspect and keep bung, lid, and access points closed. Minimise headspace and unnecessary sampling.

Step 2 — Sample correctly for sensory and analytical confirmation

1. Take a representative sample from the vessel mid-stream, not just the top or bottom layer. For barrels, gently roll or mix only if needed and if oxygen pickup can be controlled.
2. Use clean, low-odour sampling gear and pre-rinse sample bottles with the wine.
3. Evaluate sensorially at 18–22 °C. Brett-derived volatile phenols can be subtle at first; compare against a clean control wine if available.
4. Send samples for laboratory confirmation where possible, using PCR/qPCR and/or volatile phenol analysis. Prioritise testing when the sensory signal is uncertain or the wine is high value.
5. Record results and trend them against prior analyses. A single low result may still be meaningful if the sensory fault is developing.

Step 3 — Assess risk factors and likely spread

1. Check for conditions that favour Brettanomyces growth: pH above 3.5, residual sugar above 2 g/L, low free SO₂, elevated temperature, extended barrel ageing, old or porous wood, and repeated oxygen ingress.
2. Review cellar hygiene around the lot and adjacent wines. Brett often persists on dirty hoses, pumps, gaskets, valve seats, barrel bungs, and in inadequately cleaned barrels.
3. Inspect neighbouring vessels, especially where shared equipment or splash transfer has occurred.
4. If more than one lot shows risk, treat the issue as cellar-wide until proven otherwise.

Step 4 — Stabilise the wine with protection and dryness

1. If the wine is dry and commercially viable, raise free SO₂ promptly to the wine's protection target, typically 0.5–0.8 mg/L molecular SO₂ depending on pH. At pH 3.4–3.6, this usually requires roughly 25–40 ppm free SO₂; at higher pH, the target increases materially.
2. Verify dissolved oxygen is low. Keep the wine as full as practical and avoid aerative handling.
3. Reduce temperature where feasible to 12–15 °C to slow Brett activity.
4. If residual sugar remains above 2 g/L, treat the wine as higher risk and prioritise finishing fermentation, sterile filtration, or other stabilisation before any storage period.
5. Do not rely on SO₂ alone if the wine already shows strong Brett character; it is a control measure, not a cure.

Step 5 — Apply physical removal or containment options

1. Consider fine filtration only if it suits the wine style and quality target. For bottling stability, target sterile or near-sterile filtration where appropriate, using validated membrane integrity checks.
2. If the wine can tolerate it, centrifugation followed by filtration may reduce microbial load before final polishing.
3. For barrel programmes, remove affected wine from contaminated wood if the risk is significant. Reused barrels with entrenched contamination may need retirement from high-value wines.
4. If contamination is limited and the wine is commercially sound, isolate and manage rather than over-

treating and stripping the palate.

Step 6 — Sanitation and equipment control

1. Clean and sanitise all contact equipment after use: hoses, pumps, filters, couplings, sampling devices, gaskets, and filler parts.
2. Use a validated caustic wash, thorough rinse, and an approved sanitiser at the correct concentration and contact time. Do not shortcut drying or reassembly.
3. Treat suspect barrels with an approved barrel sanitation program, but recognise that some barrels cannot be fully remediated for sensitive wines.
4. Keep Brett-suspect equipment separate from clean wine gear until verified clean.
5. Document all cleaning lot numbers, concentrations, contact times, and verification results.

Step 7 — Decide on blending, release, or disposal

1. Blend only if the receiving wine can mask or dilute the fault without creating a misleading product. Check the finished blend sensorially and analytically before committing.
2. If the fault is obvious at bench scale, do not assume bottling will hide it; bottle ageing often amplifies Brett character.
3. Reject or divert any lot that cannot be stabilised to the winery's quality standard.
4. Escalate to production and quality management when the lot affects brand integrity, export compliance, or customer specifications.

Step 8 — Prevent recurrence

1. Review root cause: sanitation, barrel age, oxygen management, SO₂ control, slow processing, or residual sugar management.
2. Increase inspection frequency on vulnerable lots: monthly sensory checks for barrel-aged reds, plus periodic analytical testing where risk is elevated.
3. Use a dedicated hygiene plan for barrels, transfer lines, and filler circuits.
4. Train cellar staff to recognise early Brett markers and to report them immediately, not after the next racking.
5. Close the incident with a corrective action record, including the outcome, disposal or release decision, and preventive actions for future vintages.

DECISION LOGIC — WHY THIS APPROACH

Brettanomyces control is about early detection, then fast containment. Once the yeast is established in a wine, particularly a barrel-aged red with higher pH and lower SO₂, it is hard to remove without damaging texture or aroma. That is why this SOP starts with quarantine, sensory confirmation, and risk assessment rather than immediate aggressive treatment. Chemical control via free SO₂ is useful, but only within the bounds of the wine's pH and style; SO₂ protects, it does not reliably "fix" a contaminated wine. Physical removal by sterile filtration or removal from suspect wood is preferred when the wine can handle it. We avoid over-reliance on blending because it can hide a problem temporarily and create release risk later, especially after bottle ageing. The trade-off is always between preserving quality and preserving volume; for premium wine, quality wins. Deviate only with documented sensory and analytical evidence, and only when the treatment will not strip the wine beyond spec.

TRIBAL KNOWLEDGE

Barrel Brett often shows up first in the warmest, least topped part of the cellar, or in old barrels that "look clean" but still smell woody, sweet, and dusty when emptied. A barrel can pass a casual sniff test and still seed the next fill. Keep a dedicated set of hoses and gaskets for suspect lots; cross-contamination through a shared pump is a classic mistake. If you have to sample a suspect barrel, don't splash it into a glass and leave it sitting — the aroma can change quickly with air and mislead you. Many cellar teams prefer qPCR early because it catches problems before the wine is obviously faulty, but the panel still needs the nose to interpret significance. If the winery uses barrel steam or ozone, remember both work better on clean surfaces than on established biofilms.

Volatile Acidity (VA) & Reductive Faults

CELLAR-READY QUICK STEPS

["Isolate the lot and sample at 18–20 °C.", "Check free SO₂, pH, headspace, and temperature.", "Top up, blanket with inert gas, and cool.", "Run bench trials before any copper addition.", "Recheck aroma and chemistry within 48 hours."]]

FULL PROCEDURE

Step 1 — Confirm the fault and isolate the lot

1. Put the wine under review on hold immediately. Do not blend, fine, filter, or bottle until the fault is characterised.
2. Smell the wine at cellar temperature, then again after warming a sample to 18–20 °C. Volatile acidity (VA) usually presents as ethyl acetate/nail polish, glue, solvent, or vinegar. Reductive faults usually present as struck match, rubber, cabbage, onion, rotten egg, or burnt tyre.
3. Pull a bench sample and check: free SO₂, pH, temperature, residual sugar, and if available, malic acid and VA.
4. Record tank/barrel ID, rack history, bung condition, headspace, top-up record, aeration events, and any stuck/sluggish ferment notes. Recent nutrient issues, warm ferments, or poor cap management increase risk of both H₂S and VA.

Step 2 — Measure the risk and set the immediate controls

1. Compare free SO₂ to cellar targets. For sound red wine, hold roughly 20–35 mg/L free SO₂, adjusted for pH and storage risk. If VA is suspected, avoid over-sulfiting a wine with a clean nose already showing low oxygen exposure; excessive SO₂ can sharpen harsh volatile aromas without fixing the cause.
2. Check headspace and bung integrity. Top up barrels and tanks to minimise oxygen ingress. Use inert gas blanketing where practical.
3. If the wine is warm, cool it to 12–15 °C to slow microbial activity while diagnostics continue.
4. If reductive characters are present, do not automatically add more SO₂. First confirm whether the issue is hydrogen sulfide, mercaptans, or disulphides, because each responds differently.

Step 3 — Trace the likely cause

1. Review fermentation history. Nutrient-poor ferments, delayed yeast addition, high YAN variability, and hot ferments increase H₂S risk and can leave a wine susceptible to later reductive fault development.
2. Check for oxygen exposure at any stage. High headspace, slow topping, leaky bungs, poorly sealed pumps, and long warm storage drive acetic bacteria and VA.
3. Check for residual sugar and microbiological stability. A wine with sugar above 4 g/L, elevated temperature, or active spoilage flora is at higher VA risk.
4. If available, plate or microscope for yeast/bacteria load, and test for VA trend over 7–10 days rather than relying on one sample.

Step 4 — Treat based on the fault type

1. For confirmed or rising VA: remove the driver. Improve hygiene, eliminate oxygen exposure, top aggressively, and keep the wine cool. If the wine is still fermentable or has residual sugar, finish/secure the wine first; unstable sweet wine is highly vulnerable.
2. If VA is moderate and the wine is otherwise sound, an early rack off gross lees may help, but only if it reduces reductive load without adding oxygen. Rack under inert gas and keep post-rack SO₂ in range.
3. For H₂S: trial with copper only after bench trials and only at the lowest effective dose, usually 0.1–0.3 mg/L elemental copper equivalent. Follow with immediate fining or racking if required, and check legal limits for copper residues in your market.
4. For mercaptans/disulphides: copper may only partially help. Use bench trials first; in stubborn cases, specialist products or outside contract treatment may be needed.
5. If the wine shows both VA and reduction, address oxygen/microbial control first, then treat the reductive component. Do not chase aromas with repeated additions.

Step 5 — Verify the outcome

1. Re-sample after 24–48 h, then again at 7 days. Smell, taste, and recheck free SO₂ and, where possible, VA.
2. Confirm the fault is not intensifying before returning the lot to normal cellar schedule.
3. If aroma remains unstable, hold the wine, re-evaluate with a second panel, and consider diversion to a lower-value use rather than bottling a compromised wine.
4. Update the lot record with observations, analytical results, actions taken, and the final disposition.

Step 6 — Prevent recurrence

1. Tighten nutrient practice in future ferments with a complete nutrient program, not DAP alone, where yeast stress is likely.
2. Improve cap management, temperature monitoring, and top-up discipline.
3. Standardise bung checks, inert gas use, and barrel maintenance.
4. Flag any lot with recurring reductive history for closer SO₂, temperature, and sensory surveillance.

DECISION LOGIC — WHY THIS APPROACH

VA and reductive faults are often related but not identical, so the first job is to separate oxygen/microbial spoilage from sulfur-based reduction. VA is usually a cellar hygiene and oxygen-control issue, made worse by warm storage, headspace, and residual sugar. Reductive faults are more often a fermentation stress signal, especially where yeast were underfed or overworked. That is why this SOP starts with observation, bench analytics, and process history before any treatment. The key trade-off is intervention versus damage: copper can strip H₂S, but it is not a blanket fix and can create residue/compliance issues if overused. Likewise, adding more SO₂ may help suppress spoilage organisms, but it will not cure a reductive wine and can mask the problem. The procedure therefore prioritises controlling oxygen, temperature, and microbial load first, then applying the least aggressive corrective measure supported by bench trials.

TRIBAL KNOWLEDGE

A wine that smells faintly of struck match in the lab can become ugly in the glass after a warm bottle stand, so always test at serving temperature. Barrels with a slightly loose bung or a dry stave head will often be the repeat offenders, even if the analytical numbers look fine. If a ferment made H₂S once, assume the lot has a memory for reduction and watch it closely after racking or filtration. Copper treatments work best on fresh H₂S; if the smell has gone stale, rubbery, or onion-like, you may be chasing mercaptans or disulphides instead. Many cellar problems blamed on “VA” are really a combination of tired yeast, heat, and slow topping. Good topping discipline prevents more faults than most corrective products ever will.

Fermentation Management

Red Wine Fermentation

CELLAR-READY QUICK STEPS

Check sugar and temperature twice daily.

Punch down and wet cap evenly.

Cool promptly if ferment exceeds 28 °C.

Press at dryness, not by calendar.

Transfer cleanly and label press fractions.

FULL PROCEDURE

Step 1 — Confirm must readiness and set fermentation targets

1. Verify the red must is sound, fully destemmed or as specified by the winemaking plan, and adjusted to the correct temperature before yeast addition.
2. Record starting Brix, pH, titratable acidity, and temperature in the batch sheet. Typical red ferment targets are 22–26 °C at the cap/must interface, with a healthy, clean aromatic profile and no volatile taint.
3. If nutrient is required, plan additions before or during the first third of fermentation rather than waiting until the wine is nearly dry.
4. Check tank fill level and allow adequate headspace for cap rise and foaming. A practical working fill is 70–85% of vessel capacity depending on variety and cap volume.

Step 2 — Inoculate and mix thoroughly

1. Rehydrate or prepare yeast according to supplier instructions and inoculate at the recommended rate, commonly 10–20 g/hL depending on strain and batch condition.
2. Mix the yeast evenly through the must. Confirm a uniform temperature and no dry pockets of skins.
3. If using an initial nutrient addition, dose at the rate specified on the nutrient label; record grams added per hL and the time of addition.
4. Start fermentation observations immediately after inoculation: bubbling, cap lift, aroma development, and temperature rise.

Step 3 — Manage cap immersion and extraction

1. Begin cap management as soon as a stable cap forms, typically within the first 12–48 hours.
2. Punch down or wet the cap 1–3 times per day depending on tank size, ferment rate, fruit condition, and extraction target.
3. During each cap action, break the cap fully and rewet all skins. Avoid splashing the wine excessively into air, and avoid hard compaction that can drive harsh tannin extraction.
4. Check the cap for dryness, vinegar smell, excessive heat, or mould. Any off odour, sticking cap, or localised overheating must be reported immediately.

Step 4 — Control fermentation temperature

1. Measure must temperature at least twice daily, and more often during peak activity.
2. Keep fermentation in the target range of 22–26 °C for most standard red styles unless the wine style calls for a different regime.
3. If temperature rises above 28 °C, increase cooling and reduce extraction intensity until the batch stabilises.
4. If temperature drops below 18 °C and fermentation slows unexpectedly, check yeast viability, vessel insulation, and cooling settings before intervening.
5. Log all temperature corrections and the corresponding sensory condition of the ferment.

Step 5 — Monitor sugar depletion and ferment health

1. Test sugar daily using hydrometer, refractometer with alcohol correction, or density meter as available in the cellar.
2. Record the trend rather than a single reading. Healthy red ferments should show steady decline through the first week.

3. Watch closely once the wine reaches roughly one third sugar depletion; this is the normal window for nutrient timing if a second addition is required.
4. If sugar decline stalls, smell the ferment, check temperature, and reassess nutrient status and yeast stress before taking corrective action.

Step 6 — Finish fermentation and determine press timing

1. Test for residual sugar from day 7 onward, or earlier if the batch is running fast. Do not press solely by calendar day.
2. Press when sugar is effectively dry for the style and the wine has completed its intended skin contact. As a practical cellar target, confirm residual sugar is at or near dryness, generally below 2 g/L, before press if the objective is to finish fermentation off skins.
3. Avoid prolonged maceration once dryness is reached unless the wine style specifically calls for extended skin contact and the fruit is sound.
4. Confirm the wine is free-run and press fractions are segregated according to quality and tannin load.

Step 7 — Press and transfer for settling

1. Drain free-run wine cleanly to the receiving vessel.
2. Load fermented pomace into the press carefully to minimise oxygen pickup and physical damage to seeds.
3. Press incrementally. Combine press fractions only when sensory and analytical checks support the blend decision.
4. Transfer wine promptly to temporary storage for gross lees settling. Protect from excessive oxygen exposure during transfer.
5. Maintain batch identity, record volumes in hL, and note press fraction labels clearly for later blending.

Step 8 — Prepare for malolactic fermentation

1. Once alcoholic fermentation is complete and the wine has settled gross lees, assess the batch for malolactic readiness.
2. Check temperature, free SO₂, pH, and sensory condition before inoculating malolactic bacteria.
3. Do not inoculate MLF into obviously stuck, overheated, or contaminated wine without winemaking approval.
4. Record completion of primary fermentation, press date, settled volume, and readiness for secondary fermentation.

Step 9 — Safety, hygiene, and compliance checks

1. Keep all cellar tools sanitised before cap work, sampling, and transfer.
2. Use appropriate PPE for CO₂-rich ferment areas and press operations.
3. Ensure tank lids, hoses, pumps, and presses are secured and pressure-rated for the task.
4. Report leaks, unusual pressure build-up, or unsafe temperatures immediately and stop work if required.

DECISION LOGIC — WHY THIS APPROACH

Red wine fermentation is managed by tracking three linked variables: cap behaviour, temperature, and sugar decline. That is why the procedure prioritises frequent observation rather than fixed-day timing. Cap work drives colour and tannin extraction, but too much agitation or heat will pull harsh seed tannin and cook fruit. Temperature control sits in the middle: warm enough to keep yeast active, cool enough to preserve aroma and limit volatility. Sugar monitoring is the practical trigger for pressing, because calendar timing varies with fruit condition, yeast strain, and vessel size. The procedure sets a dryness target before press because pressing too early leaves residual sugar and risks mixed fermentation off skins; pressing too late can over-extract and increase oxidation risk. Deviate from these ranges only when the style, fruit quality, or winery direction clearly justifies it, and document the reason.

TRIBAL KNOWLEDGE

Cap management is as much about feel as schedule: a tight, dry cap usually needs more frequent wetting than a loose one. If the top smells hot or vinegary, check the whole tank, not just the surface. Small fermenters can run hotter than they read if the probe sits near the wall, so cross-check with a second thermometer. For press work, keep the first press fractions separate; they often carry the best structure,

while the later pressings can get coarse quickly. If you are using nutrients, dissolve or disperse them properly — clumps in the cap do little and can cause local stress. Always expect ferment noise to change before sugar readings tell you anything obvious.

Pump-Over Protocol

CELLAR-READY QUICK STEPS

["Check tank temperature and cap condition.", "Start pump and wet the entire cap.", "Adjust flow to avoid splashing and channels.", "Log time, temperature, and sensory observations.", "Clean and sanitise all hoses immediately."]

FULL PROCEDURE

Step 1 — Confirm fermenter readiness and safety

1. Verify the fermenter is within fermentation bay access, fitted with a clean lid or cover, and has sufficient headspace for cap rise and foaming.
2. Check pump, hoses, clamps, spray ball/return hose, and fittings are clean, sanitised, and assembled leak-free.
3. Confirm the must is in active alcoholic fermentation: signs include a working cap, CO₂ release, and a drop in Brix/SG. Pump-overs are generally required once fermentation is underway, typically from day 2 to day 14, depending on temperature and yeast activity.
4. Wear PPE: safety glasses, gloves, closed footwear, and hearing protection if required. Never lean over an open tank during pump-over because of splash and CO₂ exposure.

Step 2 — Set the schedule and intensity

1. Start with 1–3 pump-overs per day for active red ferments.
2. Increase to 3–4 pump-overs per day for hot, fast ferments or lots with thick caps, high colour potential, or elevated temperature.
3. Reduce to 1–2 pump-overs per day if the cap is thin, fermentation slows, or tannin extraction is already firm.
4. Target 10–20 minutes per pump-over on a standard 20–50 hL fermenter, or until the cap is fully wetted and the return wine runs evenly across the surface.
5. Keep ferment temperature within the winery setpoint, typically 22–30 °C for most reds, and avoid letting the cap sit dry or crusted.

Step 3 — Prepare the tank and break the cap

1. Open the tank carefully and assess cap condition before starting the pump.
2. If the cap is compacted or domed, break it mechanically with a clean punch-down tool or by directing the first return stream to penetrate the cap.
3. Check for acetobacter risk indicators: vinegar, nail polish, mould, or surface drying. If present, escalate immediately and review hygiene, cover, and pump-over frequency.
4. Record the tank temperature before the pump-over begins.

Step 4 — Execute the pump-over

1. Start the pump and recirculate wine from the bottom of the tank back over the cap.
2. Distribute the return flow across the widest possible surface area; do not jet one point only, as this creates channels and poor extraction.
3. Keep the cap fully wet for the duration of the pump-over. The surface should glisten uniformly, with no dry islands.
4. If the ferment is vigorous, pause briefly if foaming becomes excessive, then resume at a gentler flow rate.
5. Avoid excessive aeration unless a deliberate oxygenation strategy has been approved for the lot; the goal is cap wetting, extraction, and heat dispersal, not splashing.

Step 5 — Monitor extraction, temperature, and fermentation health

1. Re-check tank temperature at the end of the pump-over. If temperature has climbed above target by more than 2 °C, increase frequency, reduce pump duration, or apply cooling.

2. Smell the returning wine and cap surface. Clean ferment aromas should be fruit-driven and yeast-active; watch for H₂S, VA, or reduction.
3. Observe cap integrity and colour transfer. Good pump-overs should darken the return wine early in fermentation and keep skins suspended without drying.
4. If the lot is showing sluggish activity, review yeast nutrition and fermentation conditions before simply increasing pump-over intensity.

Step 6 — Manage nutrient additions if scheduled

1. Add any fermentation nutrient only as approved by the winemaking schedule, typically during active fermentation rather than at inoculation alone.
2. Dissolve or slurry nutrients in clean must or wine before addition to avoid clumping.
3. Add nutrients during a pump-over to ensure rapid distribution through the tank.
4. Confirm the addition is logged with product name, lot, rate, and time.

Step 7 — Clean down and document

1. Stop the pump, isolate power, and depressurise lines before disconnecting any hoses.
2. Flush equipment immediately with water, then clean and sanitise according to cellar hygiene SOP.
3. Remove any cap debris from screens, hoses, and return fittings.
4. Record date, time, tank ID, temperature, Brix/SG, pump-over duration, observed cap condition, and any sensory or corrective notes.
5. Escalate any abnormal findings: stuck ferment risk, excessive heat, contamination, or repeated poor cap wetting.

DECISION LOGIC — WHY THIS APPROACH

Pump-over is preferred over passive cap management because it gives direct control over extraction, temperature, and cap wetting in active red fermentations. Compared with leaving the cap alone, it reduces dry-cap oxidation and microbial risk while improving colour and tannin pickup. Compared with punch-down alone, it is usually faster and more scalable in boutique cellar conditions, especially in larger tanks or when staff numbers are tight. The trade-off is that pump-over can over-extract or aerate a lot if flow is too aggressive, so rate and frequency must be adjusted to grape maturity, cap density, and temperature. Nutrient additions are best folded into a pump-over because distribution is more even and there is less localised concentration. Deviate from the standard schedule if the ferment is cool and slow, if the cap is thin, if tannins are already firm, or if temperature is running high and the lot needs more heat management rather than more extraction.

TRIBAL KNOWLEDGE

A good pump-over is judged by the cap surface, not just the clock. If you can see dry patches, you're too light on wetting or you've got channeling. On cone-bottom or awkward drains, the first litre often comes out muddy; run it to waste if the lot is heavily grossed up, then return clean wine. Some hoses whip badly when the pump starts, so always secure the discharge before opening full flow. If the ferment starts foaming, back the pump off immediately rather than trying to "push through" it. Tanks with cold jackets can hide a hot core, so don't trust surface temperature alone—check the mass and keep notes by lot.

White Wine Fermentation Temperature Control

CELLAR-READY QUICK STEPS

- Measure tank temperature and Brix now.
- Set cooling to the target band.
- Check probes and glycol flow twice daily.
- Correct overshoots immediately, then recheck.
- Record every temperature intervention in the log.

FULL PROCEDURE

Step 1 — Confirm must readiness before cooling

1. Verify the juice or must has been settled, clarified as required, and is ready for inoculation or active fermentation control.

2. Measure and record starting temperature, Brix/°Baumé, pH, and YAN if available.
3. Set the fermentation target temperature based on style and aroma retention:
 - Light, fresh aromatic whites: 12–15 °C
 - Neutral whites or fuller styles: 15–18 °C
 - Avoid sustained temperatures above 20 °C unless a deliberate style decision is made.
4. Check tank volume, headspace, coolant capacity, and agitation/recirculation equipment before starting. Temperature control is most effective when the system is sized correctly and already stable.

Step 2 — Inoculate and establish the set point

1. If using cultured yeast, rehydrate and inoculate according to the winery yeast SOP.
2. Bring the must to the selected set point within 2 °C before or after inoculation. Do not shock the yeast with a rapid drop of more than 4 °C in a short period.
3. Confirm the cooling jacket, glycol loop, or tank room control is operating smoothly and not cycling excessively.
4. For tanks with poor mixing, circulate gently to eliminate hot spots, especially around the cone and top third of the tank.

Step 3 — Hold the fermentation in the target band

1. Check fermentation temperature at least twice daily during active fermentation; every 4–6 hours for small lots, warm ferments, or fast kinetics.
2. Maintain the must within ± 1 –2 °C of the target band.
3. Use cooling to pull down temperature when the ferment rises above target, rather than waiting for the tank to overshoot.
4. If the temperature falls below target and fermentation slows, reduce cooling immediately and allow the tank to drift back naturally.
5. Record temperature, Brix, tank status, and any interventions in the cellar log at each check.

Step 4 — Adjust control to match fermentation speed

1. As sugar decreases and heat generation drops, expect the ferment to require less cooling.
2. Taper set points gradually in the final third of fermentation if a cleaner finish is needed, but avoid sudden changes that can stress yeast.
3. If the ferment is sluggish, first confirm temperature, then check nutrient status, yeast health, and dissolved oxygen management under the winery's broader fermentation SOPs.
4. Do not chase temperature alone when another fault is present; temperature correction is supportive, not a cure for stuck fermentation.

Step 5 — Watch for end-of-ferment risk points

1. In the last 24–72 hours, monitor more closely because heat output becomes less predictable and residual sugar is lower.
2. Keep the wine in the lower end of the target range if retaining volatile aromatics is important.
3. Test residual sugar routinely to confirm fermentation completion rather than relying on temperature or bubbling alone.
4. Once dryness is reached, stop active cooling unless holding the wine cold for clarification or transfer.

Step 6 — Respond to deviations safely

1. If the must exceeds the target by more than 2 °C, apply cooling immediately and recheck within 30–60 minutes.
2. If coolant supply is interrupted, notify the cellar lead, move to backup cooling if available, and increase monitoring frequency.
3. If a tank is warming despite cooling, check valve function, probe placement, circulation, insulation, and glycol temperature.
4. Never use uncontrolled ice addition or direct freezing contact with the must unless approved by the winery's equipment and food safety procedures.

5. Escalate any temperature excursion that may affect fermentation completion, aroma quality, or microbial stability.

Step 7 — Verify completion and hand over

1. At dryness, confirm stable residual sugar results on consecutive checks.
2. Document final fermentation temperature profile, peak temperature, and any corrective actions taken.
3. Transfer or press according to the winery's white wine post-fermentation SOP.
4. If malolactic fermentation is planned, ensure the wine is at the appropriate temperature range for inoculation or storage as specified in the MLF SOP.
5. Clean and sanitise probes, hoses, jackets, and any circulated lines after use to prevent cross-contamination and spoilage carryover.

DECISION LOGIC — WHY THIS APPROACH

White wine fermentation is managed colder than reds because aroma retention, colour preservation, and freshness are usually the priorities. A controlled band of roughly 12–18 °C gives enough yeast activity for a complete ferment while limiting ester loss, volatile acidity risk, and unwanted oxidation-driven character changes. The lower end suits delicate aromatic styles; the upper end suits fuller, less aromatic whites that benefit from slightly faster kinetics and broader mouthfeel. Temperature control must be proactive, because once a tank overshoots, the heat lag can push it well beyond the intended range before the system catches up.

Alternatives such as letting the ferment run warm, or heavily chilling only after a spike, are less consistent and can stress yeast, particularly late in ferment when nutrient and oxygen are already limited. Deviate only when the style demands it, the yeast strain tolerates it, or the tank design cannot hold the desired band. If a ferment slows, do not assume temperature alone is the problem; check sugar, nutrients, and yeast health before changing the set point aggressively.

TRIBAL KNOWLEDGE

Probe placement matters more than most people think. A probe jammed near a cooling jacket will read falsely cold, while one sitting dead centre in a stratified tank can miss a hot top layer. On small lots, a gentle recirculation or a punch-down style mix with a pump-over loop evens the tank out fast enough to avoid chasing numbers.

Glycol performance also changes through the day. Early afternoon return temperatures can be much worse than morning, so set points that look fine at 7 am can drift by lunch. Keep an eye on solenoid valves and insulation around hoses; a small leak or warm section can cost you more than a new yeast dose. For aromatic whites, many cellar crews find it better to hold steady than to keep tightening the temperature every few hours.

Cap Management & Extraction Strategy

CELLAR-READY QUICK STEPS

["Set extraction target and punch-down schedule.", "Check cap, temperature, and Baumé twice daily.", "Punch cap 2–4 times daily during peak ferment.", "Add nutrient per protocol before yeast stress develops.", "Press when balance is reached, not just dryness."]

FULL PROCEDURE

Step 1 — Set the extraction intent before ferment starts

1. Confirm the wine style target: light, medium or full extraction, and whether tannin should be soft and early or more structured and late.
2. Record fruit condition at receipt: berry integrity, mould, pH, Baumé, temperature, and any winery-spec factor that may require a gentler cap regime.
3. Define the cap management plan in the ferment sheet: punch-down frequency, any pump-over schedule, and the review points for adjusting extraction.
4. For whole-berry or open-top ferments, make sure the cap can be safely accessed with clean tools and that there is headspace for CO₂ release and foaming.

Step 2 — Monitor ferment onset and cap formation

1. Expect activity within 1–3 days after yeast addition; colder musts will start slower, warmer musts faster.
2. Check twice daily for signs of active fermentation: bubbling, cap rise, heat gain, and increasing aroma intensity.
3. Once the skins have risen and formed a cap, start cap management immediately. Do not allow the cap to dry out or sit exposed for long periods.
4. Measure ferment temperature at least morning and afternoon. Keep the core of the ferment typically in the 20–32 °C range, depending on style; avoid uncontrolled rises above 32 °C unless there is a documented style or extraction reason and the ferment can be safely managed.

Step 3 — Punch down the cap to protect and extract

1. Punch the cap several times a day during active primary fermentation; a practical starting point is 2–4 times daily, increasing frequency if the cap is thick, hot, or drying at the edges.
2. Use a sanitised punch-down tool and break the cap fully back into the liquid. Aim to wet all surface skins and corners, not just open a central hole.
3. Observe the cap after each punch. It should sink and re-form evenly without crusting.
4. Use punch-downs to distribute heat and oxygen early in fermentation, then reduce exposure as alcohol rises and the cap becomes more fragile.
5. If the ferment smells reductive or sluggish, increase cap wetting and review nutrition before making more aggressive extraction changes.

Step 4 — Adjust extraction intensity to suit the wine

1. For delicate fruit-led reds, keep punch-downs gentle and avoid excessive mechanical shear.
2. For firmer tannin and deeper colour, use more frequent punch-downs or combine punch-downs with controlled pump-overs if the vessel and cap structure allow it.
3. Watch the taste and texture of the draining wine and cap contact. If tannin becomes gritty, bitter, or drying ahead of colour gain, back off the extraction intensity.
4. Check colour density, aroma, and mid-palate texture daily during the main extraction phase, especially from days 2 to 7.

Step 5 — Support yeast with nutrition during the ferment

1. Do not rely on starter yeast alone to carry the ferment. As alcohol rises and natural nutrients fall, yeast need additional nitrogen, amino acids, and micronutrients.
2. Add a complete fermentation nutrient in a staged approach if required by the house protocol or analysis. Typical additions are made early in active ferment and, where used, again before the ferment becomes nutrient-stressed.
3. Keep dosage within supplier instructions and cellar records; avoid blind over-dosing, which can leave nutrient residue or confuse downstream biology.
4. If hydrogen sulphide, sluggish rate, or rising volatile acidity risk appears, reassess nutrition first, then temperature and cap handling.

Step 6 — Track temperature, sugars, and press timing

1. Measure Baumé or sugar depletion daily, and more often if the ferment is running hot or fast.
2. Use temperature to shape extraction: cooler ferments generally preserve fruit and moderate tannin; warmer ferments increase extraction but raise the risk of harshness and volatile issues.
3. Plan pressing when the desired balance has been reached, not simply when dryness is achieved. The right press point is when colour, tannin and fruit are in balance for the style.
4. Taste the free-run and cap-contact fractions separately if possible. Press earlier for elegant, fruit-forward reds; extend skin contact only where the structure remains clean and the ferment is healthy.

Step 7 — Transition cleanly toward post-ferment handling

1. Once the wine is pressed, settle it promptly and protect it from oxygen.

2. Review whether the wine is suitable for malolactic fermentation. Healthy reds with correct temperature, low free SO₂, and no obvious microbial fault are preferred candidates.
3. Record cap management frequency, temperature profile, nutrient additions, pressing date, and sensory notes for each batch.
4. Clean and sanitise all punch-down and pump-over equipment immediately after use to prevent tartrate, pigment, and microbial build-up.

DECISION LOGIC — WHY THIS APPROACH

Cap management is the main tool for balancing protection and extraction during primary fermentation. Punch-downs keep the cap wet, reduce drying and spoilage risk, and release heat while driving colour and flavour from the skins into the wine. It is preferred over leaving the cap undisturbed because exposed skins can dry out and become a contamination point. More aggressive methods, such as intensive pump-overs, can improve extraction but also increase tannin and oxygen exposure; they suit firmer styles but can overwork delicate fruit. Nutrition is staged because yeast demand rises as alcohol accumulates and natural must nutrients are depleted. This helps prevent sluggish finishes, hydrogen sulphide, and elevated volatile acidity. Temperature control is the other lever: warmer ferments accelerate extraction and metabolism but need tighter oversight to avoid harsh tannin and thermal stress. Deviate from the standard plan only when fruit condition, style target, or ferment kinetics justify it, and document the reason.

TRIBAL KNOWLEDGE

A cap that looks soft can still be dry underneath, so always break it fully and check the corners of the vessel. If the ferment starts throwing heat after lunch, it usually needs attention before the evening punch, not after. Older open-top bins often keep a thicker cap at the back wall, so work that side first. Stainless punch-down poles slide better if they are rinsed clean but not left with detergent residue. For warm, fast ferments, small but frequent interventions usually give better tannin shape than one hard daily punch. Keep a tasting glass at the ferment so you can smell for reduction, sharpness, or solvent notes while the cap is still active.

Harvest & Receival

Harvest Decision & Sampling

CELLAR-READY QUICK STEPS

- Walk the block and collect representative berries.
- Measure Brix, pH, TA, and berry temperature.
- Taste skins and seeds for phenolic maturity.
- Confirm pick date, transport, and receival readiness.
- Log the decision and notify all stakeholders.

FULL PROCEDURE

Step 1 — Define the picking target

1. Set the harvest decision using vineyard samples from representative blocks, not a single “best” row or edge panel.
2. For red wine fruit, the practical trigger is ripeness plus flavour and seed maturity, not sugar alone. Use the latest lab data to confirm:
 - Brix / °Baumé in the target range for the style.
 - Titratable acidity and pH trending to cellar plan.
 - Clean, brown seed colour and softened skin tannin.
3. Record block, clone/rootstock if known, irrigation history, disease pressure, and any weather risk in the next 7 days.
4. If rain, heat or bird pressure is escalating, prioritise harvesting the most exposed or most advanced parcels first.

Step 2 — Take a representative vineyard sample

1. Sample at least 1–2 days before the intended harvest call, then resample daily if weather is changing.
2. Walk a full diagonal or “Z” path through each block. Avoid headlands only, shaded corners only, and damaged vines only.
3. Collect berries or bunch sections from at least 20–30 vines per block, more if the block is variable. Keep sample weight sufficient for all tests; 1–2 kg is usually adequate.
4. Place fruit in a clean, food-grade container. Keep samples cool and out of direct sun. Do not crush in the vineyard.
5. Label every sample with block, date, time, sampler, and intended pick window.

Step 3 — Assess the fruit on intake

1. On the same day, split the sample into two checks: laboratory-style analysis and sensory inspection.
2. Measure:
 - Brix / °Baumé.
 - pH.
 - Titratable acidity.
 - Berry temperature.
 - Any obvious rot, sunburn, mildew, or bird damage.
3. Taste berries and skins. Note flavour development, tannin ripeness, and seed bitterness. For red wine, underripe tannin usually shows as green/astringent seed bite even when sugar is acceptable.
4. If samples are inconsistent, re-sample rather than forcing a harvest call from poor data.

Step 4 — Make the harvest decision

1. Harvest when the fruit is microbiologically sound, flavour is where the style needs it, and sugar/acid balance matches winery targets.
2. Do not wait for sugar to improve if forecast heat, disease, or fruit loss will compromise quality.
3. Do not pick solely on high sugar if acidity has collapsed or fruit has become overripened and tired.
4. Confirm any special winemaking requirement before approval:
 - If using whole berries, reserve gentle fruit with sound skin integrity.

- If aiming for structure, allow modest phenolic maturity but avoid shrivel and excessive dehydration.
5. Authorise pick date, crew, bins, transport timing, and receival priority.

Step 5 — Prepare for receival and intake control

1. Communicate the estimated delivery window to the winery so press, crusher-destemmer, pumps, bins, and cold room space are ready.
2. Ensure all harvest bins, sampling tools, and containers are clean and sanitised.
3. Set a receival check to confirm:
 - Fruit condition on arrival.
 - Temperature.
 - Brix / °Baumé.
 - Bin counts and traceability.
4. Reject or segregate any lot showing obvious contamination, severe spoilage, or untraceable origin.

Step 6 — Escalate special cases

1. If fruit is borderline for ripeness but clean, consider picking early and adjusting in the winery only under approved cellar plan.
2. If fruit is high sugar and warm, schedule immediate cooling or processing to reduce oxidation and volatile loss.
3. If fruit is intended for extended maceration, be stricter on health and tannin maturity at the sampling stage.
4. Any deviation from target ranges must be signed off by winemaking management before harvest proceeds.

Step 7 — Document and communicate

1. Enter sample data, tasting notes, decision, and authoriser into the harvest log.
2. File vineyard maps, lab sheets, and any photos of fruit condition.
3. Issue the final pick instruction to the grower and cellar team, including row/block identity, expected tonnage, and transport sequence.
4. Retain all records for winery QA, audit trail, and vintage comparison.

Step 8 — Link harvest decision to downstream fermentation control

1. Use the harvest decision to set ferment expectations. Fruit picked riper and cleaner generally supports more predictable primary fermentation and cap management.
2. Confirm that must temperature, nutrition plan, and fermentation monitoring can be matched to the fruit maturity and condition.
3. If the fruit is likely to ferment hard or unevenly, plan closer sugar checks, temperature control, and yeast nutrition from the start.
4. Harvest quality decisions should always protect the ability to complete fermentation cleanly and, where required, transition smoothly toward pressing and malolactic fermentation.

DECISION LOGIC — WHY THIS APPROACH

The harvest call is based on more than sugar because red wine quality is driven by the balance of flavour maturity, tannin ripeness, acid structure, and fruit health. Sampling across the block reduces bias from edge effects, sun exposure, and variable irrigation. Lab numbers are essential, but they do not replace tasting seeds and skins; green seed bitterness and hard skin tannin often tell you more about phenolic readiness than Brix alone. This approach also protects fermentation outcomes: clean, ripe fruit ferments more predictably, needs fewer corrective actions, and is less likely to bring in spoilage or stuck-ferment risk. We deviate when weather risk, disease, or excessive dehydration makes waiting more damaging than picking slightly early. In those cases, the decision should be deliberate and documented, with winery controls ready to manage the fruit.

TRIBAL KNOWLEDGE

Always sample from the vine, not the bucket. Berries sitting in a hot ute tray can shift temperature and mask the real picture by the time you test them. If a block looks variable, take extra berries from the “worst” vines as well as the best; that is usually where the ripening story is hiding. Watch for shrivel in the middle of the canopy after a warm spell — those berries can push sugar up without improving flavour. Keep a small

refractometer in the field, but back it up at the winery with proper pH and acidity checks. Also, one grower's "ready" can mean very different things from another's, so use the same tasting and sampling routine every season.

Fruit Reception & Sorting

CELLAR-READY QUICK STEPS

["Inspect load and record source details.", "Sort out MOG, mould, and damaged fruit.", "Crush gently and destem cleanly.", "Add 50 ppm SO₂ immediately and mix well.", "Check °Brix, pH, and transfer promptly."]

FULL PROCEDURE

Step 1 — Receive, identify and inspect the fruit

1. On arrival, record the vineyard/block, cultivar, pick date and time, grower, estimated volume, transport temperature if available, and any spray or disease notes provided.
2. Visually inspect each load before tipping. Reject or quarantine fruit showing obvious rot, mould, bird damage, insect infestation, smoke taint indicators, or excessive leaf/stem trash.
3. Take a representative sample from the top, middle and bottom of the bin or trailer. Check for berry condition, colour, firmness, sunburn, and whether the fruit is intact enough to handle without excessive juice loss.
4. Confirm the fruit is cool on arrival. Aim to process at 10–18 °C where practical; above 20 °C, process immediately or hold in a cool room to limit oxidation and wild ferment activity.
5. Identify loads needing special handling: overly ripe fruit, high pH fruit, underripe fruit, fruit with botrytis, or fruit destined for whole-bunch inclusion.

Step 2 — Sort for quality and hygiene

1. Set up the sorting line, bins, and drains so fruit can move smoothly without delay. Sanitise all contact surfaces before receival begins.
2. Remove leaves, petioles, MOG (material other than grapes), unripe bunches, mouldy berries, dried berries, damaged fruit, insects, and foreign material by hand or on a sorting table.
3. Keep the sorting standard tight: no visible mould, no heavily damaged berries, and minimal stalk matter should pass through to crush.
4. For red wine fruit, maintain enough whole, sound berries to preserve colour, flavour, and tannin quality. Do not over-handle fruit; rough sorting creates juice loss and early oxidation.
5. Transfer sorted fruit promptly to the crusher/destemmer or fermenter. Do not leave fruit exposed to air for longer than necessary.

Step 3 — Decide on destemming and crush settings

1. Select the crusher/destemmer based on batch size and fruit condition. Manual units are suitable for small to moderate lots, while motorised units are preferred for larger volumes and steady throughput.
2. Adjust the crusher to achieve a gentle crush only. The aim is to split berries without shredding skins, seeds, or stalks.
3. Set destemming to remove stems cleanly with minimal berry loss. If the fruit is fragile or heat-stressed, reduce speed and feed rate to avoid excessive juice release.
4. Check the first few kilograms closely. If stems are still passing through, reduce feed rate or adjust the destemmer. If too many seeds are cracked, open the gap or reduce crushing pressure.

Step 4 — Add sulphur dioxide immediately

1. Add potassium metabisulphite at 50 ppm SO₂ equivalent to the must as soon as the fruit is crushed. A practical bench rate is 1.65 g potassium metabisulphite per 5 gal (18.9 L) must.
2. Mix thoroughly through the full volume. Do not leave the addition sitting on the top layer or in a pocket at the bottom of the tank.
3. Record the dose, batch size, time of addition, and operator name. Ensure all chemical handling follows site PPE and ventilation requirements.
4. Allow 4–24 hours contact time before yeast or enzymes are added, depending on fruit temperature, cleanliness and ferment plan.

Step 5 — Measure and correct must parameters

1. Pull a representative juice sample after crush and mixing. Measure °Brix using a refractometer or hydrometer.
2. For red wine must, target 22–25 °Brix at receival. If below range, correct with sugar addition only where permitted by site policy and local regulations; dissolve fully before rechecking.
3. If above range, dilute with acidulated water only if the winemaking plan allows it and the dilution will not compromise quality. Recheck °Brix after mixing.
4. Measure pH and titratable acidity if your cellar set-up allows. Flag fruit with high pH for closer SO₂ management and faster processing.

Step 6 — Transfer, document and hold safely

1. Move sorted must to the fermenter or holding vessel with minimal aeration. Avoid splashing, long hose runs, and unnecessary pump shear.
2. Cover tanks or bins to reduce oxidation, insect access and contamination risk.
3. Check for leaks, blockages, and residual stems or fruit in the crusher area before the next load.
4. Clean and sanitise all equipment immediately after use, including the sorting table, crusher/destemmer, hoses, pumps and floors.
5. Complete the receival record: fruit source, sort outcome, estimated rejects, additions, analytical results, and any deviations from standard.

DECISION LOGIC — WHY THIS APPROACH

This procedure prioritises fruit quality, hygiene, and controllable oxidation from the moment the grapes hit the winery. Sorting before crush is the best point to remove mould, MOG, and damaged berries because once those compounds enter the must, they are harder to manage and can lift microbial load, volatile acidity risk, and off-flavours. Gentle destemming and crush are favoured over aggressive processing because boutique lots usually benefit more from preserving intact berry material than from chasing throughput. Immediate SO₂ at 50 ppm equivalent gives a practical protection window against wild yeast and bacteria while the must settles, but it is not a substitute for clean fruit or fast processing. The 22–25 °Brix target suits most red fermentations; correcting early is preferable to trying to fix ferment issues later. Deviate when fruit is exceptionally clean and cool, or when style demands whole-bunch or carbonic handling, but only with an intentional plan and documented risk controls.

TRIBAL KNOWLEDGE

Most problems at receival start with delay, not the crusher. If fruit is sitting in bins under the sun, treat it as a hot load and move fast. Keep a hose and bucket handy for washing down the sorter between loads; dried skins and seeds become grit and contaminate the next batch. Crusher rollers that are set too tight will crack pips before you notice it in the tank, so always check the first outlet tray by hand. If you can smell mould or vinegar on the sorting line, the fruit is already telling you something useful—don't bury it under SO₂ and hope. Many cellar teams prefer to weigh rejects by bin rather than guess; it gives better grower feedback and helps spot vineyard issues early.

Cold Soak & Pre-Fermentation Hold

CELLAR-READY QUICK STEPS

- Check fruit temp, °Brix, pH and TA on receival.
- Fill tank, flush headspace, and seal under inert gas.
- Hold at 5–10 °C and log twice daily.
- Taste and smell daily for extraction and spoilage.
- Inoculate promptly once soak target is reached.

FULL PROCEDURE

Step 1 — Confirm fruit suitability and reception targets

1. Receive only sound, fully ripe red fruit. Reject or divert any parcels with rot, mould, severe sunburn, bird damage or excessive herbicide/oxidative taint.
2. Record cultivar, block, pick date, vineyard, yield, °Brix, pH, TA and fruit temperature on arrival.

3. Target fruit temperature at receipt: 10–18 °C. If fruit is hotter than 18 °C, move immediately to cooling or processing; if colder than 8 °C, expect a slower extraction phase and longer hold.
4. Sulphur dioxide additions, if used at receipt, must be approved by the winemaker and comply with site wine treatment limits and label/council requirements. Typical protective additions for cold soak are in the range 30–50 ppm free SO₂ equivalent at crush, but adjust for fruit condition and pH.

Step 2 — Crush, de-stem and protect the must

1. Crush and de-stem with minimum berry damage. Avoid splitting seeds, as seed tannin can become harsh if the hold is too long.
2. Transfer must into a clean, sanitised, sealed fermenter or cold soak tank fitted with temperature control.
3. Fill tanks to leave minimal headspace. Aim for 90–95% fill, depending on cap rise risk and tank geometry.
4. Add enzyme and tannin products only if approved by the winemaker and the cellar treatment schedule. Mix thoroughly but gently to avoid aeration.
5. Flush headspace with CO₂ or argon after filling and after any addition. Fit lids/float lids and maintain an inert gas blanket where practical.

Step 3 — Set cold soak conditions

1. Hold must at 5–10 °C for a standard cold soak. Do not exceed 12 °C unless the winemaker has approved a warmer pre-fermentation hold for a specific style.
2. Use jacket cooling, glycol, cold room or refrigerated tank control. Check tank temperature at least twice daily, and more often in warm ambient conditions.
3. Record must temperature, tank pressure or headspace condition, and any sign of spontaneous activity each check.
4. Keep the soak duration at 2–5 days for most lots. Extend only with clear intent and winemaker approval; the longer the hold, the greater the risk of microbial growth and excessive extraction.

Step 4 — Maintain hygiene, oxygen exclusion and microbial control

1. Keep all fittings, pumps, hoses and sampling tools sanitised. Any must exposed to air must be returned to inert conditions promptly.
2. Inspect the surface daily for film yeasts, gas formation, leakage, foaming or odours. A rising temperature, bubbles, or fruity gas smell indicates onset of fermentation.
3. If spontaneous activity begins early, either inoculate immediately or warm and proceed to ferment management. Do not leave active must sitting cold and exposed.
4. If any off-odour such as H₂S, VA or mouldy character develops, notify the winemaker immediately. Do not attempt correction with DAP alone; if nutrition is required, use a complete nutrient program rather than a single DAP addition.

Step 5 — Monitor extraction and prepare for yeast addition

1. Taste and assess daily for colour release, aroma development and tannin grip. The goal is improved early extraction without losing fruit freshness or creating a hard seed/tannin profile.
2. Take a small must sample daily for °Brix trend, temperature and smell. Refractometer readings become unreliable once fermentation starts, so use them only before active ferment or with the approved correction method.
3. If cap formation begins before inoculation, break it gently and re-submerge to avoid surface drying and spoilage growth.
4. End the cold soak when colour/aroma is where intended, when temperature control cannot be maintained, or when inoculation timing requires it.

Step 6 — Transition to inoculation and fermentation start

1. Bring must to the preferred yeast pitch temperature, typically 18–25 °C depending on style and yeast strain.
2. Re-check free SO₂, °Brix, temperature and nutrient plan before yeast addition.
3. Rehydrate and inoculate yeast according to the yeast supplier and winery SOP. Add a complete nutrient

regime where required, especially for low YAN fruit or high-risk ferments.

4. After inoculation, monitor for active fermentation within 1–3 days. Once CO₂ production begins, move to cap management and fermentation temperature control procedures.

Step 7 — Release and documentation

1. Release the tank only when the winemaker confirms the soak outcome, inoculation readiness and traceability records are complete.
2. File receipt records, temperature logs, SO₂ additions, sensory notes and any deviations.
3. Escalate any hold outside target time/temperature, any microbiological concern, or any unapproved addition to the winemaker before further processing.

DECISION LOGIC — WHY THIS APPROACH

Cold soak is used because it encourages early colour and aroma extraction from skins while keeping yeast activity suppressed. That gives the winemaker a more controlled extraction window before fermentation heat and carbon dioxide drive cap formation and faster phenolic movement. The main trade-off is risk: longer or warmer holds can encourage spoilage organisms, oxidation and seed-derived harshness, especially if headspace is poorly managed or fruit quality is marginal.

This SOP therefore keeps the hold short, cold and inert, with tight temperature monitoring and conservative sulphur protection. It avoids DAP-only thinking because pre-ferment nutrients are not about feeding a starving population; once yeast starts, a complete nutrient strategy is safer for avoiding H₂S. Deviate only when the fruit is exceptionally clean, the style calls for deeper extraction, or the winemaker has approved a different temperature/time regime.

TRIBAL KNOWLEDGE

Cold soak only works if the tank is genuinely cold all the way through, not just on the jacket. Wide tanks with poor glycol contact can have a warm centre and a cold edge, so stir only if the winemaker wants it and keep the mix gentle. Flat lids and old seals leak more than people think; check for CO₂ loss overnight by smell and by headspace pressure.

For red ferments, fruit from cooler blocks often needs an extra day or two to show the same extraction as warm-picked fruit. If the berries smell overly stinky or the seeds are still green, don't chase a long soak. Also, keep a dedicated sampling thief for pre-ferment lots; one dirty thief can move film yeast from one tank to the next.

Laboratory Testing

Brix / Specific Gravity Monitoring

CELLAR-READY QUICK STEPS

- Draw a representative, degassed sample.
- Check and record sample temperature.
- Measure and log Brix or SG.
- Compare against yesterday's trend.
- Escalate if readings flatten unexpectedly.

FULL PROCEDURE

Step 1 — Purpose and scope

This SOP sets out how to monitor sugar depletion during primary fermentation using Brix and/or specific gravity (SG). It applies from yeast inoculation through to completion of primary fermentation, with the highest frequency of checks in the first 3–10 days when the ferment is most active. The intent is to confirm yeast performance, identify sluggish or stuck ferments early, and support the decision to press, rack, or intervene.

Step 2 — Equipment and sampling readiness

1. Use a calibrated hydrometer or densitometer, clean test jar, thermometer, wine thief or sample tap, and a notebook or cellar software record.
2. Confirm the measuring range suits must/wine conditions. For red ferments, SG readings typically move from about 1.090–1.110 at start down to 0.998–0.995 at dryness.
3. Check the sample is representative: mix the top and bottom of the vessel as safely as possible before drawing, because sugar stratification can occur.
4. Wear PPE: safety glasses, gloves, and non-slip footwear. Keep hands clear of moving cap management tools and avoid leaning over open ferments.

Step 3 — Sampling method

1. Draw a sample of 200–300 mL from the same vessel location each time where possible.
2. If using a hydrometer, degas the sample gently by stirring or shaking in a closed container, then let foam disperse. Dissolved CO₂ can float the hydrometer and falsely lower density readings.
3. Bring the sample to a consistent temperature close to the instrument's calibration temperature, usually 20 °C.
4. Record sample temperature to the nearest 0.5 °C.

Step 4 — Measuring Brix or SG

1. Fill the cylinder with enough sample for the hydrometer to float freely, avoiding skins and seeds.
2. Spin the hydrometer lightly to dislodge bubbles, then read at the meniscus at eye level.
3. Record either Brix or SG, but keep the same measure throughout the ferment for consistency. If Brix is used, note that raw Brix during fermentation is not a true sugar value because alcohol changes the reading.
4. For cross-checking, use a temperature-corrected conversion table or cellar chart only if your winery has a standard method for that conversion.

Step 5 — Monitoring frequency and target checkpoints

1. During active fermentation, measure at least once daily; for warm, fast, or high-risk ferments, measure twice daily.
2. Record the following at each check: date/time, vessel ID, temperature, Brix/SG, cap condition, aroma, and any nutrient or press additions.
3. Typical progression should show a steady daily drop. As a practical guide, a healthy red ferment often drops several SG points per day once fully active.
4. If temperature rises above the cellar target, expect faster sugar loss but also higher extraction and potential stress. Typical active red ferment temperatures are managed around 24–30 °C, depending on style and winery policy.

Step 6 — Interpreting the trend

1. Confirm the ferment is active if Brix/SG is falling consistently and the cap is being pushed up by CO₂.
2. Treat a flat reading as a warning only after allowing for sampling error, temperature variation, and CO₂ interference.
3. Investigate immediately if there is no meaningful drop over 24 hours in an active ferment, or if the ferment slows unexpectedly before the wine is near dryness.
4. Check temperature, cap management, yeast nutrition history, and vessel mixing before declaring a problem.

Step 7 — Corrective actions and escalation

1. If the ferment is slow but still active, review temperature first. Cold musts ferment more slowly; maintain within the approved range before adding anything.
2. If there are signs of nutrient stress such as reductive aroma, sluggish kinetics, or high residual sugar, escalate to the winemaker before making additions. Use complete yeast nutrient where approved; do not default to DAP alone for a stressed ferment.
3. If the ferment appears stuck, repeat the measurement on a fresh sample, verify calibration, and test the temperature-corrected SG. If confirmed, escalate immediately for a formal intervention plan.
4. Do not press solely on a single reading. Press decisions should consider sugar depletion trend, tannin management, flavour development, and seed/skin maturity.

Step 8 — Completion criteria and records

1. Primary fermentation is considered complete when readings are stable over 2 consecutive checks and are at or near dryness, typically SG 0.995–0.998 depending on alcohol and extract.
2. Confirm the wine is not still dropping before transfer, pressing, or post-fermentation additions.
3. File all readings in the batch record, including any correction factors, instrument ID, and operator initials.
4. Retain records for traceability and quality review, especially where a deviation or remediation occurred.

DECISION LOGIC — WHY THIS APPROACH

Specific gravity is the most practical routine tool for monitoring ferment progress because it tracks density directly and is less misleading than raw Brix once alcohol starts forming. Brix is still useful for quick cellar communication, but it must be interpreted cautiously; as fermentation advances, alcohol distorts the refractometer signal, so a true trend requires a consistent method and/or correction discipline. Frequent sampling catches stalls early, which matters more than precision at any single point. The SOP emphasises temperature, degassing, and representative sampling because these are the main sources of false readings in real cellar conditions. We avoid a hard intervention rule based only on one measurement because ferment kinetics vary with cultivar, temperature, cap management, and yeast health. Deviate when the style calls for extended maceration, when a winery uses a validated correction system, or when the winemaker has a documented stuck-ferment response plan.

TRIBAL KNOWLEDGE

Hydrometers lie most often when the sample still holds CO₂ or when skins and pulp haven't been properly mixed. Let the sample sit a minute if it's foamy; otherwise you'll chase phantom low readings. In red ferments, take the reading from a consistent point in the tank and the same time of day where possible, because temperature swings and cap movement can make the numbers jump around. Keep one dedicated test cylinder for ferment work only; residue from detergents or sanitizer can change wetting and bubble behaviour. If your hydrometer has a chipped stem or a cloudy scale, bin it — they're cheap compared with a missed stuck ferment.

pH & Titratable Acidity (TA) Testing

CELLAR-READY QUICK STEPS

- Calibrate pH meter with fresh buffers.
- Degas and measure a representative sample.
- Titrate to a stable 8.2 endpoint.
- Calculate TA as tartaric acid equivalent.
- Record results and rinse all equipment.

FULL PROCEDURE

Step 1 — Confirm sample and instrument readiness

1. Collect a representative juice or wine sample in a clean, labelled beaker. Mix well before sub-sampling; solids and CO₂ can skew both pH and TA.
2. Bring the sample to **20 °C** where practical. Record temperature if testing is performed outside this range, as readings can drift with temperature.
3. Check the pH meter is clean, functioning, and calibrated with fresh buffers at **pH 7.00** and **pH 4.01**. A 3-point calibration may be used if the meter requires it. Accept only a stable reading within the manufacturer tolerance.
4. Verify the titration equipment is clean, burette is free of air bubbles, and reagents are within date. Use standard sodium hydroxide (**NaOH**) titrant, commonly **0.1 N**.
5. Wear PPE: glasses, gloves, and apron. NaOH is caustic and will burn skin and eyes.

Step 2 — Prepare the sample

1. Degas sparkling or fermenting samples before testing. Excess dissolved CO₂ will depress the apparent pH and increase titration error.
2. For wines with high turbidity, settle or filter a small portion if needed, but do not strip the sample unnecessarily. Use the same method consistently across the vintage.
3. Measure a fixed aliquot, typically **10.0 mL** for TA, into a titration beaker. For very high-acid musts or limited sample volume, **5.0 mL** may be used, but keep calculations adjusted.
4. Record sample identification, date, lot, temperature, and whether the sample is juice, must, or finished wine.

Step 3 — Measure pH

1. Rinse the electrode with deionised water and blot dry. Do not wipe the bulb; this can create static and slow response.
2. Immerse the electrode in the sample and stir gently until the reading stabilises.
3. Record pH to **two decimal places**. If the reading drifts more than **0.02 pH units** over 30 seconds, check calibration, electrode condition, and sample temperature.
4. Rinse the electrode immediately after use and store it in appropriate storage solution, not distilled water.

Step 4 — Determine titratable acidity

1. To the measured sample, add approximately **50 mL** deionised water to allow adequate stirring and endpoint visibility.
2. Add **2–3 drops** of phenolphthalein indicator, or use the meter endpoint if your cellar SOP is set up for potentiometric titration.
3. Titrate with **0.1 N NaOH** while swirling continuously.
4. Approach the endpoint slowly. The endpoint is reached when a **faint pink colour persists for 30 seconds** at **pH 8.20–8.30** for visual titration, or when the meter reaches the set endpoint and stabilises.
5. Record the volume of NaOH used to the nearest **0.01 mL**.

Step 5 — Calculate and report TA

1. Calculate TA as tartaric acid equivalent using the cellar's standard formula:
 - **TA (g/L) = (mL NaOH × N NaOH × 7.5) / mL sample**
2. Example for a **10.0 mL** sample with **0.1 N NaOH**:
 - **TA (g/L) = mL NaOH × 0.75**

3. Report TA to **one decimal place** in **g/L tartaric acid equivalent**.
4. If your winery also reports in **g/L as sulfuric acid** or **meq/L**, convert only using approved internal conversion factors and keep one primary format for all routine records.

Step 6 — Interpret results against winemaking stage

1. For **grape juice and must**, compare pH and TA with the intended style, harvest timing, and fermentation plan. Higher pH with low TA usually needs acid adjustment or a different lot allocation.
2. For **red ferments**, track pH and TA through fermentation and after pressing to support yeast health, colour extraction, and later malolactic decisions.
3. For **finished wine**, use pH and TA together: pH drives microbial stability and SO₂ effectiveness; TA shapes freshness and balance.
4. Flag any of the following for winemaking review: pH above **3.65** in dry table wines, TA below **5.0 g/L** where freshness is critical, or abrupt vintage-to-vintage changes greater than **0.5 g/L TA** or **0.10 pH**.

Step 7 — Record, clean, and store

1. Enter results immediately into the laboratory log or winery software with analyst initials and any deviations from the method.
2. Rinse all glassware thoroughly. Neutralise NaOH spills with plenty of water and follow site chemical spill procedures.
3. Switch off, clean, and store the pH meter per manufacturer instructions. Replace electrode storage solution as required.
4. Review abnormal results before releasing them. If pH and TA do not make sense together, repeat the analysis on a fresh sample before taking cellar action.

DECISION LOGIC — WHY THIS APPROACH

pH and TA are measured together because they answer different questions. TA shows how much acid is present overall, while pH shows how available that acid is for microbial stability, SO₂ efficacy, and colour behaviour. A wine can have decent TA and still sit at a risky pH, particularly if potassium is high; conversely, a sharp TA at low pH may still feel hard and unbalanced. The method uses a fixed aliquot and 0.1 N NaOH because it is robust, inexpensive, and repeatable in a busy cellar. Visual endpoint titration is acceptable, but potentiometric endpointing improves consistency when colour is dark or samples are cloudy. Deviate only when sample matrices are difficult, such as highly carbonated ferments, highly pigmented reds, or low-volume experimental lots; in those cases, degas well, consider filtration, or use a meter endpoint rather than relying on colour alone.

TRIBAL KNOWLEDGE

Fresh calibration buffers matter more than people think; old pH 4.01 buffer drifting by a few tenths will send every result sideways. Keep the electrode wet in proper storage solution and never leave it dry on the bench overnight. With reds, a dark sample can hide the pink endpoint, so many cellars prefer potentiometric titration for consistency. If the sample is still gassy, give it more time to degas or warm gently to room temperature; CO₂ is a common cause of noisy pH and inflated TA. For NaOH, use a small dispensing bottle or burette tip that does not carbonate the titrant from the air, and re-standardise regularly if the lab runs a lot of samples.

Bench Trials — Fining & Acid Adjustment

CELLAR-READY QUICK STEPS

- Confirm MLF status before acid trials.
- Prepare labelled aliquots and a control.
- Dose, mix, and settle each treatment.
- Taste, compare, and record the winner.
- Convert the selected dose to tank addition.

FULL PROCEDURE

Step 1 — Define the bench-trial objective and sampling plan

1. Confirm whether the aim is to trial **fining**, **acid adjustment**, or both. Do not run mixed trials unless

the cellar has enough sample volume to keep the matrix clear.

2. Collect a representative tank sample after racking and before any final stabilisation additions. For red wines undergoing or completing malolactic fermentation (MLF), only bench-trial acid changes once **MLF is complete**; do not acid-adjust early or you risk chasing a moving target.

3. Take enough sample for all treatments, rinses, and repeats. As a guide, collect **2–3 L** per wine lot for a standard set of fining and acid trials.

4. Record current data: wine temperature, pH, titratable acidity (TA), free SO₂, alcohol if available, turbidity, colour, and sensory notes. If MLF is pending, note malic acid status and hold off final acid decisions.

Step 2 — Prepare glassware, controls, and trial set-up

1. Use clean, odour-free bench-trial bottles or cylinders, ideally **100–250 mL** per treatment. Label each clearly with treatment, dose, date, and operator.

2. Include an untreated control in every series. Without a control, you cannot separate treatment effect from natural wine character.

3. Match all aliquots as closely as possible by volume. Keep headspace consistent to reduce oxidation bias.

4. For fining work, pre-hydrate or pre-make fining solutions exactly as the product requires. Use the same water source and temperature for all samples.

5. For acid trials, prepare a stock solution so dosing is accurate. Typical stock is **10 g/L tartaric acid** in clean water. Mix until fully dissolved.

Step 3 — Set fining dose ranges and run the trial

1. Start with a practical dose ladder that brackets the expected range. Common starting points are:

- Bentonite: **0.5, 1.0, 1.5, 2.0 g/L**
- Gelatin or protein fining: **5, 10, 15, 20 mg/L**
- PVPP: **0.05, 0.10, 0.15, 0.20 g/L**
- Casein/caseinate: **0.05, 0.10, 0.15 g/L**

2. Add the fining agent to each aliquot, then mix consistently for **30–60 seconds** or invert several times to ensure full dispersion without excess oxygen pickup.

3. Allow samples to settle for the product's expected contact time. As a general guide:

- Bentonite: **24–72 h**
- Protein finings and PVPP: **4–24 h**
- Complex blends: follow supplier guidance, then confirm by taste and clarity.

4. Evaluate clarity, aroma, palate impact, and colour. Look for improved stability without stripping fruit, structure, or mid-palate.

5. Measure turbidity if available. Select the **lowest effective dose** that achieves the required result.

Step 4 — Set acid adjustment dose ranges and run the trial

1. Only trial acid changes after MLF is confirmed complete. If there is any doubt, run chromatography or enzymatic malic testing first.

2. Prepare a dose ladder around the expected target. For tartaric acid, practical additions are typically **0.5, 1.0, 1.5, 2.0 g/L** depending on the wine style and current pH/TA.

3. Add acid stock carefully to each sample, mix thoroughly, and allow the sample to equilibrate for **12–24 h** before tasting. Recheck pH and TA where possible.

4. Assess the impact on freshness, structure, and finish. Avoid pushing pH too low for the style; in red wines, watch for harshness, reduced mouthfeel, and over-bright acidity.

5. Select the dose that brings the wine into target range with the least sensory distortion.

Step 5 — Record results and recommend the cellar action

1. Rank treatments against the untreated control using a simple score for clarity, aroma, flavour, and overall balance.

2. Confirm the chosen treatment by repeat tasting from at least two operators where possible.

3. Convert the winning bench-trial dose to tank addition using the exact wine volume and actual product strength.

4. Document the calculation, proposed addition rate, lot ID, date, and any compliance notes.
5. If the wine has completed MLF and still needs acid adjustment, schedule the addition only after final confirmation and before stabilisation.

Step 6 — Apply safety and compliance checks

1. Use gloves, eye protection, and dust control when handling powdered fining agents and acids.
2. Keep all additions within approved winery specifications and local food safety requirements.
3. Retain trial records, tastings, and calculations in the batch file.
4. If the wine shows instability, refer it back for repeat bench trials rather than forcing a full-scale addition.
5. Clean and dry all glassware immediately after use to prevent cross-contamination between lots.

DECISION LOGIC — WHY THIS APPROACH

Bench trials are the safest way to keep additions precise and avoid expensive mistakes in tank. Finings and acid both have outsized sensory impact, so the lowest effective dose matters more than a “standard” addition. A control sample is essential, because even a good trial can look better or worse simply from time, aeration, or temperature. Acid work is deliberately placed after MLF because malic acid is still changing during secondary fermentation; adjusting too early can leave the wine out of balance once malate converts to lactate. Dose ladders give a clear decision window without overhandling the wine. Tasting remains central because analytical targets alone do not tell you whether a wine has become stripped, thin, or too sharp. In practice, you deviate only when the wine is obviously unstable, the style demands a different pH/TA outcome, or supplier guidance requires a narrower product-specific range.

TRIBAL KNOWLEDGE

Most bench trials fail from sloppy sample handling, not from the chemistry. Keep the aliquots the same temperature and give them enough time to settle; cold samples often hide haze and make finings look better than they are. For acid trials, a stock solution saves you from weighing milligrams on a shaky bench scale. Mark one glass as the control and never taste it last every time — palate fatigue can make a poor dose look “good enough”. Some fining products are unforgiving if they’re under-dispersed, so mix harder than you think, but don’t whip in air, especially on delicate reds. If you’re trialling post-MLF reds, confirm malic is truly gone before you commit; a small leftover malate load can make the finished wine taste flatter than the bench sample.

Malolactic Fermentation

Malolactic Fermentation Management

CELLAR-READY QUICK STEPS

["Confirm wine is dry, topped, and sorbate-free.", "Warm lot to 18–22 °C.", "Inoculate selected ML culture at label rate.", "Check malic acid weekly until complete.", "Adjust SO₂, only after MLF finishes."]]

FULL PROCEDURE

Step 1 — Confirm the wine is suitable for MLF

1. Only proceed once alcoholic fermentation is complete or essentially complete, and the wine has been pressed/racked off gross lees.
2. Check the wine is stable enough for bacterial culture: free SO₂ should be low, ideally <10 ppm at inoculation, and pH should generally be above 3.10. Wines below pH 3.20 or above 15% v/v alcohol may be slow or difficult; plan for a strong, selected strain and warmer temperatures.
3. Do not add sorbate or any other malolactic inhibitor if MLF is intended. This is a hard stop: sorbate can prevent MLF and create major spoilage risk if bacteria are present.
4. Confirm the lot is protected from oxygen exposure and has adequate headspace management. MLF is not a "set and forget" stage.

Step 2 — Prepare the cellar conditions

1. Set tank or barrel temperature to 18–22 °C. The practical target is 20 °C for reliable completion.
2. Keep the wine topped and under clean inert gas where possible. Maintain sanitary airlocks or sealed bunging for barrels.
3. If the wine was cold after primary fermentation, warm it gradually. Avoid sudden heating above 24 °C, which can stress bacteria and increase volatility.
4. Record the starting values: temperature, pH, free SO₂, total SO₂, alcohol, and lot identification.

Step 3 — Rehydrate and inoculate the ML culture

1. Select a commercial ML culture suitable for the wine style and condition. Use a strain rated for the wine's alcohol, pH, and temperature range.
2. Rehydrate and acclimatise the culture exactly to supplier instructions. Do not shock the bacteria with a large temperature or pH jump.
3. Typical inoculation rates are 1 g/hL unless the supplier specifies otherwise. For difficult wines, some products allow an increased dose; follow the manufacturer's label.
4. Add the culture evenly to the wine with gentle mixing or rack-back circulation. Avoid aggressive aeration.
5. If co-inoculating during late primary fermentation, only do so where the yeast strain, alcohol, and temperature are known to be compatible. Otherwise, inoculate after alcoholic fermentation and first racking.

Step 4 — Manage the fermentation during conversion

1. Monitor the wine at least twice weekly for temperature, aroma, and pressure/activity in barrel or tank.
2. Keep the wine between 18–22 °C. Below 16 °C MLF often stalls; above 25 °C bacterial performance becomes less predictable and spoilage risk rises.
3. Minimise free SO₂ additions during active MLF. If protection is required, keep additions conservative and only when bacterial activity is clearly established.
4. Prevent oxygen pick-up. MLF can run with some CO₂ release, but the wine still needs good cellar hygiene and topping discipline.
5. Watch for problems: sharp volatile acidity, mousy, cheesy, or sweaty aromas, unexpected haze, or sluggish activity. If any appear, check pH, temperature, and hygiene before intervening.

Step 5 — Check for completion

1. Do not assume completion based on time alone. Normal MLF takes 2–6 weeks, with many lots finishing around 4–6 weeks.
2. Test malic acid directly using paper chromatography or another approved malic acid assay.

Chromatography remains a practical cellar check, but it is a screening tool, not a laboratory substitute.

3. Confirm malic acid is absent or below the winery's completion threshold before considering the lot finished. Record the result date and method.

4. If malic acid remains detectable, continue temperature control and recheck in 7–10 days.

Step 6 — Stabilise and protect the wine after completion

1. Once MLF is complete, rack off lees as required for style and cleanliness.

2. Measure pH and free SO₂, immediately, then adjust protection to the winery target for the finished wine. As a guide, many red wines require free SO₂ in the 25–35 ppm range depending on pH and storage conditions.

3. Wait 24–48 hours after any SO₂ addition, then verify MLF has not restarted.

4. Proceed to clarification, topping, and storage under standard cellar protection. Keep full records of inoculation date, checks, completion test, and final SO₂ adjustment.

5. If MLF fails to complete, hold the lot warm, maintain hygiene, and reassess the strain, pH, alcohol, and SO₂ history before re-inoculating.

DECISION LOGIC — WHY THIS APPROACH

Selected-strain inoculation gives the best balance of reliability, timing, and flavour control in boutique cellar conditions. Wild MLF can happen, but it is slower, less predictable, and more exposed to spoilage and volatile acidity. Controlled inoculation is especially valuable where lots vary in pH, alcohol, or temperature, or where tank space and barrique turnover are tight.

Temperature is the main lever. Around 18–22 °C gives consistent conversion without pushing the wine into bacterial stress or oxidation risk. Free SO₂ must be low at the start because SO₂ is the biggest practical inhibitor of MLF. Sorbate is excluded because it can lock up bacterial activity and create geranium taint if any ML bacteria are present.

Completion must be verified analytically. Time alone is not enough, because MLF can appear active while malic acid remains. Paper chromatography is a useful cellar screen, but confirm with a proper assay where possible, especially for high-value lots.

TRIBAL KNOWLEDGE

A lot that looks “quiet” can still be working slowly, so always test before adding SO₂. Barrels often finish earlier than tanks because they sit warmer, but they also hide problems; sniff each barrel, don't just rely on a spreadsheet. If a culture seems sluggish, the first thing to check is temperature, then pH, then whether the wine was overdosed with SO₂ after pressing.

Different ML strains behave very differently in cool rooms and high-alcohol reds, so keep a shortlist of proven suppliers rather than chasing the cheapest packet. When rehydrating, use clean, chlorine-free water and match temperatures carefully; rough handling at this stage is a common reason for poor starts. Also, don't over-mix once conversion is underway — just enough movement to distribute the culture is plenty.

MLF Inoculation & Co-Inoculation

CELLAR-READY QUICK STEPS

- Confirm wine is dry, sound, and within target pH.
- Prepare culture exactly to supplier instructions.
- Inoculate during good tank movement.
- Hold wine at 18–22 °C and low free SO₂.
- Test malic acid until below the acceptance limit.

FULL PROCEDURE

Step 1 — Confirm the wine is suitable for MLF

1. Only proceed once alcoholic fermentation is complete or clearly in terminal decline, unless co-inoculating by design.

2. Check the wine is sound: no acetic taint, no severe volatile acidity, no obvious refermentation risk, and free SO₂ is within the winery's MLF target window.

3. Record wine temperature, pH, alcohol, free SO₂, and residual sugar. As a rule, MLF is easier and more reliable at pH 3.25–3.70 and 18–22 °C.
4. For red wines, favour MLF unless there is a specific style or stability reason not to.

Step 2 — Choose inoculation timing: sequential or co-inoculation

1. Sequential inoculation: add bacteria after alcoholic fermentation and once gross lees have begun to settle. Use this when the wine is hot, high alcohol, low pH, or when yeast management needs to be finished first.
2. Co-inoculation: add bacteria near the end of alcoholic fermentation, typically when sugar is below about 5 g/L and the must/wine is still warm and active. Use this when you want faster turnaround and reduced risk of spoilage from a long lag phase.
3. Do not co-inoculate if the fermentation is stressed, stuck, excessively cool, or if free SO₂ has not been controlled.

Step 3 — Prepare the malolactic culture

1. Pull the chosen culture from cold storage and temper it to cellar temperature before rehydration or acclimatisation.
2. Use only clean, chlorine-free water and follow the supplier's hydration instructions exactly; do not guess the contact time or temperature.
3. If a nutrient or activator is supplied, add it at the stated rate only. Avoid copper, peroxide, sanitiser residues, or any contact with chlorinated water.
4. Keep the prepared culture protected from temperature shock. The goal is to get it into the wine promptly and evenly.

Step 4 — Inoculate accurately

1. Mix the prepared culture into a small volume of wine first, then return it to the main tank to improve distribution.
2. Dose according to supplier specification. Typical commercial direct-inoculation rates are 1 sachet per 1,000–5,000 L depending on strain strength and product form; record the exact lot and calculated dose used.
3. Add the inoculum at a point of good tank movement or during a pump-over/rouse, so the culture disperses through the full volume.
4. If co-inoculating, add after the yeast population is established but before alcohol rises to the strain's tolerance limit. If sequential, add once alcoholic fermentation is complete and the wine has been racked off heavy lees.

Step 5 — Manage the wine during MLF

1. Hold the wine at 18–22 °C where practical. Avoid prolonged storage below 16 °C unless the strain is known to perform cold.
2. Keep free SO₂ low enough for bacterial activity; as a practical operating target, stay below 10 ppm free SO₂ at inoculation unless the supplier or winemaking plan specifies otherwise.
3. Minimise headspace and maintain vessel closure. MLF produces CO₂, but the wine is still vulnerable to oxygen pick-up and spoilage organisms.
4. Taste and check the wine weekly for reduction, VA increase, sluggish activity, or any sulphidic notes. If activity stalls, confirm temperature, pH, SO₂, and whether the culture was compromised by heat or old stock.
5. Do not add any SO₂ or lysozyme during active MLF unless there is a documented corrective reason and the winemaker authorises it.

Step 6 — Monitor completion

1. Test for malic acid depletion using paper chromatography, enzymatic analysis, or another approved method.
2. Do not rely on taste alone. MLF can be mostly complete before the wine tastes fully integrated.
3. Consider MLF complete only when malic acid is below the winery's acceptance limit, typically <0.1 g/L, and no residual activity is detected on repeat testing.
4. Retest after 5–7 days if there is any doubt, especially in cool cellars or high alcohol wines.

Step 7 — Stabilise after completion

1. Once complete, rack off lees if required by style and cellar policy.
2. Add SO₂ to the post-MLF protection target promptly, using pH-based dosage calculations.
3. Verify dryness, microbiological stability, and vessel integrity before moving to maturation or filtration.
4. Document strain, dose, inoculation date, tank, temperature, and completion result in the cellar record.

Step 8 — Safety and compliance checks

1. Wear standard cellar PPE when handling cultures, acids, and SO₂.
2. Keep sanitiser residues out of the tank; rinse and drain lines properly before inoculation.
3. Dispose of used culture packs and samples according to site hygiene procedures.
4. Escalate any stuck, contaminated, or unexpectedly volatile MLF to the winemaker immediately.

DECISION LOGIC — WHY THIS APPROACH

MLF is best managed as a controlled bacterial conversion, not a passive wait-and-see step. Sequential inoculation is safer when the primary ferment is finished and conditions need to be reset for bacteria; co-inoculation is faster and can reduce the window for spoilage, but only works when yeast are healthy and the wine is not under stress. Temperature, pH, alcohol, and free SO₂ are the main levers because they directly control bacterial survival and rate. The SOP prioritises low free SO₂, moderate warmth, and good hygiene because these give the highest chance of clean completion. We avoid blanket additions of SO₂ or other inhibitors during active MLF because they can stop the culture and create a stuck, unstable wine. Completion is confirmed analytically because taste can be misleading, especially with partial conversion or in structured red wines.

TRIBAL KNOWLEDGE

Fresh culture matters. Old sachets from the back of the fridge are a common cause of laggy MLF, especially late in vintage when the cellar is warm and everyone is rushing. If you are co-inoculating, add the bacteria into active tank movement or right after a pump-over; still wine in a dead corner of the tank is where MLF goes patchy. Keep an eye on wine temperature overnight, not just during the day — many cellar rooms swing cooler than expected and MLF quietly stalls. Paper chromatography is useful for trend checks, but if a wine must be released or blended, enzymatic confirmation is worth the extra cost. After completion, get SO₂ on quickly; wines left open and unprotected after MLF are the ones that drift volatile first.

Pressing & Free-Run

Wine Press Operation & Pressing

CELLAR-READY QUICK STEPS

["Confirm dryness and lot readiness.", "Sanitise press and set up fractions.", "Load evenly and start low pressure.", "Taste each cut and label volumes.", "Transfer clean wine to settling tank."]]

FULL PROCEDURE

Step 1 — Confirm readiness to press

1. Verify the ferment is at dry or near-dry status before loading the press. Target residual sugar should be <2 g/L; if using hydrometer/Brix, confirm the wine is at or below 0.995 SG or 0.0 to -1.0 °Brix equivalent, allowing for instrument error.
2. Taste and smell the must/wine. Pressing too early leaves higher sugar in the skins and increases the risk of stuck or sluggish completion; pressing too late can raise extraction of harsh seed tannin.
3. Check fermentation temperature is under control before handling. Keep red fermentations typically in the 20–30 °C range during maceration; do not press hot, active wine if avoidable.
4. Confirm pressing plan: lot ID, destination tank(s), press fractions, and whether free-run will be separated from pressings.

Step 2 — Prepare and verify the press

1. Inspect the basket or membrane press for cleanliness, integrity, and correct assembly. All wine-contact surfaces must be sanitised before use.
2. Check drains, hoses, gaskets, and collection receivers for leaks and blockages.
3. Set up two or more receiving options: one for free-run and one for press fractions. Use labelled, food-grade hoses and vessels.
4. Confirm that the receiving tank is ready for gross lees settling and has sufficient headspace. Wine should be transferred promptly after pressing to a temporary storage vessel for settling.

Step 3 — Load the press safely

1. Stop cap management and drain the fermenter as needed so the pomace can be transferred efficiently.
2. Transfer the fermented grapes/skins to the press in manageable loads. Avoid compacting the mass unevenly; level the load to promote even pressure.
3. Do not overload the press beyond manufacturer capacity. Overfilling reduces press efficiency and increases the risk of channeling.
4. Close guards and verify all operators are clear of moving parts before starting.

Step 4 — Run the free-run and first press fraction

1. Apply very gentle pressure first and collect free-run separately.
2. Monitor flow rate, colour, and aroma. Free-run should be clean and bright for the lot style, with less seed-derived bitterness than later fractions.
3. If the press is releasing excessively cloudy or gritty juice, stop and reassess. Heavy solids may indicate too much force or poor drain pattern.
4. Record volume, lot ID, time, and fraction cut points.

Step 5 — Increase pressure gradually and monitor quality

1. Raise pressure in small increments only. Allow each stage to drain before increasing again.
2. Taste each increment and assess tannin quality, dryness, and astringency. Stop separating pressings into a higher-quality stream if bitterness or harsh seed tannin becomes obvious.
3. Maintain hygienic handling throughout. Minimise oxygen pickup by keeping hoses full and transfers smooth.
4. If the press cake channels or collapses unevenly, pause, break up the cake if needed, and reload or redistribute.

Step 6 — Cut fractions and decide on inclusion

1. Separate press fractions by quality, not only by volume. Early press fractions may be acceptable for blending; later fractions are usually more tannic and oxidatively sensitive.
2. Keep a clear record of each cut: free-run, light press, medium press, hard press, and reject if applicable.
3. Blend decisions should be based on tasting and analysis. If the fraction shows harsh phenolics, overly dry finish, or gritty texture, divert it away from premium blend components.
4. For red wines intended for malolactic fermentation, aim to transfer the cleanest suitable fractions promptly to the holding vessel.

Step 7 — Transfer and settle

1. Pump or drain wine to the temporary storage vessel immediately after pressing.
2. Allow gross lees to settle out before racking or inoculating for malolactic fermentation.
3. Protect wine from excessive oxygen and temperature swings during transfer and settling.
4. Check tank fill, headspace, and identification once the lot is secured.

Step 8 — Clean down and document

1. Rinse the press promptly after use to prevent seed, skin, and tartrate build-up.
2. Wash and sanitise all wine-contact parts according to cellar hygiene procedure.
3. Log press date, lot, yield, free-run volume, press fraction volumes, sensory notes, and any deviations.
4. Record incidents, repairs, or unusual press behaviour for maintenance follow-up.

DECISION LOGIC — WHY THIS APPROACH

Pressing is best done when the ferment is effectively dry because the skins still hold valuable wine, but the risk of carrying residual sugar forward is reduced. Separating free-run from later press fractions gives the winery control over tannin and colour: free-run is generally cleaner, while hard press fractions trend more bitter, seed-driven, and oxidation-prone. A gradual pressure curve is preferred over a quick ramp because it limits channeling and harsh extraction, and it preserves the option to divert poorer fractions away from the premium blend. Prompt transfer to a settling vessel supports clean gross lees removal and sets up malolactic fermentation with less sediment load. Deviate only when style, vineyard tannin profile, or fermentation timing demands it; for example, a more structured red may accept a broader press cut, while delicate fruit should be pressed earlier and more conservatively.

TRIBAL KNOWLEDGE

Keep a chalkboard or lot sheet beside the press and write fractions as they come off; memory is unreliable when the cellar is busy. A press that has been rinsed immediately is far easier to clean than one left overnight, especially around basket seams, drain channels, and membrane folds. If the press cake starts squealing or the flow turns patchy, stop and break it up before forcing more pressure — that usually saves quality. Many cellar teams keep a separate hose for pressings because the later fractions can stain and carry more solids. Taste often, not just at the end; the difference between acceptable and harsh is usually only one pressure increment.

Free-Run vs Press-Run Separation

CELLAR-READY QUICK STEPS

["Check sugar and temperature before pressing.", "Separate free-run into a clean, protected vessel.", "Taste each press cut and keep it separate.", "Stop pressing when bitterness starts to climb.", "Record volumes, fractions, and destination vessels immediately."]

FULL PROCEDURE

Step 1 — Confirm press timing

1. Start assessing separation from day 7 after fermentation start, and continue daily through day 15.
2. Check sugar level before any press decision. Press only when fermentation is effectively complete or nearly complete, with residual sugar at dry or near-dry target for the style.
3. Record must temperature at each check; active ferments often run 10–15 °F (about 6–8 °C) above ambient, so do not judge progress by room temperature alone.
4. If the ferment is still clearly active, keep cap management going and delay press unless the wine style or

fruit condition requires early separation.

Step 2 — Assess free-run quality

1. Open the fermenter and evaluate the top and drain valve/door first. Free-run is the first liquid released before heavy pressing begins.
2. Take a small sample of free-run wine and check aroma, colour, tannin feel, and reduction notes.
3. Look for clean fruit, bright colour, and a softer tannin profile than press fractions.
4. If you detect H₂S, volatile reduction, heavy greenness, or seed harshness, isolate that lot for separate handling.
5. Keep free-run in a clean, labelled temporary vessel or tank, protected from oxygen as soon as practical.

Step 3 — Separate and collect the free-run fraction

1. Drain free-run slowly and avoid disturbing skins and seeds more than necessary.
2. Transfer free-run directly to a temporary storage vessel for settling gross lees.
3. Protect the transfer with inert gas where available, especially if the wine is warm or the transfer is slow.
4. Label the vessel with variety, parcel, date, fermenter number, fraction, and estimated volume.
5. Record the volume collected as free-run in hL and note sensory observations.

Step 4 — Load and press the marc

1. Move the remaining skins and seeds into the press without delay once free-run is removed.
2. Load evenly to avoid channeling and uneven pressure build-up.
3. Press gently at first to recover the cleanest press juice, then increase pressure in stages only as needed.
4. Stop and reassess if the press fraction turns bitter, astringent, or seed-derived.
5. Keep press fractions separate from free-run until blending decisions are made.

Step 5 — Manage press cuts

1. Split the press juice into cuts if the lot, quality, or press style warrants it.
2. Keep early press fractions separate from hard press fractions; quality usually drops as pressure rises.
3. Taste each cut promptly for fruit purity, tannin dryness, bitterness, and colour stability.
4. If a cut shows coarse structure or earthy/seed character, divert it to a lower-value blend or hold back for later assessment.
5. Record each cut volume, press pressure setting, and sensory outcome.

Step 6 — Settle and protect each fraction

1. Transfer free-run and any selected press fractions to temporary storage vessels to settle gross lees.
2. Allow settling under cool, hygienic conditions; keep the vessel topped and protected from oxygen.
3. Check for spontaneous fermentation restart, temperature rise, and obvious solids accumulation.
4. Rack off gross lees when settled sufficiently for the next cellar operation.
5. Maintain full traceability between fractions, including any later blending or rejection.

Step 7 — Hygiene, safety, and compliance

1. Wear PPE when handling fermenters and presses: gloves, eye protection, safety footwear, and hearing protection if required.
2. Treat press areas as confined, wet, and slip-prone work zones.
3. Keep equipment cleaned and sanitised before use; dirty presses carry seed bitterness, spoilage risk, and microbial load into the next lot.
4. Do not leave exposed marc or warm free-run wine sitting open; oxidation and microbial spoilage accelerate quickly.
5. Complete all batch records immediately: dates, volumes, sugar check, temperature, fraction splits, and destination vessels.

DECISION LOGIC — WHY THIS APPROACH

Free-run and press-run are separated because they are not the same wine in quality or structure. Free-run is typically cleaner, softer, and more fruit-forward, while press fractions bring extra colour and tannin but also

more bitterness, seed extract, and phenolic harshness. Keeping them separate preserves blending flexibility: you can build the final wine to style rather than being forced to accept whatever came off the press together. Timing the press off sugar level reduces the risk of trapping residual sweetness or pushing the press too early while yeast are still active. I prefer staged press cuts over a single mixed run because it lets you stop before coarse phenolics dominate. Deviation makes sense in very light, fruit-driven styles where minimal tannin is desired, or in compromised fruit where early separation helps limit spoilage or heat damage.

TRIBAL KNOWLEDGE

In the cellar, the first litre or two from the bottom of the fermenter often tells you more than the lab sheet. If it smells clean and tastes supple, keep that fraction pristine; if it smells reduced or has a seed bite, quarantine it early. A floating thermometer is worth its weight in gold during ferment — the cap can mask a hot core, and press timing gets messy when the lot is warmer than expected. On basket presses, the first gentle cycles usually give the best juice; once you hear the press bed start to squeal and the run turns gritty, you are into the hard press. Always tag fractions immediately; once they are all in similar stainless bins, memory becomes unreliable very quickly.

Whole-Bunch & Whole-Cluster Pressing

CELLAR-READY QUICK STEPS

["Confirm dryness and sensory readiness.", "Label free-run and press-run vessels.", "Load clusters evenly and avoid overpacking.", "Increase pressure slowly; taste each fraction.", "Transfer promptly and clean the press."]

FULL PROCEDURE

Step 1 — Confirm readiness to press

1. Check fermenter for dryness before any solid-liquid separation. Target residual sugar is effectively dry: <2 g/L, with no perceptible sweetness.
2. Confirm cap management and fermentation notes. Press only when extraction is complete and the wine has achieved the desired tannin colour and flavour profile.
3. Taste the must/wine. If the berry/skin character is fully extracted and the seed tannin is moving toward dry, dusty astringency, do not delay.
4. Keep the timeframe tight. Once fermentation is finished, aim to press within 24–48 hours. Prolonged contact with skins and seeds after dryness increases the risk of harsh phenolics and off-flavours.

Step 2 — Prepare equipment and receiving vessels

1. Clean and sanitise press, hoses, buckets, drain trays, fittings, and receiving vessels before use.
2. Set up separate, labelled vessels for free-run and press-run. If blending later, separation allows proper quality control.
3. Confirm the press is mechanically sound: basket staves secure, bladder intact if using a bladder press, pressure gauge functioning, and drain channels clear.
4. Pre-chill receiving vessels if practical for aromatic whites or delicate whole-cluster material; otherwise maintain cellar temperature at 12–18 °C to reduce oxidation risk.
5. If using inert gas, purge press headspace and receiving lines with nitrogen or argon before transfer.

Step 3 — Load whole clusters or whole bunches carefully

1. Handle fruit gently to preserve berry integrity and limit extraction of seed and stem bitterness.
2. For whole-bunch or whole-cluster pressing, load directly into the press basket or bladder chamber in even layers.
3. Do not overfill. Leave enough free space for drainage and even pressure distribution; a compacted, overpacked charge channels poorly and presses unevenly.
4. If whole clusters are being pressed after cold soaking or sorting, remove damaged, mouldy, or excessively lignified stems before loading.
5. Record the batch identification, load weight, press start time, and vessel allocation.

Step 4 — Collect and isolate free-run juice

1. Allow free-run to drain before applying significant pressure.
2. Collect the first juice separately. This fraction is generally the cleanest, lowest in phenolic load, and best suited to premium blending or direct settling.
3. Monitor clarity, aroma, and taste as the flow begins. Free-run should smell clean and fruit-driven, without green or phenolic bite.
4. Keep vessels topped or blanketed with inert gas where possible to minimise oxygen pick-up.

Step 5 — Apply pressure gradually

1. Start with very light pressure and allow the press cake to settle.
2. Increase pressure in small increments only after drainage slows. Avoid fast ramping, which can shred skins, force fine solids through the drain paths, and increase harsh tannin extraction.
3. Taste the run-off frequently during pressing. Check for transition from ripe, supple fruit to a thinner, more astringent profile.
4. Typical stopping point is when the wine begins to taste thin, gritty, dry, and markedly astringent. That is the signal to stop or isolate the press fraction.
5. For whole-cluster material, be especially alert for stem influence: green, woody, or tea-like characters indicate the press is going too far.

Step 6 — Separate quality fractions

1. Keep free-run and press-run in separate containers until bench trials or blending decisions are made.
2. If the press-run remains balanced, it may be retained for blending. If it is coarse, bitter, or overly phenolic, divert it to a lesser parcel or exclude it.
3. Label each fraction with batch, vessel, and time stamp. This is essential for later blending and traceability.
4. If the press fraction is notably different, consider different settling and oak treatment to manage texture and phenolic load.

Step 7 — Transfer promptly to settling or storage

1. Once pressing is complete, move wine to a temporary settling vessel with minimal delay.
2. Let gross lees settle under cool cellar conditions, ideally 10–15 °C, before racking.
3. Protect from oxygen during transfer. Use closed transfer, inert gas, and full vessels wherever practical.
4. Clean the press immediately after unloading. Dried pomace is harder to remove and increases microbial risk.

Step 8 — Record, assess, and release the press

1. Document total press cycle time, pressure profile, free-run volume, press-run volume, sensory observations, and any deviations.
2. Note whether the batch was whole-bunch or whole-cluster, as stem inclusion changes the press endpoint.
3. Confirm all vessels are labelled and samples retained for cellar records.
4. Release the press only after the final tasting check, ensuring no fraction is held beyond the point of acceptable phenolic quality.

DECISION LOGIC — WHY THIS APPROACH

Whole-bunch and whole-cluster pressing is about preserving delicacy while still extracting enough juice for a clean, structured wine. The key advantage is lower mechanical damage at load-in, which usually means less seed breakage, less bitter press character, and better aromatic retention than aggressive destemming and crushing. The trade-off is that intact stems can improve drainage but can also contribute green, woody, or tea-like notes if pressure is pushed too hard or if the material is not ripe. That is why the SOP insists on early sensory checks, slow pressure increases, and separate collection of free-run and press-run. Keeping fractions apart gives the winemaker flexibility: the cleanest wine stays protected, while the firmer press portion can be blended only if it tastes fit for purpose. If fruit is very ripe and stems are lignified, a little more press-run may be acceptable; if fruit is delicate, underripe, or aromatic, stop earlier and protect quality.

TRIBAL KNOWLEDGE

Whole-cluster loads can bridge and channel if you dump them in too fast. Feed the press evenly and give the cake time to settle between pressure increments. On basket presses, the first litre or two after each tighten

often looks cleaner than it really is, so don't judge on colour alone—taste it. Bladder presses are gentler, but if the bladder is overinflated too quickly, they can still create harsh fractions and plug the drain slots. Keep a dedicated hose for free-run if you can; once it has carried press-run, it tends to drag solids into the clean fraction. A lot of good cellar work here is simply restraint: stop earlier than you think, then blend later if the press fraction earns its place.

Racking & Clarification

Racking off Gross Lees

CELLAR-READY QUICK STEPS

["Confirm dry finish and settled gross lees.", "Sanitise hoses, pumps, and receiving vessel.", "Rack gently, avoiding lees disturbance and splashing.", "Top immediately; minimise headspace.", "Record volumes, SG, temperature, and faults."]

FULL PROCEDURE

Step 1 — Confirm the wine is ready to rack

1. Verify alcoholic fermentation is complete or essentially complete before racking: SG should be at or below target dry finish, typically **0.995–0.999** for dry red wine, with no active CO₂ release beyond the expected light residual off-gassing.
2. Taste and smell the wine in a representative sample. A clean, dry, slightly yeasty or muddy note is acceptable; **sulfur, rotten egg, vinegar, or solvent notes require immediate escalation**.
3. Assess lees depth in the vessel. Gross lees should appear as a heavy layer of spent yeast, pulp, skins, seeds, tartrates, and solids settled at the bottom. As a working rule, rack once the wine has had enough time to settle but **before compacted gross lees begin to autolyse and taint the wine**.
4. Check vessel inventory and headspace. Prepare a clean receiving tank or vessel with enough capacity, allowing for transfer losses and minimal ullage management.

Step 2 — Prepare equipment and sanitation

1. Clean and sanitise all hoses, pumps, taps, valves, fittings, thieves, sight glasses, and receiving vessels before use.
2. Confirm the receiving vessel is inert, sound, and protected from oxygen pickup. Where possible, purge headspace with inert gas and keep the vessel topped to reduce air exposure.
3. Confirm filtration or clarification aids are **not** being introduced at this stage unless specified by the winemaking plan. Racking off gross lees is a mechanical separation step, not a fining step.
4. Stage all equipment so the transfer can be completed smoothly and without interruption. Delays increase oxidation risk and allow sediment to resuspend.

Step 3 — Rack carefully off the gross lees

1. Position the racking point above the sediment line, leaving the compacted gross lees undisturbed. Use a clean racking cane, drain port, or pump inlet positioned to avoid disturbing the lees bed.
2. Start transfer slowly and maintain a steady flow. Avoid splashing, free-falling wine, or agitation that can entrain oxygen.
3. Monitor the clarity of the outgoing wine continuously. As soon as the flow begins to draw cloudy with sediment, slow the transfer and stop before the lees layer is disturbed.
4. If needed, recover a small final portion separately and decide whether to keep it with the main lot or divert it based on sensory quality and intended wine style.
5. Keep the receiving vessel topped as the transfer proceeds. Target minimal headspace; where practical, leave no more than **1–2% ullage** at completion, then top or inert as required.

Step 4 — Manage the wine after racking

1. Measure and record the transferred volume, SG, temperature, and any notable sensory observations.
2. Review the wine for freshness, reduction, and volatile acidity. Gross lees removal should reduce the risk of reductive and sulphidic faults associated with heavy solids contact.
3. Decide whether the wine requires further settling, a second racking, or immediate progression to the next cellar operation.
4. If malolactic fermentation is planned, confirm the wine is at the intended temperature and SO₂ status before inoculation or encouragement. Gross lees removal usually improves control by removing unstable solids and excess yeast debris.
5. Maintain hygiene throughout storage. Any vessel left with lees residue should be cleaned promptly to

prevent microbial load, crusting, and difficult-to-remove deposits.

Step 5 — Record, review, and close out

1. Enter batch ID, date, vessel numbers, volumes transferred, and operator name in the cellar log.
2. Record the condition of the lees, including colour, odour, and approximate depth. Note any warning signs such as sulphide, mousiness, or oxidation.
3. Flag any deviations from the standard process, including cloudy transfer, excessive lees loss, unexpected sweetness, or need for emergency sulphur dioxide management.
4. Release the wine for the next stage only when the tank is cleanly separated from gross lees and the lot is stable enough for the intended pathway.
5. Dispose of gross lees in accordance with site waste and environmental procedures. Heavy sediment can harbour high microbial load and must not be left standing in open containers.

DECISION LOGIC — WHY THIS APPROACH

Racking off gross lees early is a practical quality safeguard. Heavy solids carry dead yeast, skins, pulp, and microflora that can drive reductive aromas, bitterness, and instability if left in contact too long. The aim is not to strip the wine clean, but to separate the bulk sediment while preserving fresh fruit and texture. This is why the SOP focuses on careful drainage, minimal agitation, and immediate headspace control rather than aggressive clarification. Alternatives like extended lees contact can suit some styles, but they increase risk unless the lot is clean, healthy, and specifically managed. Likewise, filtering at this stage is usually unnecessary and can remove desirable character. Deviate only when sensory assessment supports it: if the wine is still turbid but clean, give it more settle time; if sulphidic, oxidised, or microbiologically suspect, act quickly and document the fault.

TRIBAL KNOWLEDGE

Gross lees behave differently in every tank. Conical fermenters, shallow bins, and pressed wine all settle with their own quirks, so don't trust the volume estimate alone—always look at the lees line before drawing wine. A slow start is safer than a fast one; once you pull a muddy tail, you've bought yourself extra handling and more oxygen exposure. Keep a spare hose clamp, gasket, and reducer on hand because a small leak at racking time can turn into a major oxidation issue. If the lees smell clean and bready, you've probably waited long enough; if they smell sulphidic, don't park the wine on them any longer than needed. Many cellars prefer a dedicated coarse racking line for dirty work, then a separate clean line for the main transfer.

Fine Lees Stirring (Bâtonnage)

CELLAR-READY QUICK STEPS

- Check dryness, MLF status, and free SO₂.
- Sanitize baton, thief, and bung gear.
- Stir lees gently to homogeneous suspension.
- Top immediately and reseal vessel.
- Taste, log, and adjust the schedule.

FULL PROCEDURE

Step 1 — Confirm the wine is eligible for bâtonnage

Fine lees stirring is only started after the wine is dry and has finished, or is very close to finishing, malolactic fermentation. Check residual sugar is <2 g/L and that the wine is microbiologically stable enough to handle oxygen exposure. Typical timing is after gross lees have been removed and the wine has settled in tank, barrel, or puncheon. Do not stir wines with obvious reduction, volatile acidity issues, or excessive turbidity. Record vessel ID, wine style, volume, temperature, and current SO₂ status before starting.

Step 2 — Prepare the vessel and stirring tools

Use a clean, sanitised bâtonnage tool, lees wand, or inert mixing paddle suitable for the vessel size. Confirm bungs, lids, and airlock systems are intact and that the vessel is topped within 2–5 cm of the fill point. If the wine is in barrel, check headspace is minimal and the bung can be refit immediately after stirring. Ensure transfer hoses, sampling thief, and measuring equipment are clean and ready. In barrel work, have spare bungs and a flashlight available for leak checks.

Step 3 — Assess lees condition before the first stir

Pull a small sample and assess aroma, turbidity, and mouthfeel. Fine lees should be a light, creamy sediment, not chunky gross lees. The wine should smell clean, with fresh fruit, yeast, and subtle savoury notes; avoid stirring if there is sulphidic, mouldy, or acetic character. Taste for texture and balance. If the lees smell reductive, increase monitoring and consider a small controlled aerative rack rather than immediate bâtonnage.

Step 4 — Stir gently and evenly

Stir enough to lift the fine lees fully into suspension, but not so hard that you create vortexing or excessive oxygen pickup. In barrels, rotate the baton through the lees layer and lift from the base without scraping staves. In tanks, recirculate or mix with a clean inert stirrer, keeping the contact time short. Aim for a homogeneous, milky suspension rather than froth. For standard white and barrel-fermented wines, one complete stir per vessel is usually sufficient per session.

Step 5 — Set the frequency and duration

Start with 1 stirring per week for the first 2–4 weeks, then reduce to every 2 weeks if the wine shows good texture and no reductive signs. Some fuller-bodied wines may benefit from weekly stirring for up to 8 weeks, but delicate aromatic wines often need less. Do not stir on a fixed calendar if the wine is already showing harmony; taste guides the program. Stop bâtonnage once the palate has enough volume and the wine no longer gains complexity from the lees.

Step 6 — Monitor oxygen, SO₂, and flavour development

After each session, check free SO₂, dissolved oxygen risk, and sensory change. Fine lees stirring can soften palate and build mid-palate weight, but it also increases oxygen exposure and can reduce effective SO₂ protection. Maintain free SO₂ according to winery policy and style target; in practical terms, do not let free SO₂ drift low after repeated stirring. Taste for improved creaminess, reduced hardness, and integrated oak or fruit sweetness. If oxidation markers appear, cut the frequency immediately and top up/adjust SO₂.

Step 7 — Re-top, seal, and document

Once stirring is complete, top the vessel immediately to the correct fill level, reseal, and confirm no leakage. In barrel, wipe the bung area clean and replace any damaged bung. Enter the date, vessel, number of stirs, sensory notes, free SO₂, temperature, and any corrective action in the cellar log. If the wine is to remain on lees, review weekly; if it is ready, rack off fine lees before the next operation in the clarification plan.

Step 8 — Stop conditions and corrective action

Cease bâtonnage if the wine becomes reductive, develops a flat or sulphidic nose, shows instability, or becomes overly cloudy for the style. Also stop if the lees are becoming compact and inactive, or if the wine no longer improves after two consecutive stir sessions. Where needed, switch to a settling period, light rack, or targeted sulphur management. Any deviation from the planned stirring schedule must be approved by the winemaker and recorded as a quality-system change.

DECISION LOGIC — WHY THIS APPROACH

Fine lees stirring is used because it gives texture, volume, and integration without stripping the wine the way heavier handling can. Compared with aggressive aeration, bâtonnage is more controlled: it can soften acidity, lift mid-palate, and improve mouthfeel while keeping fruit and oak in balance. The trade-off is oxygen exposure and the risk of reduction if the lees are unhealthy or the wine is under-protected. That is why the SOP starts with dryness, clean sensory status, and post-gross-lees settling. Frequency is deliberately flexible because different wines respond differently: aromatic, delicate styles usually need less, while barrel-fermented, fuller-bodied wines can tolerate more. Deviate from routine when the wine is already balanced, when reduction appears, or when free SO₂ is trending too low after repeated stirring. The key rule is sensory-led, but always supported by records and SO₂ checks.

TRIBAL KNOWLEDGE

A quiet barrel makes better lees wine than an overfilled, gurgling one — keep headspace tight and bung

seals sound. In barrel, don't jab the baton straight down into compacted lees; work it up from the bottom and avoid scraping oak staves, which can throw up rough tannin and chips. If the wine smells a bit struck-matchy after stirring, give it time before reaching for more oxygen — many wines blow off a light reduction after topping and settling. Some suppliers' fine lees are creamier and more stable than others, so taste each lot before committing to a long bâtonnage program. Barrels that have been sitting cold often need a slower, gentler first stir than the spreadsheet suggests.

Cold Stabilisation

CELLAR-READY QUICK STEPS

- Confirm batch is finished and tank is gas-tight.
- Chill wine to -2 °C to 2 °C.
- Hold 5–10 days and sample daily.
- Verify stability with approved test.
- Rack, top, recheck SO₂, and clean down.

FULL PROCEDURE

Step 1 — Confirm wine is ready for cold stabilisation

1. Verify the wine is finished fermenting, or has been chemically arrested and is microbiologically stable before chilling. Do not cold stabilise active ferment unless the process is part of a controlled arrest plan authorised by the winemaker.
2. Check and record:
 - Alcohol and residual sugar status
 - Free SO₂ and total SO₂
 - pH and titratable acidity (TA)
 - Turbidity/clarity, gross lees load, and tank fill level
3. Ensure the wine has been racked off gross lees where practical. Leave only fine lees if the winemaking plan requires them. Heavy lees at low temperature can create reductive faults and make later filtration harder.
4. Confirm the tank, chiller, hoses, and seals are clean, sanitised, and fit for chilled operation. Cold tanks draw in air through faulty fittings, so gas tightness matters.

Step 2 — Set the stabilisation target

1. Set the target wine temperature between -2 °C and 2 °C, with 0 °C the usual cellar target for tartaric stabilisation.
2. For white, rosé, and lighter reds, hold at target for 5–10 days, depending on wine composition and tank volume.
3. For higher alcohol, high potassium, or high acid wines, allow the full hold period and do not shorten the time unless bench and field checks confirm stability.
4. Maintain headspace protection with inert gas (nitrogen, CO₂, or argon) throughout chilling. Do not rely on air contact suppression alone.

Step 3 — Chill down progressively

1. Drop the temperature gradually at a controlled rate. A practical target is 2–4 °C per day for full tanks, slower for fragile wines or where tartrate precipitation is uneven.
2. Circulate chilled glycol through the jacket or heat exchanger and monitor both supply and return temperatures.
3. Mix the tank gently if required to avoid stratification, but do not over-agitate. Excess movement can entrain oxygen and resuspend sediment.
4. Record wine temperature at least twice daily during pull-down, and check for leaks, frosting on fittings, or pump cavitation.

Step 4 — Hold at stabilisation temperature

1. Once target temperature is reached, hold the wine at -2 °C to 2 °C for the planned duration.
2. Check the tank daily for the following:
 - Stable temperature within ±1 °C of target

- No visible air ingress
- No excessive tartrate crusting around fittings
- No sulphide or reduction issues developing from prolonged cold lees contact

3. If signs of H₂S or reduction appear, rack promptly, aerate only if the winemaking plan allows, and reassess sulphur protection before continuing.

4. Take a sample mid-hold for a quick cold test or conductivity test if your winery uses that method. Use the winery's validated pass/fail criteria.

Step 5 — Verify stability before release

1. Before warming the wine, confirm cold stability by one of the winery's approved methods:

- Mini-contact test on a chilled sample
- Conductivity test
- Microscopic or visual tartrate check after a controlled cold hold

2. The wine is acceptable for release from cold stabilisation when no new tartrate crystals form under the validated test conditions and the sample remains clear.

3. If instability remains, extend the hold by 24–72 hours and retest. Wines with high potassium, high acidity, or lower alcohol may need a longer or colder treatment.

4. Document the result, test method, date, operator, and any deviations.

Step 6 — Warm up and rack/clarify

1. Raise the tank temperature slowly back to cellar temperature, typically no faster than 2–3 °C per day, to reduce shock and minimise haze formation.

2. Rack the wine off precipitated tartrate deposit and any settled solids once back above freezing risk.

3. If polishing filtration is planned, assess filterability after racking. Cold stabilised wine is often clearer, but fine crystals can still block pads or cartridges if racking is rushed.

4. Top the tank immediately after racking and re-check free SO₂. Adjust to the winery's target based on pH and wine style.

Step 7 — Clean down and close out

1. Drain and clean the tank, lines, and chiller circuit promptly. Tartaric scale is easier to remove while fresh.

2. Inspect gaskets, sample valves, and pump seals for damage from cold service.

3. Record final parameters in the batch record:

- Start and finish dates
- Temperature profile
- Hold time
- Stability test result
- Racking date and volume loss

4. Release the wine only when all checks are complete and the batch is signed off by the responsible winemaker.

DECISION LOGIC — WHY THIS APPROACH

Cold stabilisation is used to force out potassium bitartrate before bottling, which reduces the risk of crystals appearing later in bottle or tank. Holding the wine near 0 °C for several days is the most reliable boutique-winery approach because it is simple, predictable, and easy to verify with a field test. It is gentler than aggressive chemical correction and avoids changing the wine's sensory profile.

The main trade-off is time and cold-room capacity. Faster, more forceful chilling can save days, but it increases the risk of uneven precipitation, haze, oxygen pickup, and filter blockage. Very low temperatures can also stress wines that are still carrying fine lees or developing reductive character, so the SOP requires racking, gas protection, and daily checks. Where a wine is naturally stable, the hold can be shortened after a validated test passes; where composition is challenging, the safer move is to extend the hold rather than bottle a marginal wine.

TRIBAL KNOWLEDGE

In practice, the ugliest tartrate deposits often build on the coldest spots first: bottom plates, jacket zones,

sample ports, and tank seams. If you skip a pre-check on seals, you usually find the leak after the tank has already pulled a vacuum. Glycol systems also lie a bit — the supply line might read perfect while the wine core is still several degrees warmer, so always trust the wine temperature, not just the jacket reading.

If you are using conductivity as the stability check, make sure the sample and reference have both been conditioned the same way; warm samples give false confidence. Wines with high potassium or lots of colour and extract often need the full hold period, even if the first sample looks clean. And once tartaric sludge dries on stainless, it becomes a chipping job, not a wash job — clean it straight after racking.

Protein Stability & Bentonite Fining

CELLAR-READY QUICK STEPS

- Confirm dry fermentation and take a representative sample.
- Run heat stability bench trials and select the lowest passing dose.
- Hydrate bentonite at 1:10 to 1:15 before dosing.
- Mix thoroughly, then settle 3–7 days.
- Re-test stability and rack off lees before release.

FULL PROCEDURE

Step 1 — Confirm fermentation status and readiness to treat

1. Confirm primary fermentation is complete or nearly complete before any fining for protein stability. Do not fine a wine that is still actively fermenting unless there is a documented production reason and the winemaker approves it.
2. Check residual sugar with a hydrometer, enzymatic test, or refractometer with alcohol correction. Target is dry or effectively dry: <2 g/L residual sugar, or stable readings over 24–48 hours with no visible CO₂ activity.
3. Record wine temperature. Bentonite works best when the wine is cool and settled, typically 10–18 °C. If the wine is warmer than 20 °C, plan to cool or hold before treatment where practical.
4. Rack gross lees off the wine before bench trials if fermentation has finished and sediment is heavy. Heavy lees skew protein tests and can hide the true bentonite demand.

Step 2 — Run a protein stability test

1. Take a representative sample from a well-mixed tank or barrel batch. If the lot has multiple parcels, sample each parcel separately and combine proportionally.
2. Prepare a heat stability test on the untreated sample and on candidate bentonite doses. A practical cellar test is to heat sealed samples to 80 °C for 30 minutes, then cool to 20 °C and inspect for haze or flocculation.
3. If the wine shows protein haze after heating, it is not stable and requires bentonite. If no haze forms, no bentonite is needed unless the wine has known instability history or a finishing requirement.
4. Keep notes on sample pH, temperature, and the exact bentonite dose used in the bench trial. This becomes the production reference.

Step 3 — Conduct a bench trial to set the dose

1. Use a clean, odour-free bentonite suitable for wine fining. Hydrate only the amount needed for the trial.
2. Trial a dose range of 0.5 to 2.0 g/L in increments of 0.25 or 0.5 g/L, depending on haze severity and wine style.
3. For each trial jar, add the required bentonite slurry at a consistent concentration, mix thoroughly, and allow settling for 24–48 hours at cellar temperature.
4. Rack or decant the clear portion and repeat the heat stability test on each trial.
5. Select the lowest dose that passes heat stability with acceptable wine loss, aroma retention, and colour impact. Avoid overfining; excess bentonite strips aroma and can reduce volume unnecessarily.

Step 4 — Prepare the bentonite slurry correctly

1. Measure the production volume accurately in hL or L. Calculate total bentonite required from the chosen bench dose.
2. Pre-hydrate bentonite in clean water at a ratio of 1:10 to 1:15 by weight to volume. Example: 1 kg bentonite in 10–15 L water.

3. Add bentonite slowly to water with strong agitation; never add water to dry bentonite. Allow it to swell fully for 2–4 hours, or overnight if the product instructions require it.
4. Re-mix immediately before dosing to ensure a uniform slurry. A lumpy slurry will not disperse evenly and causes local overfining.
5. Confirm the slurry is free of dry clumps before transfer to the tank.

Step 5 — Dose the wine and mix thoroughly

1. If the wine has been racked, return it to a clean tank with sufficient headspace for mixing.
2. Add the bentonite slurry slowly into a moving wine stream or via recirculation loop. Dose evenly across the batch; do not dump into one point.
3. Mix for 15–30 minutes minimum, or until the tank is fully homogeneous. In barrels, circulate gently or roll where safe and practical.
4. Check that the tank is sealed or protected from oxygen ingress during the mixing period. Bentonite treatment itself does not need air exposure.
5. Record the exact dose, lot number, timing, and mixing method for traceability.

Step 6 — Settle, rack, and verify stability

1. Allow the wine to settle for 3–7 days, longer for cool cellar temperatures or low-conviction settling. Monitor turbidity and lees formation.
2. Rack off the bentonite lees carefully to minimise losses. Avoid aggressive suction that can pull fine sediment back into the clarified wine.
3. Re-test heat stability on the treated wine before release to the next process step.
4. If haze remains, make a second smaller trial and add only the minimum additional dose needed, typically 0.25 to 0.5 g/L.
5. Confirm the wine remains sound with no sensory stripping, no unusual reduction, and no excessive solids carryover.

Step 7 — Safety, compliance, and records

1. Wear dust protection when handling dry bentonite, plus eye protection and gloves when mixing slurries.
2. Keep floors dry and clean up bentonite spills immediately; slurry is slippery and a serious slip hazard.
3. Record all fining materials, supplier batch numbers, doses, test results, and final approval in the cellar log and QA system.
4. Dispose of bentonite lees according to site waste procedures and local regulations.
5. Only release the wine for downstream clarification, filtration, or bottling once protein stability has been verified and documented.

DECISION LOGIC — WHY THIS APPROACH

Bentonite remains the standard tool for protein stability because it is reliable, inexpensive, and scalable from barrel to tank. The key is not to treat by habit: first prove instability with a heat test, then determine the minimum effective dose by bench trial. That approach protects aroma, colour, and yield better than defaulting to a high blanket rate. Cool, settled wine gives more consistent settling and less re-suspension of lees, while racking off gross lees beforehand improves the accuracy of the trial and the outcome of the treatment. Alternatives such as enzyme-only approaches or filtration alone do not reliably remove heat-unstable proteins; filtration can also blind too quickly if protein load is left untreated. Deviate only when the style or downstream process changes the risk profile, such as wines destined for sterile filtration or wines with a known history of instability. In those cases, confirm the final stability target used by the winery and document the rationale.

TRIBAL KNOWLEDGE

Bentonite quality matters more than people think: fresh, low-odour, fine-grade products hydrate more predictably and settle cleaner. If the slurry looks like wallpaper paste, it is too thick; thin it out and mix again before dosing. Most cellar headaches come from poor dispersion, not the clay itself, so dose into a moving stream or a proper recirculation loop rather than into dead wine. Cold wine settles better, but very cold wine can make the bentonite act sluggishly, so don't rush the rack. Barrel lots often need gentler handling because bentonite sediment compacts differently and can drag more volume than expected. Always re-test

after treatment; one good-looking tank does not guarantee bottle stability.

SO₂ , Management

SO₂ , at the Crush

CELLAR-READY QUICK STEPS

- Measure must volume, pH, and temperature.
- Calculate target SO₂ , dose from fruit condition.
- Add promptly and mix thoroughly.
- Confirm free SO₂ , after homogenisation.
- Log dose, time, lot ID, and operator.

FULL PROCEDURE

Step 1 — Confirm the fermentation plan before dosing

1. Assess the fruit on receipt: variety, condition, level of rot, skin damage, and intended style.
2. Measure °Brix, pH and temperature of the incoming must as soon as practicable.
3. Decide whether the lot is destined for a clean, fruit-forward wine or whether it requires a more restrained, lower-SO₂ , approach because of planned wild ferment, extended skin contact, or high MLF risk.
4. Confirm the inoculation plan. If yeast are to be added, prepare the yeast hydration and nutrition program separately; do not use SO₂ , as a substitute for proper yeast management.

Step 2 — Set the target SO₂ , addition at crush

1. Use SO₂ , at crush to suppress spoilage organisms and oxidation before yeast take over.
2. Target free SO₂ , in the must at a level that is effective but not excessive: generally 25–50 mg/L (ppm) for sound fruit.
3. Use the lower end of the range for ripe, clean, cold fruit with low pH and fast inoculation.
4. Use the higher end for warmer fruit, fruit with any Botrytis or sour rot pressure, delayed processing, or where a crush delay is likely.
5. Avoid over-application. Excessive SO₂ , can slow or stress yeast, increase fermentation risk, and complicate later malolactic fermentation.

Step 3 — Calculate and prepare the addition

1. Measure the must volume in hL or L before adding SO₂ ,.
2. Calculate the exact dose based on the product strength and the target mg/L.
3. Use approved cellar SO₂ , products only, and confirm whether the material is potassium metabisulphite solution, powder, or gaseous SO₂ ,.
4. Wear PPE: gloves, eye protection and, where required, respiratory protection. Keep the area ventilated.
5. Prepare a small liquid slurry or stock solution where practical to improve mixing consistency and reduce local hotspots.

Step 4 — Add SO₂ , promptly and mix thoroughly

1. Add SO₂ , as early as possible once the fruit is in the receiving vessel, crusher, destemmer hopper, press, or must pan.
2. For whole bunches, treat at the earliest safe mixing point; for destemmed fruit or must, add during transfer or immediately after.
3. Distribute the addition evenly across the must or fruit stream. Do not leave sulphite concentrated in one zone.
4. Mix thoroughly using pump-over, punch-down, or a dedicated recirculation loop.
5. Recheck that the addition has been fully incorporated before any yeast or nutrient additions are made.

Step 5 — Verify the chemistry and adjust for conditions

1. Record the addition rate, time, lot ID, and operator name in the cellar log.
2. For cold, low-pH lots, confirm the free SO₂ , target with an appropriate analysis once homogenised.
3. For high-pH lots, damaged fruit, or fruit with elevated microbial load, consider a modestly higher initial dose, but do not use SO₂ , to mask poor fruit quality.

4. If MLF is intended, keep the SO₂ addition conservative so the bacteria are not suppressed later.
5. If wild ferment is planned, apply only enough SO₂ to manage oxidation and gross spoilage pressure; avoid doses that sterilise the must.

Step 6 — Proceed to yeast preparation and fermentation control

1. Once the SO₂ is incorporated, proceed with yeast hydration and inoculation according to the yeast SOP.
2. Add nutrients during fermentation as required by the yeast nutrition plan; do not rely on SO₂ for microbial control beyond crush.
3. Monitor temperature, cap behaviour and fermentation kinetics closely in the first 24–72 hours.
4. If signs of sluggish start appear, check free SO₂, temperature, and inoculation viability before making any corrective additions.
5. Maintain full traceability and chemical compliance records for audit and product safety purposes.

Step 7 — Safety, compliance and storage

1. Store sulphite products in a dry, labelled, ventilated chemical area away from acids and incompatible materials.
2. Keep SDS documents available to all cellar staff.
3. Never add SO₂ in a confined, unventilated space without suitable controls.
4. Dispose of empty containers according to local chemical waste requirements.
5. Review addition records weekly during vintage to ensure targets are being met consistently.

DECISION LOGIC — WHY THIS APPROACH

SO₂ at crush is used as an early control step because the must is at its most vulnerable before yeast establish dominance. A moderate initial dose suppresses spoilage flora, slows oxidative browning, and gives the winemaker time to manage the lot properly. The preferred range is deliberately conservative: enough to protect the fruit, but not so much that it damages yeast performance or makes later malolactic fermentation difficult. That trade-off matters, especially in red ferments where fermentation should start cleanly and proceed without stress.

This SOP avoids a one-size-fits-all fixed dose because fruit condition, temperature, pH, and intended style all change the risk profile. Higher doses are justified for damaged or delayed fruit; lower doses suit clean, cold, quickly inoculated parcels. The procedure also separates SO₂ management from yeast nutrition and hydration, because those are distinct controls. Good cellar practice is to protect the must, not to use SO₂ as a blunt instrument for poor fruit handling or weak fermentation management.

TRIBAL KNOWLEDGE

SO₂ only works well if it is mixed properly. A splash on top of the must does very little; a thorough pump-over or recirculation is what gives you even protection. Watch for dead corners in fermenters and bins, especially with heavy, sticky reds and small tanks where mixing can be patchy.

In warm vintages, fruit can carry a lot more microbial load than it looks like it does, so rely on condition and smell as much as the numbers. If the fruit is borderline, get the dose in early rather than waiting until after the crush queue backs up. Also, keep sulphite stock solutions fresh and clearly labelled; weak or old stock is a common reason cellar additions miss target.

Post-MLF SO₂ Adjustment

CELLAR-READY QUICK STEPS

["Confirm MLF is finished by chromatography.", "Calculate target free SO₂ from pH.", "Mix a fresh metabisulphite solution.", "Add with full-tank recirculation.", "Retest free SO₂ after equilibration."]

FULL PROCEDURE

Step 1 — Confirm MLF completion and readiness for SO₂

1. Do not dose SO₂ until malolactic fermentation (MLF) is confirmed complete. Verify by paper chromatography, enzymatic assay, or another validated lab method; a simple sensory check is not enough.
2. Acceptable endpoint: malic acid is not detectable or is below your winery's internal threshold (typically

<0.1 g/L). If results are borderline, re-test in 3–5 days before adding SO₂.

3. Check the wine is clear enough for reliable analysis and that active MLF signs have ceased: no persistent CO₂ release, no rising temperature trend, and no fresh buttery/diacetyl build-up from ongoing bacterial activity.
4. Record current wine temperature, free SO₂, pH, alcohol, and any remaining VA or microbial risk notes. This determines the dose and whether the wine needs protective handling before sulphuring.

Step 2 — Calculate the target free SO₂

1. Set the target free SO₂, based on pH and wine style. For red wine post-MLF, aim for 25–35 mg/L free SO₂, at pH 3.20–3.40, and 30–40 mg/L where pH is above 3.40 or the wine has higher spoilage risk.
2. If pH is very low (below 3.20), the lower end of the range is usually sufficient; if pH is high, do not underdose. Higher pH means less molecular SO₂, protection.
3. If the wine will remain in tank for extended storage, or if there is any residual sugar, oxygen pickup, or elevated microbial load, choose the upper end of the range.
4. Never rely on a fixed ppm addition without a calculation. Dose must be adjusted to current free SO₂, pH, temperature, and vessel fill status.

Step 3 — Calculate the addition accurately

1. Determine the deficit: Target free SO₂, minus measured free SO₂, = required increase in mg/L.
2. Use a calibrated SO₂ product strength and calculate for the full wine volume in hL or L.
3. As a guide, 1 g/hL of potassium metabisulphite contributes approximately 0.57 mg/L total SO₂, equivalent in wine. Adjust for product analysis and winery dosing policy.
4. Example: 50 hL wine, target increase 15 mg/L. Required addition is calculated from the product specification and winery calculator, then rounded to the nearest practical dosing increment. Do not guess.
5. Double-check calculations with a second person before adding chemical. Treat this as a critical control point.

Step 4 — Prepare the wine and the addition

1. Ensure the vessel is full or topped to working level to minimise headspace. SO₂ is a protectant, not a substitute for sound ullage control.
2. If using potassium metabisulphite, make up a fresh solution in clean wine or cool potable water according to winery practice. Use immediately; do not store mixed solution.
3. Add with good mixing: pump-over, recirculation, or tank agitation. Aim for even distribution through the full volume. Avoid localised over-dosing near the draw-off point.
4. Keep the wine cool if possible, ideally 12–18 °C for handling and binding stability. Warm wine consumes SO₂ faster and gives less reliable protection.
5. Maintain ventilation and PPE: gloves, eye protection, and respiratory awareness. SO₂ dust and vapour are hazardous.

Step 5 — Verify the result and document

1. After mixing, allow sufficient equilibration time before re-testing: 12–24 h for tank lots, longer if the wine is cold, dense, or only lightly mixed.
2. Re-measure free SO₂, and confirm it has landed in the target range. If still low, make a small corrective addition and recheck.
3. Sensory check for excessive sulphur character. A correctly dosed wine should not show burnt match, harsh sulphur, or muted aroma from overdosing.
4. Update the batch record with date, lot ID, volume, pH, malic status, product used, grams added, calculated ppm, final free SO₂, and operator sign-off.
5. Lock in the storage plan: keep topped, minimise air exposure, and schedule the next SO₂ check based on vessel type and risk level.

Step 6 — Escalate exceptions

1. If MLF is not complete, stop and continue MLF management; do not sulphur prematurely.
2. If free SO₂ will not rise as expected, investigate high binding demand, oxygen ingress, microbial activity, or

faulty analysis.

3. If pH is above 3.60, residual sugar is present, or the wine is to be racked/filtered soon, advise the winemaker before proceeding with a routine target.

4. Any unexplained decline in free SO₂, after dosing is a QA issue and must be documented and investigated.

DECISION LOGIC — WHY THIS APPROACH

Post-MLF SO₂ is timed after malic acid conversion is complete because sulphur dioxide can suppress or stop lactic bacteria, leaving the wine part-finished and unstable. The target is driven by pH rather than a single universal ppm because molecular SO₂, protection drops as pH rises; a high-pH wine needs more free SO₂, to achieve the same antimicrobial effect. We also avoid dosing too early because active MLF and warm, unsettled wine will consume SO₂, quickly and create misleading test results. The practical trade-off is between microbial security and sensory impact: enough SO₂, to protect against *Brettanomyces*, acetic bacteria, and oxidation, but not so much that aroma is stripped or the wine becomes harsh. Deviate upward for high-pH, residual sugar, oxygen exposure, or long storage; hold lower if pH is low, the wine is fresh, and prompt bottling is planned.

TRIBAL KNOWLEDGE

Most post-MLF SO₂, problems come from bad mixing, not bad maths. If you only stir the top third of a tank, the lab result will lie to you and the bottom of the vessel will still be vulnerable. Potassium metabisulphite solutions should be made fresh; old solutions drift and can precipitate. On small lots, a recirculation loop through the pump is usually more reliable than hand-rousing. Be careful with wines that have lots of lees fines or high colour extraction — they can bind more SO₂, than expected, so the first dose may look 'lost' on paper. If the wine smells tight or sulphury after dosing, give it time before panicking; if it still smells burnt or dead after equilibration, you likely overdid it.

Free vs Molecular SO₂, Calculation

CELLAR-READY QUICK STEPS

["Measure pH and free SO₂ .", "Set molecular target by wine type.", "Calculate required free SO₂ .", "Dose, mix, and retest.", "Record results and any deviations."]

FULL PROCEDURE

Step 1 — Confirm the wine status before calculating

1. Record the wine ID, vintage, vessel, volume (hL or L), and intended stage: post-fermentation, pre-bottling, or in barrel/tank during storage.
2. Measure **free SO₂**, using the winery's approved method and note wine **pH** at the same time. Free SO₂, alone is not enough; pH drives the molecular fraction.
3. Confirm whether **malolactic fermentation (MLF)** is complete. Do not target bottling SO₂, until MLF is finished and the wine is microbiologically stable. If MLF is active or planned, keep SO₂, low enough not to suppress it.
4. Check wine temperature. For consistent interpretation, use the actual wine temperature, not room temperature.

Step 2 — Set the target molecular SO₂,

1. Use molecular SO₂, as the control point for microbial protection, especially for white and rosé wines and any wine at risk of spoilage.
2. Standard winery targets are:
 - **White/rosé wines:** 0.8 mg/L molecular SO₂,
 - **Red wines:** 0.5 mg/L molecular SO₂,
3. Deviate only with winemaking approval:
 - Lower target if the wine is delicate, low pH, or intended for immediate bottling with minimal sulphur.
 - Higher end of the range if the wine has residual sugar, higher pH, or elevated spoilage risk.
4. Keep in mind that lower pH means more molecular SO₂, at the same free SO₂, level. A wine at pH 3.10 needs much less free SO₂, than a wine at pH 3.60 to achieve the same protection.

Step 3 — Calculate the required free SO₂,

1. Use the winery calculation sheet, lab software, or approved SO₂ table to convert pH and target molecular SO₂ into a required free SO₂ (mg/L, ppm).
2. As a practical guide, approximate targets are:
 - pH 3.10: about 19 mg/L free SO₂ for 0.8 mg/L molecular
 - pH 3.20: about 24 mg/L free SO₂ for 0.8 mg/L molecular
 - pH 3.30: about 30 mg/L free SO₂ for 0.8 mg/L molecular
 - pH 3.40: about 38 mg/L free SO₂ for 0.8 mg/L molecular
 - pH 3.50: about 47 mg/L free SO₂ for 0.8 mg/L molecular
3. For red wines targeting 0.5 mg/L molecular SO₂, use the same calculator/table and record the resulting free SO₂ target.
4. If pH is very high or the calculated free SO₂ is impractical, escalate to the winemaker. Consider pH adjustment, tighter hygiene, topping, filtration, or early packaging rather than chasing excessive sulphur.

Step 4 — Compare actual free SO₂ to target and decide the action

1. Subtract current free SO₂ from target free SO₂.
2. If the result is 0 to 5 mg/L short, retest after blending, topping, or normal cellar handling before dosing.
3. If the wine is more than 5 mg/L short, calculate the addition.
4. If the wine is above target, do not add SO₂. Recheck the analysis and confirm the sample was representative.

Step 5 — Calculate and add SO₂ safely

1. Calculate addition using the product strength and vessel volume.
2. Typical dosing references:
 - Potassium metabisulphite powder (57% SO₂): add only the amount needed to reach target.
 - As a guide, 10 g/hL potassium metabisulphite delivers about 5.7 mg/L SO₂.
3. Dissolve powder fully in clean wine or water before addition; never dump dry powder directly into a vessel.
4. Add to a mixing stream, pumpover, or with vigorous tank circulation. Barrel additions must be split and rolled or inert-gas mixed.
5. Taste and smell after mixing. Check for reduction, oxidation, or H₂S before releasing the wine for the next stage.

Step 6 — Verify and document

1. Recheck free SO₂ after adequate mixing: same day for tank additions, next day for barrels if circulation is limited.
2. Record pH, free SO₂, target molecular SO₂, calculated free SO₂ target, addition rate, product lot number, date, operator, and final result.
3. File any deviation notes, especially where pH, residual sugar, or MLF status altered the target.
4. Review the result before bottling, transfer, or long storage. SO₂ management is only effective when the calculation and the follow-up check both happen.

DECISION LOGIC — WHY THIS APPROACH

Molecular SO₂ is the control variable because it represents the antimicrobial fraction that actually protects the wine. Free SO₂ by itself is misleading: at low pH, a modest free level gives strong protection; at high pH, the same free level is far less effective. That is why pH must be measured every time the calculation is done. The SOP uses standard molecular targets of 0.8 mg/L for white/rosé and 0.5 mg/L for red wines because those levels balance protection with sensory impact in most boutique cellar conditions. We deviate downward for fragile wines and upward only when spoilage risk, residual sugar, or packaging delays justify it. Alternatives like “always dose to a fixed free SO₂ number” are simpler but unsafe across varying pH. If the calculated free SO₂ is excessive, the correct response is usually to improve hygiene, reduce pH, or hold the wine in a safer state rather than force a large sulphur addition.

TRIBAL KNOWLEDGE

Always use the same lab method for free SO₂ through a batch; switching methods will make your trend line useless. In barrels, SO₂ can stratify or take time to equilibrate, so a one-off test right after dosing can read falsely high or low depending on how well you mixed. If the wine has any sign of reduction, don’t “fix” it with

more SO₂, until you've confirmed the cause — sulphur can make a rotten-egg issue harder to shift. Keep an eye on headspace and topping; a clean, topped vessel often needs less SO₂, than a neglected one. Most failures are not calculation errors, they're sampling errors or poor mixing.

Sanitation & Equipment

Tank Cleaning & Sanitisation

CELLAR-READY QUICK STEPS

- Remove gross solids and pre-rinse the tank.
- Circulate detergent wash through all surfaces.
- Rinse until clear and neutral.
- Sanitise immediately before filling.
- Record clean, inspect, and sign off.

FULL PROCEDURE

Step 1 — Prepare the tank and isolate the work area

1. Confirm the tank is empty, depressurised and tagged out for cleaning. No person is to enter a tank unless confined-space controls are in place and signed off.
2. Remove all product, lees, tartrate build-up, gaskets, manways and fittings that can be taken off for separate cleaning.
3. Check the tank internally with a torch: look for skins, tartrate crystals, dried wine film, mould, fruit flies, sludge and any visible corrosion or pitting.
4. Secure drainage so all washings are captured and routed to the effluent system. Prevent cleaner runoff entering stormwater.

Step 2 — Dry remove gross soil

1. Scrape or shovel out solids first. Do not rely on chemicals to remove thick lees or dried residue.
2. Vacuum loose material where available.
3. Pre-rinse with potable water at ambient to 40 °C to loosen remaining soil. Use enough flow to remove debris, but avoid atomising residue into the air.
4. Inspect drain points, valve seats and lower seams for trapped solids.

Step 3 — Apply detergent wash

1. Prepare an alkaline winery detergent at the supplier's recommended rate, typically 1–2% v/v or 10–20 g/L, unless the tank soil load requires otherwise.
2. For heavily soiled fermentation vessels, circulate hot wash water at 50–60 °C where tank linings and seals permit. Do not exceed equipment temperature limits.
3. Run sprayball or CIP for 20–40 minutes, ensuring complete coverage of roof, side walls, cone, seams, manway and fittings.
4. Confirm return water becomes visibly cleaner and no foam or colour is trapped in dead legs.
5. If tartrate scale or mineral film remains, follow with an approved acid rinse at the manufacturer's dosage.

Step 4 — Rinse to no-residue standard

1. Rinse with potable water until detergent and soil are fully removed.
2. Check final rinse water for clarity, absence of foam, and neutral pH where a meter or strips are used.
3. Smell the tank interior and return line: there should be no alkaline, sour or musty odour.
4. Repeat rinse if any film, slipperiness or colour persists.

Step 5 — Sanitise immediately before use

1. Sanitise only after the tank is visibly clean. Sanitiser is not a cleaner.
2. Use an approved winery sanitiser at label rate. Common options are peracetic acid at 100–200 ppm active or equivalent no-rinse sanitiser, subject to site approval and material compatibility.
3. Wet all internal surfaces, fittings, gaskets, valves and hose ends. Maintain contact time as per label, typically 5–15 minutes.
4. Drain fully. Do not rinse unless the product label specifically requires it.
5. Keep the tank closed once sanitised. Do not leave it open to dust, insects or airborne contamination.

Step 6 — Sanitise ancillary equipment

1. Clean and sanitise all hoses, pumps, racking lines, gaskets, sample taps, valves, thermowells and fittings that touch product.
2. Pay special attention to hose ends and valve threads, as these are common contamination points.
3. Replace cracked seals, swollen gaskets and damaged O-rings immediately.
4. Store cleaned items off the floor and protected from splash and dust.

Step 7 — Verify and release the tank

1. Complete a visual final check under strong light. The tank must be clean, odour-free, and free of standing water before release.
2. Record: tank ID, date, operator, detergent used, sanitiser used, contact times, issues found, and corrective actions.
3. Release the vessel only when it passes inspection and the cellar supervisor signs off.
4. If the tank will sit unused, keep all ports closed and re-sanitise before filling if there has been exposure longer than the site's holding time policy.

Step 8 — Manage cleaning frequency and risk

1. Clean immediately after emptying. Do not allow must, wine or lees to dry on surfaces.
2. After red ferments, clean promptly because skins, tartrate and heat-set deposits become harder to remove once dried.
3. If any H₂S, spoilage, mould, or taint is suspected in the vessel, extend wash time and treat the tank as high risk.
4. Escalate persistent odours, staining or corrosion to the winemaker or maintenance team before reuse.

DECISION LOGIC — WHY THIS APPROACH

This SOP uses a clean-then-sanitise sequence because sanitiser cannot penetrate heavy soil, tartrate or dried wine film effectively. A strong physical wash first gives the best result and reduces chemical load. Hot detergent circulation is preferred for tanks because it reaches seams, sprayballs and dead legs more reliably than hand washing, but temperatures must stay within liner, gasket and seal limits. Peracetic acid or an approved equivalent is practical in wineries because it is fast, leaves minimal residue and suits no-rinse use when applied correctly. The key trade-off is contact time versus turnaround speed: longer contact improves assurance, but over-wetting or leaving tanks open increases contamination risk. Deviate when you have stubborn tartrate, mould, or taint, in which case you need a more aggressive clean, possible acid descale, and supervisor sign-off before release.

TRIBAL KNOWLEDGE

Sprayballs often look like they are working when they are not; check the return pattern and make sure the whole roof is being hit. A partially blocked sprayball will leave a dirty ring that shows up only after the tank dries. Valve seats and sample taps are common hideouts for spoilage bugs, so strip them down rather than just hosing over them. If a tank has held red ferment, expect a stubborn skin line at the liquid level and around the cone welds. Warm water helps, but overly hot water can bake proteins and make the job worse on some surfaces. Keep a dedicated set of clean hoses for post-sanitised work only, and store them capped, not coiled wet on the floor.

Barrel Preparation & Topping Schedule

CELLAR-READY QUICK STEPS

- Inspect barrel condition and ullage.
- Top slowly with clean matching wine.
- Set bung tight; wipe all spills.
- Check weekly, then fortnightly when stable.
- Log volumes, SO₂, and any faults.

FULL PROCEDURE

Step 1 — Prepare the barrel and work area

1. Confirm the barrel identity, wine lot, fill history, and intended use before opening the bung.

2. Move the barrel to the topping area or rack on a secure cradle. Chock both sides to prevent rolling.
3. Check the exterior for leaks, loose hoops, damaged staves, and bung thread condition.
4. Ensure all hoses, bungs, thieves, pumps, and measuring tools are clean and sanitised. Use only food-grade materials.
5. Record barrel number, wine, date, operator, and last topping date in the cellar log.

Step 2 — Inspect the barrel internally

1. Remove the bung and smell immediately. Note any volatile acidity, sulphur, vinegar, mould, wet cardboard, or oxidation.
2. Take a small sample from the top of the barrel. Check clarity, aroma, and temperature.
3. If the wine shows fault, isolate the barrel and notify the winemaker before topping.
4. Measure ullage by gauge stick or calibrated topping rod. Target headspace after topping: minimal, with bung hole safely wetted but not overfilled.
5. If the barrel is dry or severely low, assess soak-up risk and whether an intermediate fill is needed before final top.

Step 3 — Prepare topping wine

1. Use wine of the same variety, vintage, lot, and stage of maturation wherever possible.
2. Topping wine must be microbiologically stable, clean, and sensorially sound.
3. Verify free SO₂ target before use. As a practical cellar target, maintain the receiving wine within the site's protection program, typically 25–35 mg/L free SO₂ depending on pH and cellar policy.
4. Match temperature within 2–3 °C of the barrel contents to reduce contraction/expansion losses and oxidative pickup.
5. If topping wine has been stored in tank, mix thoroughly before use to avoid stratification.

Step 4 — Top the barrel

1. Add wine slowly using a clean, sanitised hose fitted to minimise splashing.
2. Stop 1–2 cm below the bung hole. Do not overfill into the bung cavity.
3. Allow 5–10 minutes for soak-up in older or drier barrels, then recheck level and top again if needed.
4. Wipe any spilt wine from the head and chime to avoid contamination and stickiness.
5. Replace the bung tightly but without forcing or damaging threads/seals.

Step 5 — Protect the wine after topping

1. Minimise headspace immediately after topping. If the barrel cannot be filled properly, transfer to a smaller vessel or use inert gas blanketing until the next fill opportunity.
2. If the wine is intended for no malolactic fermentation, sulphite promptly and thoroughly mix where practical.
3. If malolactic fermentation is planned, do not add SO₂ unless directed by the winemaker.
4. Check for leaks at the bung, stave joints, and croze after 15–30 minutes.
5. Refit or replace any bung that weeps, smells of oxygen ingress, or does not seat properly.

Step 6 — Set the topping schedule

1. Newly filled barrels: inspect and top weekly for the first month.
2. Active ageing barrels: inspect and top every 2 weeks, or weekly in warm weather or low-humidity cellars.
3. Older barrels, porous oak, or barrels showing repeated loss: inspect weekly until stable.
4. After any racking, transportation, or temperature swing, inspect within 24 hours.
5. Increase frequency when evaporation losses, ullage, or oxidation risk rise.

Step 7 — Record, review, and escalate

1. Document fill level, volume added, SO₂ addition if applicable, sensory observations, and any faults.
2. Flag barrels that require repeated top-up volumes greater than normal, as this may indicate leakage, excessive evaporation, or microbial spoilage.
3. Isolate and report any barrel with persistent off-odours, excessive ullage, or structural defects.
4. Review topping records monthly to identify trends by barrel age, cooperage, room temperature, and humidity.

5. Replace or retire barrels that cannot be kept sound with reasonable topping frequency.

DECISION LOGIC — WHY THIS APPROACH

This SOP keeps barrel ageing simple and defensive: minimise oxygen, maintain microbial stability, and catch faults early. Frequent inspection and prompt topping are preferred over letting ullage build and then correcting it later, because headspace oxygen accelerates oxidation and can encourage spoilage organisms. Using same-variety, same-lot topping wine reduces flavour dilution and avoids blending in instability or mismatched maturation characters. Matching temperature helps prevent volume contraction from giving a false fill level and reduces dissolved oxygen pickup during topping.

SO₂ is handled conditionally because the correct action depends on whether malolactic fermentation is desired. If MLF is planned, sulphiting too early can suppress it; if not, prompt protection is warranted. Weekly checks for new or leaky barrels and 2-week checks for stable ageing barrels strike a practical balance between labour and risk. Deviate upward in frequency in hot, dry cellars, with older barrels, or after movement/racking, and deviate downward only when a barrel has proven exceptionally stable and the cellar environment is controlled.

TRIBAL KNOWLEDGE

Older barrels often need a second top after 10–15 minutes because the oak can drink the first fill. Don't trust the first sight-glass or dip if the wine has just come from a cool tank into a warmer cellar; let it settle before finalising the level. A bung that looks fine can still weep under slight vacuum, so always feel around the bung and croze after filling. In practice, barrels with recurring loss usually tell you something: cracked stave, loose hoop, or a room that's too warm and dry.

Keep a dedicated topping hose for finished wine only, and don't use the same line that handles dirty lees or racking waste. If the topping wine smells tired, oxidised, or bretty, reject it rather than "hiding" the problem in a barrel. Good topping is about consistency: same temperature, same lot, clean gear, and no splashing.

Hose, Pump & Transfer-Line Hygiene

CELLAR-READY QUICK STEPS

["Drain and rinse lines immediately after transfer.", "Wash with 1–2% alkaline cleaner at 40–55 °C.", "Sanitise at label rate before next use.", "Inspect gaskets, clamps, and hose liners.", "Log defects and quarantine suspect equipment."]

FULL PROCEDURE

Step 1 — Define when transfer-line hygiene is required

Carry out hose, pump and transfer-line cleaning before first use each day, between lots, after any interruption where wine sits in the line for more than 30 minutes, and immediately after moving juice or wine from fermenter to press, tank, or temporary vessel. Treat any line that has carried grape solids, press-run wine, lees, or spoiled wine as contaminated until cleaned. Do not leave wine in hoses or pumps overnight. Residual product will oxidise, support microbial growth, and carry taint into the next lot.

Check that the transfer is actually needed: confirm the lot, vessel ID, and destination before connecting. Use only food-grade hoses with intact seals, sound camlocks/clamps, and no cracked liners. Remove any hose with abrasion, swelling, odour, or sticky film from service.

Step 2 — Disassemble, drain, and segregate equipment

Isolate the line and depressurise it fully before opening any connection. Wear gloves and eye protection when handling caustic, peroxide, or hot water. Drain all product into the correct receiving vessel; do not flush product to waste unless the lot is unfit for use.

Break the transfer set into practical sections: hose, gaskets, pump head, strainer, sight glass, and fittings. Separate hoses used for free-run wine from those used for press-run, lees, or spoilage work. Where possible, dedicate hoses to clean wine, press work, and cellar drains. Keep fittings off the floor and protect open ends with clean caps.

Step 3 — Rinse out all visible solids

Pre-rinse immediately after transfer with clean water at 30–40 °C. Use enough flow to clear the full bore of the hose and pump casing. Continue until discharge runs free of visible solids, seed fragments, skins, tartrate, or colour. For sticky residues, pulse the pump briefly with water to move trapped material, but do not run the pump dry.

Check the rinse water at the outlet. If it remains coloured or cloudy after the first flush, repeat until the line is visibly clean. Do not proceed to chemical wash while solids remain; organic load will consume the cleaner and reduce sanitation.

Step 4 — Wash with the correct chemical cycle

Use a cellar-approved alkaline wash for wine stone, protein, and general organic soil. Typical working concentration is 1–2% w/v, or per supplier instruction. For stainless steel pumps and hard lines, circulate the wash for 10–15 minutes at 40–55 °C, ensuring full contact with all internal surfaces. If using a diaphragm, lobe, or impeller pump, prime fully and circulate through the entire transfer path, including bypass loops and dead legs where fitted.

For heavily coloured or phenolic residues from press work, extend contact time to 20 minutes and manually brush external fittings if required. Avoid excessive caustic on soft plastic hoses that are not rated for it. Verify compatibility before use. Never mix cleaning chemicals.

Step 5 — Rinse to neutral and sanitise before reuse

After the alkaline wash, rinse with potable water until runoff is clear and free of cleaner. Confirm there is no slippery feel or alkaline odour. If local water is hard or scales form readily, a periodic acid rinse may be needed to remove mineral deposits, but only on a scheduled basis and never as a substitute for daily washing.

Sanitise the clean line immediately before reuse with an approved winery sanitiser at label rate. Keep the full contact time specified by the product, typically 5–10 minutes. Drain thoroughly. For lines carrying clean wine, a final sterile or filtered rinse is only used where the cellar SOP specifically requires it.

Step 6 — Inspect, reassemble, and protect the line

Inspect hose interiors, seals, and pump internals before reassembly. Replace damaged O-rings, perished gaskets, and tired hose clamps. Confirm all connections are tight but not over-compressed. Check that the pump rotates correctly and that there are no leaks at the shaft, housing, or fittings.

Cap or hang hoses off the floor in a clean, dry area with ends protected from dust and insects. Label hoses by service type if the winery uses multiple line sets. Store pump covers, gaskets, and quick-connects in a clean crate or designated cupboard.

Step 7 — Record the clean and release for service

Enter the cleaning event in the cellar log: date, operator, equipment ID, cleaner used, contact time, and any faults found. Note whether the line handled free-run, press-run, lees, or any suspect wine. Do not release equipment back into service until the cleaning record is complete and the supervisor has signed off on any defect or replacement need.

If a line has held wine with visible spoilage, VA taint, or microbial instability, flag it for a more aggressive clean and possible inspection before reuse. When in doubt, take it out of service until it passes inspection.

DECISION LOGIC — WHY THIS APPROACH

This procedure prioritises fast removal of organic residue, then chemical cleaning, then sanitation. That sequence matters because grape solids, lees, tartrate and wine stone shield microbes from sanitiser and make pumps harder to clean. Warm water and immediate rinsing are the simplest, safest first step; they

prevent residue from drying and reduce chemical demand. An alkaline wash is the main workhorse because it removes the oily, phenolic and proteinaceous films left in hoses and pump heads after transfer or pressing. Acid is kept as an occasional descaling tool rather than a daily cleaner, which avoids unnecessary corrosion and hose damage. The emphasis on segregation between free-run, press-run and spoilage work reflects practical risk: a clean wine line can be tainted by one contaminated transfer if hoses are shared casually. Deviate only if the product manufacturer specifies a different concentration, if hose material limits chemical choice, or if a contaminated line needs escalation to a more intensive clean and inspection.

TRIBAL KNOWLEDGE

The biggest mistake is letting hoses sit dirty until after knock-off; by then the film has set and the smell gets into the rubber. A short immediate rinse saves a lot of caustic later. Keep one set of hoses for clean wine and another for press and lees if the budget allows — it pays back in fewer taint scares. With diaphragm and lobe pumps, the dead spots are usually at the ports and around the seals, not the obvious straight-through section. Camlocks and gaskets fail quietly; if you are tightening harder each week, the seal is already done. A clean hose should smell like nothing at all. If it has any sour, mouldy, or phenolic note after washing, pull it from service.

Yeast & Fermentation

Yeast Rehydration & Inoculation

CELLAR-READY QUICK STEPS

Sanitise tools and confirm yeast lot.

Rehydrate in 35–40 °C water, 1:10.

Temper slurry to within 10 °C.

Inoculate with thorough tank mixing.

Log temperatures, nutrients, and start time.

FULL PROCEDURE

Step 1 — Confirm readiness and hygiene

1. Verify the must/juice is at the intended inoculation temperature: typically 18–28 °C for red ferments, unless a site-specific recipe states otherwise.
2. Check tank/barrel cleanliness, drain clearances, temperature probe calibration, and that all additions are recorded on the batch sheet.
3. Prepare a sanitised rehydration vessel, mixing paddle, thermometer, and a floating thermometer or probe for ferment monitoring.
4. Confirm the yeast strain, lot number, and expiry date. Do not use damaged or expired sachets.
5. Wear PPE: gloves, eye protection, and dust mask when handling dry yeast and nutrients.

Step 2 — Rehydrate the yeast

1. Use clean, chlorine-free water at 35–40 °C. Do not exceed 40 °C.
2. Add water at 10 times the yeast weight (1:10 w/v). Example: 500 g yeast into 5 L water.
3. Sprinkle yeast evenly onto the water surface without clumping. Allow 15 minutes undisturbed.
4. Gently stir to fully disperse. The slurry should be smooth, cream-like, and free of dry pockets.
5. If using a protective rehydration nutrient, add only as per supplier instructions. Do not add DAP at this stage.
6. Let the yeast stand no longer than 30 minutes total from initial wetting to inoculation.

Step 3 — Equalise temperature before inoculation

1. Measure slurry temperature and must temperature.
2. Keep the difference to less than 10 °C. If the gap is greater, temper the slurry by adding small portions of must to the yeast over 10–15 minutes.
3. Avoid thermal shock; a sudden temperature drop or rise reduces viability and can delay start-up.
4. Record both temperatures on the batch sheet.

Step 4 — Inoculate the must

1. Add the rehydrated yeast slurry to the fermenter during a pump-over, mixing, or vigorous hand stir so it disperses evenly.
2. Target complete inoculation coverage across the vessel; avoid dumping into one corner or onto a dry cap.
3. Rinse the rehydration vessel with a small amount of must and add all rinsings to the tank.
4. Close the fermenter in a way that allows CO₂ release but limits oxygen ingress and contamination.
5. Mark the inoculation time as time zero.

Step 5 — Start fermentation monitoring

1. Expect visible activity within 1–3 days, faster in warmer musts and slower in cooler musts.
2. Check sugar, temperature, aroma, and cap condition at least twice daily, and more often for warm or high-Brix ferments.
3. Note that active fermentation can run 10–15 °F (about 6–8 °C) above ambient due to heat generation.
4. If temperature climbs above the site limit, initiate cooling promptly to protect yeast and aroma.
5. If fermentation remains sluggish after 48 hours, confirm temperature, viable yeast count, and nutrient status before making corrective additions.

Step 6 — Manage nutrient support during early fermentation

1. Use a complete yeast nutrient according to the winery protocol and supplier dosage, typically split during early fermentation rather than all at once.
2. Add nutrient when the ferment is active, not into a stalled or nutrient-starved tank without diagnosis.
3. Prefer complete nutrient blends over straight DAP where H₂S risk is present, as nitrogen alone will not correct a broad nutrient deficiency.
4. Record product, lot number, dose, and addition time. Observe for reduction in sulphidic or reductive aromas within 24 hours.

Step 7 — Maintain cap management and observations

1. Punch down or otherwise re-submerge the cap several times per day during active red fermentation.
2. Break up the cap fully and push skins back into contact with the liquid to assist colour/flavour extraction and heat dispersal.
3. Watch for drying, cracking, or surface mould risk if the cap is left exposed to air too long.
4. At each cap management event, measure and record temperature, smell, colour, and bubble activity.
5. Escalate any off-odours, slow starts, or temperature spikes to the winemaker immediately.

Step 8 — Document completion of inoculation phase

1. Confirm ferment has started when CO₂ release, foam, and temperature rise are evident.
2. File all rehydration, inoculation, nutrient, and temperature records in the batch log.
3. Review whether the start-up met target timing and make notes for future yeast selection and temperature control.
4. Dispose of waste yeast slurry and packaging according to winery waste and biosecurity requirements.

DECISION LOGIC — WHY THIS APPROACH

Rehydration is done in warm, clean water because dry yeast cells need controlled hydration before they face sugar, alcohol, and acidity. Keeping the slurry within 10 °C of the must prevents shock and loss of viability. A 1:10 water-to-yeast ratio is standard, practical, and gives enough volume to disperse the cells evenly. Complete nutrients are preferred over DAP alone because fermentation problems usually come from a broad nutrient deficit, not just low nitrogen; DAP can worsen cell imbalance if used as the sole fix. Monitoring temperature early matters because fermentation generates heat and can move well above ambient, changing yeast kinetics and the risk profile. Multiple daily cap management actions are used on reds because they protect against drying, improve extraction, and control heat. Deviate only when the yeast supplier's protocol, vintage conditions, or a specific ferment plan calls for a different approach.

TRIBAL KNOWLEDGE

Use non-chlorinated water for rehydration; chlorine and chloramine can quietly knock yeast around before they even hit the tank. If the slurry looks grainy after 15 minutes, it was usually sprinkled too fast or the water was too cold. A small tempering addition of must works better than a big splash — keep it gentle and staged. In many cellars, the first sign of a problem is smell, not numbers: a faint rotten-egg note after inoculation is a cue to check nutrient timing and temperature immediately. Floating thermometers are handy, but on deep tanks a calibrated probe gives a truer read near the cap.

Yeast Strain Selection

CELLAR-READY QUICK STEPS

["Confirm fruit chemistry and target style.", "Select strain for temperature, alcohol, and nutrient fit.", "Check supplier rehydration and dosing instructions.", "Record inoculation rate, timing, and batch rationale.", "Review performance at press and retain best performer."]

FULL PROCEDURE

Step 1 — Confirm the wine style and fermentation target

1. Review the fruit condition, intended style, and batch size before selecting a yeast strain.
2. Match the strain to the job: colour and tannin extraction for reds, aroma retention for whites/rosés, tolerance for high sugar or high alcohol lots, and any requirement for low-temperature kinetics.

3. Set the target fermentation range for the lot and record it on the batch sheet. As a practical guide, choose a strain that will comfortably operate within the planned must temperature, not just one that looks good on paper.
4. Check must parameters before inoculation: Brix, pH, temperature, and any obvious nutrient limitation or rot pressure. Do not select a strain that is marginal for a problematic lot.

Step 2 — Match strain performance to the must

1. Confirm the strain's alcohol tolerance exceeds the expected finished alcohol by at least 1.0% v/v.
2. For cool ferments, select a strain known to start reliably at 12–18 °C. For warmer red ferments, use a strain that remains clean through 24–32 °C, depending on house style and cap management.
3. If the lot is high in sugar, low in nitrogen, or from compromised fruit, favour a robust strain with strong nutrient efficiency and low hydrogen sulphide risk.
4. Avoid “blind” use of a favourite strain across all parcels; the same yeast can perform well in one lot and stress badly in another.
5. Where aromatic expression is critical, select a strain that preserves varietal character rather than one that pushes overt fermentation-derived character.

Step 3 — Check compatibility with cellar constraints

1. Confirm the strain is available in the required pack size and in-date.
2. Check supplier notes for rehydration temperature, nutrient requirements, and recommended inoculation rate.
3. Verify the strain is compatible with any planned co-inoculation, sequential inoculation, or malolactic timing.
4. If a batch will be fermented under difficult thermal conditions, ensure the cellar can hold the must within the strain's preferred range for the full peak fermentation.
5. Record any deviations from the standard strain choice and note the reason on the batch record.

Step 4 — Define the inoculation plan

1. Set a yeast dose according to supplier guidance and house practice, commonly 20–30 g/hL for active dry yeast unless the lot requires a different rate.
2. For high-risk ferments, use the upper end of the recommended range to improve population density and reduce lag time.
3. Plan hydration water at the supplier's stated temperature range, typically 35–40 °C, with protective nutrient if required by the strain.
4. Prepare the nutrient strategy at the same time as strain selection. A selected yeast that is underfed is still a poor choice in practice.
5. Ensure all additions are documented with timing, rate, and operator initials.

Step 5 — Monitor the start of fermentation

1. Expect visible activity within 1–3 days after inoculation, with temperature being the main driver of lag time.
2. Check the must temperature at least once per shift during the first 72 hours; fermentation itself can run 5–8 °C above ambient in active red ferments.
3. Confirm the cap is forming and that CO₂-driven rise is occurring normally.
4. If fermentation is slow to start, do not rush to change yeast strain mid-lot; first check temperature, hydration practice, and nutrient status.
5. Escalate any sign of H₂S, sluggish kinetics, or stuck fermentation risk to the winemaking lead immediately.

Step 6 — Record, review, and improve future strain choice

1. At press, review the strain's performance against start time, temperature behaviour, aroma, colour extraction, and completion.
2. Note any off-aromas, excessive heat, cap behaviour, or nutrient interventions required.
3. Compare the result with other strains used on similar parcels and retain the best performers for repeat use.
4. Update the winery strain library with practical outcomes, not just technical datasheets.
5. Keep batch notes for future vintage planning, especially where a strain proved too clean, too aggressive,

or too fragile for the site.

DECISION LOGIC — WHY THIS APPROACH

Yeast strain selection is about matching biology to the job, not choosing the most fashionable culture. The best strain is the one that can finish the ferment cleanly at the planned temperature, tolerate the expected alcohol, and suit the fruit condition and wine style. A robust strain with sensible nutrient needs is usually the safer choice for high-Brix, low-nitrogen, or marginal fruit, because stressed yeast are more likely to throw H₂S, VA, or stall. That is why strain selection is paired with temperature and nutrition planning rather than treated as a separate purchasing decision.

The main trade-off is character versus reliability. Highly expressive strains can lift aromatics but may be less forgiving, while neutral workhorse strains often give better kinetics and fewer surprises. Deviate only when the fruit profile, cellar temperature control, or style target clearly justifies it. If a ferment is already under stress, fixing temperature and nutrient availability is more effective than hoping a different strain alone will rescue the lot.

TRIBAL KNOWLEDGE

In the cellar, the datasheet rarely tells you how a strain behaves when the room is hot, the fruit is tired, and the pump is already booked. Keep a short list of strains that have proved dependable on your own site, especially for cold rooms, warm reds, and difficult high-Brix parcels. Some strains look brilliant on aroma in a trial but become sluggish once the cap gets thick and the must heats up.

Watch supplier storage and expiry dates closely; old yeast can still work, but lag time creeps up. If you are splitting a parcel, don't assume every tank will behave the same, because temperature gradients and solids load matter. Practical winemakers also keep notes on hydration quirks, because a strain that needs a gentle rehydration and timely nutrient addition can outperform a more "robust" culture that was handled badly.

Yeast Nutrition & DAP Additions

CELLAR-READY QUICK STEPS

["Check Brix, temperature, and ferment aroma.", "Confirm nutrient plan before adding DAP.", "Weigh dose accurately and dissolve fully.", "Add during punch-down for even dispersion.", "Record dose, temperature, and sensory response."]

FULL PROCEDURE

Step 1 — Confirm fermentation stage and set the target

1. Apply this SOP from the start of visible fermentation through the middle of primary ferment, typically days 2–14.
2. Confirm the ferment is active: visible CO₂ release, cap formation, and/or a drop in Brix/SG.
3. Record must temperature at the time of assessment. Primary fermentation can run 5–8 °C above ambient in a vigorous ferment, so do not rely on room temperature alone.
4. Check whether yeast nutrition is already being supplied via a complete nutrient regime. If Fermaid K or another complete nutrient is in use, do not default to separate DAP.

Step 2 — Assess yeast health before adding anything

1. Smell and taste the ferment if safe to do so. Healthy ferment characters should be clean, bready, fruity, or neutral.
2. Watch for warning signs: sluggish activity, stuck or slowing ferment, reduction in CO₂, rotten egg character (H₂S), rubbery/sulphidic notes, or volatile acidity lift.
3. Take a Brix or SG reading and compare with the previous record. A ferment that is falling normally may not need any intervention.
4. Check the current nutrient plan against the yeast strain, must composition, and YAN history if available. Low-nitrogen fruit and high-solids ferments are higher risk.

Step 3 — Decide whether DAP is appropriate

1. Use DAP only as a targeted nitrogen supplement, not as a fix for an unhealthy ferment.

2. Do not use DAP alone to correct H₂S or a clearly nutrient-starved ferment. In a deficiency state, DAP can increase yeast population without providing the broader nutrients needed for healthy metabolism.
3. Prefer a complete nutrient product where possible, especially early in fermentation or when restarting a sluggish ferment.
4. If Fermaid K is already in the addition schedule, review the label before adding extra DAP; there is usually little reason to duplicate nitrogen.

Step 4 — Calculate the addition rate

1. Base additions on winery trial data, juice analysis, and supplier instructions.
2. As a working cellar guide, keep DAP additions modest and split where possible. Common dosing sits in the range of 10–30 g/hL per addition, with total additions rarely exceeding 40–50 g/hL unless approved by the winemaker.
3. If nutrient status is uncertain, add the lower end first and reassess 12–24 hours later.
4. Record the exact batch volume in hL, the DAP rate in g/hL, and the total mass weighed in grams.

Step 5 — Prepare and add the DAP safely

1. Weigh DAP accurately on a calibrated scale.
2. Dissolve in a small volume of clean must or warm water if approved by site practice; mix until fully dispersed. Avoid adding dry crystals directly into a fermenting cap.
3. Sanitation matters: use clean, food-grade vessels and tools, and keep additions off the floor and away from ignition sources.
4. Add the solution during active fermentation, ideally when the cap is being punched down or otherwise mixed, so the nutrient disperses evenly.
5. If adding more than one nutrient component, split the addition across two doses rather than dumping the full load at once.

Step 6 — Mix thoroughly and monitor response

1. Punch down the cap immediately after addition, or recirculate if that is the approved cellar method, to distribute the nutrient through the must.
2. Record the temperature at the same time as the punch-down.
3. Expect a positive response within 12–24 hours in a healthy ferment: steadier CO₂ evolution, improved aroma, and more consistent temperature movement.
4. Recheck Brix/SG, smell, and temperature at the next scheduled monitoring point.
5. If H₂S persists after a single targeted DAP addition, stop and escalate. Do not keep feeding DAP into a ferment that is obviously struggling.

Step 7 — Reassess and document

1. Update the batch record with date, time, lot, volume, nutrient product, dose, and operator initials.
2. Note the must temperature, Brix/SG, aroma observations, and any cap-management action.
3. If the ferment is nearing dryness, reduce intervention frequency but continue close checks until sugar depletion is confirmed.
4. Escalate to the winemaker if the ferment is hot, stalled, sulfurous, or not responding within 24 hours.
5. Retain records for traceability and compliance with the winery's quality system.

DECISION LOGIC — WHY THIS APPROACH

DAP is a useful tool, but only when the ferment genuinely needs accessible nitrogen and the rest of the nutrient picture is sound. It is not a universal cure for H₂S or slow fermentation. If a ferment is already deficient in broader nutrients, vitamins, or micronutrients, DAP can simply increase the number of stressed cells rather than improve yeast performance. That is why a complete nutrient regime is preferred for prevention and for problematic ferments. Splitting additions is safer than one large hit because it reduces osmotic shock, improves uptake, and gives the cellar time to observe the response. Temperature monitoring matters because fermentation rate and yeast demand change quickly with heat; a warm ferment may not need more nitrogen, just better cap management and attention. Deviate from this SOP only with winemaker approval, especially for stuck ferments, high pH lots, or wines already showing sulphide or VA risk.

TRIBAL KNOWLEDGE

Most nutrient problems show up in the smell before the numbers tell you anything. If the cap smells a bit struck or the ferment goes flat after a hot day, check temperature first, then nutrition. DAP clumps easily in damp air, so keep the bag sealed and weigh it fresh. A small slurry mixed in clean must usually disperses better than dry powder tipped straight in. In practice, Fermaid K or a similar complete nutrient is the better default for most red ferments; separate DAP is usually only for fine-tuning, not rescue work. Also, don't chase a slow ferment with repeated DAP doses if the yeast are already stressed from heat or alcohol — you'll often make the aroma worse, not better.
